O ‘ QUVQO ‘ LLANMA · 2 0 2 6 - YILKUZ

Chiziqli algebra I

5–7 haftalar

Har bir kurs fayli va tayinlangan darslik bo‘limlaridan barcha savollar va formulalar — geometriya va chiqarish, matritsa operatsiyalari va ajratishlar, transponirlash, almashtirishlar va vektorli fazolar — har bir masala to‘liq bosqichma-bosqich yechim bilan. Mustaqil ishlash va Midterm imtihoniga tayyorgarlik uchun tuzilgan Toshkent texnologiya universiteti ·2026-yil Kuz Manbalar: kurs konspektlari, darslar, ISA, amaliy mashqlar; G. Strang, Introduction to Linear Algebra $(4e)$ va Linear Algebra and its Applications $(4e)SAVOLLARBAZASI \cdot FORMULALARKATALOGI \cdot TO‘LIQYECHIMLAROKTABR2026$

Manbalar: kurs konspektlari, darslar, ISA, amaliy mashqlar; G. Strang, Introduction to Linear Algebra $(4e)$ va Linear Algebra and its Applications $(4e)$

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1 Kurs xaritasi va hujjatdan qanday foydalanish

Hafta	Chiziqli algebra I
5	Geometriya va chiqarish: row picture va column picture, Gauss chiqarishi, teskari almashtirish, chiqarish matritsalari E, almashtirish matritsalari P, rang bilan bog‘lanish, $A = LU (Strng §§2.1–2.3)$
6	Matritsa amallari: ko‘paytirish qoidalari va AB ning to‘rt talqini, maxsus matritsalar, teskari matritsalar va ularning chiqarilishi, chiqarish matritsalari, LU/LDU-ajralishlari, transponirliklar, almashtirishlar $(Strng §§2.4–2.7)$
7	Transponirliklar va almashtirishlar: transponirlik identifikatorlari, simmetrik matritsalar, 3×3 oltita almashtirish; vektorli fazolar va kichik fazolar, ustun fazosi $C(A)$, nol fazosi $N(A) (Strng §§2.7, 3.1)$

Ushbu qo‘llanma «Chiziqli algebra I — 1–4 haftalar» rasmiy to‘plamini davom ettiradi va 2026-yil kuz semestrining 5–7 haftalarini qamrab oladi. Unda har bir kurs fayli va tayinlangan darslik bo‘limlaridagi barcha savollar va formulalar to‘plangan — har bir masala to‘liq bosqichma-bosqich yechim bilan, asosiy to‘plamdagi kabi.

Qo‘llanma qamrab olgan kurs fayllari (5–7 haftalar): ma’ruzalar; yechimlari bilan amaliy mashg‘ulotlar va dars masalalari; mustaqil ish topshiriqlari (ISA); 5-hafta kengaytirilgan Beamer nazariy konspekti; EXTRA material (6-hafta misollar bilan takrorlash). Darsliklar: G. Strang, Introduction to Linear Algebra (4-nashr) va parallel Linear Algebra and Its Applications (4-nashr).

Har bir hafta qanday tuzilgan. Avval — formulalar katalogi: haftaning barcha ta’riflari, teoremalari va qoidalari izohlar bilan. So‘ngra har bir manba fayli uchun «savollar va to‘liq yechimlar» bo‘limi: dars masalalari, amaliy mashg‘ulot, mustaqil ish topshiriqlari (ISA) va qo‘shimcha material. Masalalar so‘zma-so‘z keltirilgan; yechimlar to‘liq, bosqichma-bosqich, tekshiruvlar bilan.

Belgilash. Matritsalar qatorma-qator yoziladi: $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ — birinchi qator ab, ikkinchi qator cd; ustunli vektor $\begin{bmatrix} x \\ y \end{bmatrix}$, qatorli vektor $\begin{bmatrix} x & y \end{bmatrix}$ deb yoziladi. Darajalar va indekslar ^{} va _ orqali: A^{-1}, A^T, a_{11}. 5–7 haftalar kursning birinchi yarmidan hisoblanadi: chiqarish va matritsa amallari, $A = LU = LDU = PA$ ajratishlari, transponirlash, almashtirishlar va vektorli fazolar. 1–4 haftalar materiali bilan birga ular 8-hafta Midterm uchun ma’lum deb hisoblanadigan bazani tashkil qiladi.

2 5-hafta — Geometriya va chiqarish (Chiziqli algebra I)

5-kurs haftasi Chiziqli algebra $I (TT2026) Ax = b$ tenglamasining geometriyasi va uning yechimi — Gauss usuli bilan bog‘laydi. Manbalar: Strang, Introduction to Linear Algebra, 4th ed., §§2.1–2.3; Linear Algebra and Its Applications, 4th ed., §§1.2–1.4; Beamer-konspekt «Geometry and Elimination» (80 daqiqa ma’ruza, 90

daqiqa masalalar). Hafta reji: rang va teskari matritsalarni takrorlash $(10min)$, qator va ustunlar tasviri $(20min)$, eliminatsiya va teskari almashtirish $(25min)$, eliminatsiya E va almashtirish P matritsalari $(20min)$, xulosa $(5min)$. Maqsadlar: $Ax = b$ ni to‘g‘ri chiziqlar va tekisliklarning kesishishi (qator tasviri) va ustunlarning chiziqli birikmasi (ustun tasviri) sifatida ko‘rish; ko‘paytuvchilar bilan eliminatsiyani va teskari almashtirishni amalga oshirish; qator operatsiyalarini E matritsalari bilan yozish, ularni E^{-1} matritsalari bilan bekor qilish, tenglamalarni P matritsalari bilan almashtirish. Belgilash: matritsa satrlar bo‘yicha yoziladi — $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ (birinchi satr ab, ikkinchi cd); ustun-vektor $\begin{bmatrix} x \\ y \end{bmatrix}$; kengaytirilgan matritsa $[A|b]$.

2.1 Chiziqli algebra I, 5-hafta: formulalar katalogi

[Manba: 5-hafta «Geometry and Elimination» ma’ruza konspekti (Chiziqli algebra $I, TT2026$): Strang, Introduction to Linear

Algebra, 4th ed., §§2.1–2.3; Linear Algebra and Its Applications, §§1.2–1.4]

2.6.1 Rang bilan bog‘liqlik: n ta asoschi elementlar teskari matritsa mavjudligini bildiradi

Takrorlash: $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ rangi 1 va teskari matritsaga ega emas; $Ax = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$ cheksiz ko‘p yechimga, $Ax = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$ esa hech qanday yechimga ega emas. Asosiy fakt: $n \times n$ kvadrat matritsa uchun n ta asoschi element (pivot) mavjudligi teskari matritsa mavjudligiga teng — bu holda $Ax = b$ barcha b uchun yagona yechimga ega; eliminatsiya topilgan asoschi elementlar soni rangdir. Ikkala algoritm ham $[A|b]$ da ishlaydi: Gauss–Jordan matritsani qisqartirilangan bosqichli shakl (reduced echelon form) yetkazadi; Gauss eliminatsiyasi yuqori uchburchak U da to‘xtaydi va teskari almashtirish bilan tugaydi.

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} : rankA = 1, A^{-1} yo'q;$$

$$Ax = \begin{bmatrix} 3 \\ 6 \end{bmatrix} — cheksiz ko'p yechimlar, Ax = \begin{bmatrix} 3 \\ 7 \end{bmatrix} — bittahamyo'q$$

2.6.2 Qator tasviri: qatorlar to‘g‘ri chiziqlar va tekisliklar sifatida

$$2x - y = 1, x + y = 5;$$
$$\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$$

Har bir qator — bitta geometrik cheklov: 2D da to‘g‘ri chiziqlar, 3D da tekisliklar; yechim — ularning umumiy nuqtasi. Misol to‘g‘ri chiziqlari $(x, y) = (2, 3)$ da kesishadi; 3D da uchta tekislik:

$$\begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix} \cdot \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 9 \end{bmatrix} — tekisliklar(u,$$
$$v, w) = (1, 1, 2) dakesishadi$$

2.6.3 Ustun tasviri: Ax — ustunlarning chiziqli birikmasi

$$x \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} + y \cdot \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix};$$
$$tekshirish: 2 \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 3 \cdot \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$$

Strang ta’kidlaydi: Ax — A ustunlarining chiziqli birikmasidir — $x_1 \cdot a_1 + \ldots + x_n \cdot a_n = b$ bo‘lishi uchun koeffitsientlarni tanlaymiz. Uch o‘lchovli variant (ogirliklar (1, 1, 2)):

$$u \cdot \begin{bmatrix} 2 \\ 4 \\ -2 \end{bmatrix} + v \cdot \begin{bmatrix} 1 \\ -6 \\ 7 \end{bmatrix} + w \cdot \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 9 \end{bmatrix}$$

Ikkala tasvir ham bir xil $Ax = b$ tenglamaga: qatorlar — kesishishlar (tenglamalar), ustunlar — chiziqli birikmalar (vektorlar).

2.6.4 Uchta mumkin bo‘lgish natija

- • Bitta yechim — mustaqil cheklovlar. • Cheksiz ko‘p yechimlar: $x + y = 2$ va 2x + 2y $= 4$ — ikkinchi tenglama

birinchisining ikki barobar katta, to‘g‘ri chiziqlar ustma-ust tushadi.

- • Yechim yo‘q: $x + y = 2$ va 2x + 2y $= 5$ — chap tomonlar proporsional, o‘ng

tomonlar emas, to‘g‘ri chiziqlar parallel.

$$istisnochiqarishnatijaniko'rsatadi: [00 \ldots 0|0] \to 0 = 0 — cheksizko'pyechimlar;$$
$$[00 \ldots 0|c], c$$

$$\neq 0, \to 0 = c — yechimyo'q$$

2.6.5 Asoschi element, ko‘paytuvchi va to‘g‘ridan-to‘g‘ri yurish

$$\lambda_{ij} = \frac{a_{ij}}{a_{jj}}, amaliyotR_i \leftarrow R_i - \lambda_{ij} \cdot R_j$$

a_{ij} elementini nol bo‘lmagan asoschi element a_{jj} ostidan chiqarish uchun (kopaytuvchi bu yerda λ_{ij} bilan belgilanadi), i-satrga j-satrni λ_{ij} ga ko‘paytirib ayiriladi. To‘liq misol: $\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 4 & -6 & 0 & -2 \\ -2 & 7 & 2 & 9 \end{array}\right]$; asoschi element 2, ko‘paytuvchilar $\lambda_{21} = \frac{4}{2} = 2$ va $\lambda_{31} = -\frac{2}{2} = -1$:

$$R_2 \leftarrow R_2 - 2R_1,$$
$$R_3 \leftarrow R_3 + R_1 : \left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 8 & 3 & 14 \end{array}\right]$$

Ikkinchi asoschi element -8, ko‘paytuvchi $\lambda_{32} = 8/(-8) = -1$, operatsiya $R_3 \leftarrow R_3 + R_2$. Uchburchak tizma va to‘g‘ridan-to‘g‘ri yurish natijasi:

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 0 & 1 & 2 \end{array}\right],$$
$$U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{bmatrix}, asosiy elementlar 2, -8, 1$$

2.6.6 Teskari almashtirish

$$w = 2; -8v - 2w = -12 \to v = 1;$$
$$2u + v + w = 5 \to u = 1; x = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

Uchburchak tizma pastdan yuqoriga echiladi: oxirgi tenglamada faqat w, oldingisida v va w bor. Javobni asl tenglamaga almashtirib tekshiriladi: $\begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 9 \end{bmatrix}$. Satrni almashtirish teskari amalga ega ($2R_1 niqo$shib qo‘ying), shuning uchun eliminatsiya yechimlarni saqlaydi; lekin operatsiya chap va o‘ng tomonlarga ham qo‘llaniladi, aks holda tenglamalar ekvivalent bo‘lmaydi.

2.6.7 Eliminatsiya E matritsalari

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

$$E_{21} \cdot A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ -2 & 7 & 2 \end{bmatrix}, E_{21} \cdot b = \begin{bmatrix} 5 \\ -12 \\ 9 \end{bmatrix}$$

Chapdan ko‘paytirish qator operatsiyasini amalga oshiradi va o‘ng tomon ham xuddi shu matritsa bilan o‘zgaradi: $Ax = b(E_{21}A)x = E_{21}b$ ga aylanadi. U gacha to‘liq yurish ($o$ngdagi ko‘paytuvchi birinchi ishlaydi — ko‘paytirish tartibi muhim):

$$E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix},$$

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, E_{32} \cdot E_{31} \cdot E_{21} \cdot A = U$$

Bosqichni bekor qilish: ayirish o‘rniga qo‘shish kiritiladi — $E = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$ uchun teskari $E^{-1} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$ va $E^{-1} \cdot E = I$.

2.6.8 Nol asoschi element: almashtirish P matritsalari

$$\left[\begin{array}{cc|c} 0 & 2 & 4 \\ 3 & 1 & 7 \end{array}\right] \to \left[\begin{array}{cc|c} 3 & 1 & 7 \\ 0 & 2 & 4 \end{array}\right] (qatorlarnialmashtirish),$$

$$keyiny = 2, x = \frac{5}{3}$$

Nol asoschi element mutlaqo aynanlikni bildirmaydi: pastda nol bo‘lmagan element bo‘lsa, satrlar almashtiriladi. Almashtirish almashtirish matritsasi bilan amalga oshiriladi va uo‘ng tomonni ham o‘zgartiradi:

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

$$P \cdot \begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}, P \cdot \begin{bmatrix} 4 \\ 7 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \end{bmatrix}, ekvivalentsistemaPAx = Pb$$

P xossalari: har bir qator va ustunda bitta 1, qolganlari nol; $P^{-1} = P^T$; bitta almashtirish uchun $P^2 = I$; 1 va 3-qatorlarni almashtirish — $P_{13} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$; chapdan

ko‘paytirish tenglamalarni, emas, noma’lumlarni o‘zgartiradi. Eliminatsiyadagi muvaffaqiyatsizliklar: vaqtinchalik — nol asosiy elementda uning pastida nol bo‘lmagan element (almashtirish yordam beradi); doimiy — ustunda mavjud asosiy element yo‘q(singanlik: yechim yo‘q yoki cheksiz ko‘p yechim).

2.6.9 Oldinga qarash: $A = LU$ va $PA = LU$

$$almashtirishsiz A = LU (U — yuqoriuchburchak,$$
$$Lchiqaruvchilarnisaqlaydi); almashtirishlarbilan$$

$$PA = LU$$

Hafta yakuni: qatorlar — kesishishlar, ustunlar — kombinatsiyalar; eliminatsiya — asosiy elementlar ostidagi nollar va teskari almashtirish; E — qatorni almashtirish, P — qatorlarni almashtirish. LU-ajratish $(A = LU, PA = LU)$6-haftada quriladi.

2.2 Chiziqli algebra I, 5-hafta: dars mashqlari — savollar va to‘liq yechimlar

[Manba: 5-hafta dars masalalari (Proyeksiya), masala 1–15($law5$classg01–$g11$ va 23–31-bet matn qatlami)]

Mashq 1. Qatorlar va ustunlar tasviri. Tizimni $\begin{bmatrix}2 & -1\\ 1 & 1\end{bmatrix}\cdot\begin{bmatrix}x\\ y\end{bmatrix}=\begin{bmatrix}1\\ 5\end{bmatrix}$ yeching va ikkala geometrik tasvirlarni tavsiflang.

Yechim. Tenglamalarni qo‘shamiz: 3x $= 6, x = 2$, undan $y = 5 - 2 = 3$. 2x $- y = 1$ va $x + y = 5$ to‘g‘ri chiziqlari (2, 3) nuqtada kesishadi; ustunlar: $2\cdot\begin{bmatrix}2\\ 1\end{bmatrix}+3\cdot\begin{bmatrix}-1\\ 1\end{bmatrix}=\begin{bmatrix}4 & & 3\\ 2 & + & 3\end{bmatrix}=\begin{bmatrix}1\\ 5\end{bmatrix}$. Javob: $x = 2, y = 3$.

Mashq 2. Uch holat. Tizimni $\begin{bmatrix}1 & 1\\ 2 & 2\end{bmatrix}\cdot\begin{bmatrix}x\\ y\end{bmatrix}=\begin{bmatrix}2\\ c\end{bmatrix}$ $c = 4$ va $c = 5$ uchun tasniflang.

Yechim. $R_2 \leftarrow R_2 - 2R_1$ [00|$c - 4$] qatorni beradi. $c = 4$ da bu $0 = 0$: $x + y = 2$ qoladi, cheksiz ko‘p yechimlar — $\begin{bmatrix}2 & & t\\ t & & \end{bmatrix}$. $c = 5$ da bu $0 = 1$ — ziddiyat, yechim yo‘q.

Mashq 3. Ko‘paytchini toping. Kengaytirilgan matritsada $\left[\begin{array}{cc|c}2 & 3 & 7\\ 8 & 13 & 29\end{array}\right]$ birinchi asosiy element ostidagi hadni chiqaring va ikkala asosiy elementni toping.

Yechim. Ko‘paytchi $\lambda_{21} = \frac{8}{2} = 4$. $R_2 \leftarrow R_2 - 4R_1$: [8, 13 | 29] $-4\cdot$[2, 3 | 7] = [0, 1 | 1]. Natija: $\left[\begin{array}{cc|c}2 & 3 & 7\\ 0 & 1 & 1\end{array}\right]$; asosiy elementlar 2 va 1.

Mashq 4. Orqaga almashtirish. Tizimni $\begin{bmatrix}2&1&-1\\0&3&2\\0&0&4\end{bmatrix}\cdot\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}3\\7\\8\end{bmatrix}$ **yeching.**

Yechim. Pastdan yuqoriga: 4z $=8, z=2$; 3y $+2\cdot 2=7, y=1$; 2x $+1-2=3, x=2$. Javob: $x=\begin{bmatrix}2\\1\\2\end{bmatrix}$. Tekshirish: $2\cdot 2+1-2=3, 3+4=7, 8=8$ — to‘g‘ri.

Mashq 5. To**‘liq chiqarib tashlash. Tizimni**

$$\left[\begin{array}{ccc|c}1&1&1&6\\2&3&1&10\\1&2&3&14\end{array}\right]$$

chiqarib tashlash usuli bilan yeching.

Yechim. $R_2\leftarrow R_2-2R_1\to$ [0 1 −1 | −2]; $R_3\leftarrow R_3-R_1\to$ [0 1 2 | 8]; $R_3\leftarrow R_3-R_2\to$ [0 0 3 | 10], ya’ni 3z $=10$. Orqaga almashtirish: $z=\frac{10}{3}$; $y-z=-2\to y=\frac{4}{3}$; $x+y+z=6\to x=\frac{4}{3}$. Javob: [4/3; 4/3; 10/3]. Tekshirish: qatorlar 6, 10, 14 beradi — to‘g‘ri.

Mashq 6. Elementar matritsani aniqlang. 3×3 **matritsida** $R_3\leftarrow R_3-4R_1$ **amalini amalga oshiruvchi** E **ni toping.**

Yechim. E — birlik matritsida $(3,1)o$‘rinda −4 bo‘lgan matritsa: $E=\begin{bmatrix}1&0&0\\0&1&0\\-4&0&1\end{bmatrix}$; EA ning uchinchi qatori (3-qator) $-4\cdot$(1-qator), 1–2-qatorlar o‘zgarmaydi. Aks operatsiya — qo‘shish: $E^{-1}=\begin{bmatrix}1&0&0\\0&1&0\\4&0&1\end{bmatrix}$ da $(3,1)o$‘rinda $+4$.

Mashq 7. EA **ko‘paytmasi.** $E=\begin{bmatrix}1&0\\-2&1\end{bmatrix}$ **va** $A=\begin{bmatrix}3&1\\6&5\end{bmatrix}$ **bo‘lsin.** EA **ni hisoblang va amalni tushuntiring.**

Yechim. Ko‘paytmaning 1-qatori: $\begin{bmatrix}3&1\end{bmatrix}$; 2-qatori: $-2\cdot\begin{bmatrix}3&1\end{bmatrix}+\begin{bmatrix}6&5\end{bmatrix}=\begin{bmatrix}0&3\end{bmatrix}$. Shunday qilib, $EA=\begin{bmatrix}3&1\\0&3\end{bmatrix}$: E2-qatordan 1-qatarning ikkilanganini ayiradi — $\frac{6}{3}=2$ ko‘paytchili chiqarib tashlash.

Mashq 8. O**‘zgartirilgan** o**‘ng tomon.** $A=\begin{bmatrix}3&1\\6&5\end{bmatrix}$ **va** $b=\begin{bmatrix}4\\17\end{bmatrix}$ **uchun** $E=\begin{bmatrix}1&0\\-2&1\end{bmatrix}$ **dan foydalanib, tizimni yeching.**

Yechim. $Eb=\begin{bmatrix}4\\-8+17\end{bmatrix}=\begin{bmatrix}4\\9\end{bmatrix}$; mashq 7 dagi $EA=\begin{bmatrix}3&1\\0&3\end{bmatrix}$ bilan birga 3y $=9, y=3$ va 3x $+3=4, x=\frac{1}{3}$. Asl tizimda tekshirish: $3\cdot(\frac{1}{3})+3=4, 6\cdot(\frac{1}{3})+5\cdot 3=17$ — to‘g‘ri.

Mashq 9. Nol asosiy element. Tizimni $\begin{bmatrix}0&2\\3&1\end{bmatrix}\cdot\begin{bmatrix}x\\y\end{bmatrix}=\begin{bmatrix}4\\7\end{bmatrix}$ **qatorlarni almashtirib yeching;** P **ni toping.**

Yechim. $(1,1)o$‘rinda nol — vaqtacha muvaffaqiyatsizlik: $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $PA = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$, $Pb = \begin{bmatrix} 7 \\ 4 \end{bmatrix}$. Unda 2y $= 4, y = 2$; 3x $+y = 7, x = \frac{5}{3}$. Tekshirish: $2 \cdot 2 = 4, 3 \cdot (\frac{5}{3}) + 2 = 7$ — to‘g‘ri; nol asosiy element aynanlikni anglatmadi.

Mashq 10. Uch qatorni almashtirish. $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 4 & 0 & 2 \end{bmatrix}$ **matritsasining 1 va 3-qatorlarini almashtiruvchi** P **ni toping;** PA **ni hisoblang.**

Yechim. $P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ — birlik matritsining qatorlari 3, 2, 1 tartibida; PA ning 1-qatori A matritsasining 3-qatori va $h.k.$: $PA = \begin{bmatrix} 4 & 0 & 2 \\ 0 & 1 & 3 \\ 1 & 2 & 0 \end{bmatrix}$. $1 \leftrightarrow 3$ almashtirishi — involutsiya: $P^{-1} = P$ va $P^2 = I (P^{-1} = P^T$ bilan mos keladi).

Mashq 11. Aynan, lekin mos keladigan. Tizimni $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 0 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$ **yeching.**

Yechim. 2-qatori o‘ng tomoni $(6 = 2 \cdot 3)1$-qatordan ikki barobar katta, shuning uchun $R_2 \leftarrow R_2 - 2R_1 0 = 0$ beradi. $x+$ 2y $+z = 3$ va $y + z = 2$ qoladi; erkin o‘zgaruvchi $z = t$: $y = 2 - t, x + 2(2 - t) + t = 3 \to x = t - 1$. Javob: $\begin{bmatrix} t & 1 \\ 2 & t \\ t & \end{bmatrix}$ — cheksiz ko‘p yechimlar. $t = 1$ da tekshirish: $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ $3, 6, 2$ beradi — to‘g‘ri.

Mashq 12. Aynan va mos kelmaydigan. Oldingi tizimda o**‘ng tomonni** $b = \begin{bmatrix} 3 \\ 7 \\ 2 \end{bmatrix}$ **ga** o**‘zgartiring. Tizim mos keladimi?**

Yechim. Endi $7 \neq 2 \cdot 3$. $R_2 \leftarrow R_2 - 2R_1$ [0 0 0 | 7 − 6] = [0 0 0 | 1] qatorni beradi, ya’ni $0 = 1$ — ziddiyat. Tizim mos kelmaydi: o‘ng tomon zarur shart $b_2 = 2b_1$ ni buzadi.

Mashq 13. Ustunli mos kelish talqini. $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ **uchun qaysi** $b =$ **[**b_1**;** b_2**] vektorlari** $Ax = b$ **tenglamani qondiradi?**

Yechim. Ikkinchi qator birinchisining ikki barobar katta: $R_2 \leftarrow R_2 - 2R_1$ amalidan keyin $[00|b_2 - 2b_1]$ qoladi, shuning uchun moslik $b_2 = 2b_1$ ni talab qiladi. $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ va $\begin{bmatrix} 2 \\ 4 \end{bmatrix} =$

$2 \cdot \begin{bmatrix}1\\2\end{bmatrix}$ ustunlari $\begin{bmatrix}1\\2\end{bmatrix}$ ga karrali, shuning uchun Ax har doim $b_2 = 2b_1$ to‘g‘ri chiziqda yotadi. Javob: $b = [b_1;\ 2b_1]$ — b ustunlar A orqali o‘tadigan to‘g‘ri chiziqda yotishi kerak.

Mashq 14. Matritsa amallari tartibi. $E = \begin{bmatrix}1 & 0\\-2 & 1\end{bmatrix}$ **va** $P = \begin{bmatrix}0 & 1\\1 & 0\end{bmatrix}$ **deb belgilaymiz.** EP **va** PE **ni solishtiring.**

Yechim. EP: 1-qator $= \begin{bmatrix}0 & 1\end{bmatrix}$, 2-qator $= -2 \cdot \begin{bmatrix}0 & 1\end{bmatrix} + \begin{bmatrix}1 & 0\end{bmatrix} = [1,\ -2]$, ya’ni $EP = \begin{bmatrix}0 & 1\\1 & -2\end{bmatrix}$. PE: 1-qator $= 1 \cdot \begin{bmatrix}-2 & 1\end{bmatrix} = \begin{bmatrix}-2 & 1\end{bmatrix}$, 2-qator $= \begin{bmatrix}1 & 0\end{bmatrix}$, ya’ni $PE = \begin{bmatrix}-2 & 1\\1 & 0\end{bmatrix}$. Ko‘paytmalar har xil: qator amallari, umuman olganda, kommutativ emas.

Mashq 15. Murojaat: tizim qurish. Qatorlarni almashtirishni talab qiladigan, uchta asosiy elementga va $x = \begin{bmatrix}1\\2\\3\end{bmatrix}$ **yechimga ega bo‘lgan** 3×3 **tizimini qurib ko‘ring.**

Yechim. $(1,1)o$‘rningidagi nol va undan pastdagi nol bo‘lmagan element: $A = \begin{bmatrix}0 & 1 & 0\\1 & 0 & 0\\0 & 0 & 1\end{bmatrix}$; o‘ng tomonni ustun tasviri bilan belgilaymiz: $b = A \cdot \begin{bmatrix}1\\2\\3\end{bmatrix} = \begin{bmatrix}2\\1\\3\end{bmatrix}$. 1 va 2-qatorlarni almashtirish $(P = \begin{bmatrix}0 & 1 & 0\\1 & 0 & 0\\0 & 0 & 1\end{bmatrix})A$ ni birlik matritsiga aylantiradi: asosiy elementlar 1, 1, 1, yechim $x = \begin{bmatrix}1\\2\\3\end{bmatrix}$. Tekshirish: $A \cdot \begin{bmatrix}1\\2\\3\end{bmatrix} = \begin{bmatrix}2\\1\\3\end{bmatrix}$ — to‘g‘ri.

2.3 Chiziqli algebra I, 5-hafta: amaliy mashq — savollar va to‘liq yechimlar

[Manba: 5-hafta amaliy sessiya masalalari yechimlari (Week5TA Problems Projection), masalalar 1–15; matritsat satrli matn

qatlamidan tiklangan]

Mashq 1. Ikkita to‘g‘ri chiziq sistemasini yeching $\begin{bmatrix}1 & 2\\3 & 1\end{bmatrix} \cdot \begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}7\\8\end{bmatrix}$**.**

Yechim. $R_2 \leftarrow R_2 - 3R_1$: $(1-6)y = 8-21$, ya’ni $-5y = -13, y = \frac{13}{5}$; undan $x = 7 - 2\cdot(\frac{13}{5}) = \frac{9}{5}$. Javob: (9/5, 13/5). Tekshirish: $\frac{9}{5} + \frac{26}{5} = 7, \frac{27}{5} + \frac{13}{5} = 8$ — to‘g‘ri.

Mashq 2. Parallel yoki ustma-ust tushadimi? Sistemaning $\begin{bmatrix}2 & 4\\1 & 2\end{bmatrix} \cdot \begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}6\\c\end{bmatrix}$ **ni** $c = 3$ **va** $c = 4$ **uchun tasniflang.**

Yechim. 1-satr 2-satrdan ikki marta katta chap tomonda; moslik o‘ng tomonda ham xuddi shuni talab qiladi: $6 = 2c$, ya’ni $c = 3$. $c = 3$ bo‘lganda, $x + 2y = 3$ — cheksiz

ko‘p yechimlar $\begin{bmatrix} 3 & 2t \\ t & \end{bmatrix}$ ($togri$ chiziqlar ustma-ust tushadi). $c = 4$ bo‘lganda, ikkinchi tenglamaning ikkilanganidan 2x + 4y $= 8 \neq 6$ — yechim yo‘q($togri$ chiziqlar parallel).

Mashq 3. Bitta eliminatsiya bosqichi. Matritsadagi $\begin{bmatrix} 3 & 2 \\ 9 & 8 \end{bmatrix}$ birinchi asosiy element ostidagini yo‘qing.

Yechim. $\lambda_{21} = \frac{9}{3} = 3$ ko‘paytuvchisi; $R_2 \leftarrow R_2 - 3R_1$: [9 − 9, 8 − 6] = [0, 2]. Natija: $\begin{bmatrix} 3 & 2 \\ 0 & 2 \end{bmatrix}$; asosiy elementlar 3 va 2.

Mashq 4. Uchburchakli tizim. $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 3 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ 0 \\ 6 \end{bmatrix}$ tizimini yeching.

Yechim. Pastdan yuqoriga: 3z $= 6, z = 2$; 2y $-2 \cdot 2 = 0, y = 2$; $x + 4 + 2 = 8, x = 2$. Javob: $\begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$. Tekshirish: $2 + 4 + 2 = 8, 4 - 4 = 0, 3 \cdot 2 = 6$ — to‘g‘ri.

Mashq 5. Uch noma’lumli to‘liq tizim.

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 3 \\ 2 & 3 & 1 & 9 \\ 1 & 0 & 1 & 4 \end{array}\right]$$

tizimini yeching.

Yechim. $R_2 \leftarrow R_2 - 2R_1 \rightarrow$ [0 1 1 | 3]; $R_3 \leftarrow R_3 - R_1 \rightarrow$ [0 −1 1 | 1]; $R_3 \leftarrow R_3 + R_2 \rightarrow$ [0 0 2 | 4], ya’ni 2z $= 4, z = 2$. Orqaga: $y + z = 3 \rightarrow y = 1$; $x + y = 3 \rightarrow x = 2$. Javob: $\begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$.

Tekshirish: $2 + 1 = 3, 4 + 3 + 2 = 9, 2 + 2 = 4$ — to‘g‘ri.

Mashq 6. E_{32} ni tuzing. $R_3 \leftarrow R_3 + 5R_2$ amali uchun 3×3 matritsani yozing.

Yechim. E_{32} — bu $(3,2)o$‘rinda 5 bo‘lgan birlik matritsa: $E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 5 & 1 \end{bmatrix}$; 1–2-qatorlar o‘zgarmaydi, 3-qator ko‘paytmasi esa (3-qator) $+ 5 \cdot$ (2-qator) ga teng.

Mashq 7. Amalni teskarisiga o‘tkazish. $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix}$ matritsasi uchun teskarisini toping.

Yechim. $ER_3 \leftarrow R_3 - 3R_2$(2−qatorning uch baravarini ayirish) amalini amalga oshiradi; teskari amal — qo‘shish: $E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$ (belgisi o‘zgardi). Tekshirish: $E^{-1} \cdot E = I$.

Mashq 8. O‘ng tomonni o‘zgartirish. $E = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ **ni** $b = \begin{bmatrix} 3 \\ 5 \\ 7 \end{bmatrix}$ **ga qo‘llang.**

Yechim. $Eb = [3;\ 5 - 2 \cdot 3;\ 7] = \begin{bmatrix} 3 \\ -1 \\ 7 \end{bmatrix}$: ikkinchi komponent $5 - 2 \cdot 3 = -1$ bo‘ladi. Ex ni ikkinchi tenglamadan 2 koeffitsient bilan chiqarib tashlaydi — o‘ng tomon bir xil amal bilan o‘zgaradi.

Mashq 9. Qatorlarni almashtiruvchi matritsa. 3×3 **matritsaning 2 va 3-qatorlarini o‘rnini almashtiruvchi P ni toping.**

Yechim. $P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ — bu 2 va 3-qatorlari almashtirilgan birlik matritsa. Ikkita almashtirish boshlang‘ich tartibni qaytaradi: $P^2 = I$, shuning uchun $P^{-1} = P (P^{-1} = P^T$ bilan mos keladi); chapdan ko‘paytirish tenglamalarni, noma’lumlarni emas almashtiradi.

Mashq 10. Almashtirishni talab qiladigan tizim. $\begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 14 \end{bmatrix}$ **tizimini yeching.**

Yechim. $(1,1) o$‘rnidagi asosiy element nol — satrlarni almashtiramiz: 2x + 3y = 14 va $y = 4$. Unda 2x $+12 = 14, x = 1$. Javob: (1, 4). Tekshirish: $0 \cdot 1 + 4 = 4, 2 + 12 = 14$ — to‘g‘ri.

Mashq 11. Erkin o‘zgaruvchi. Tizimni yeching $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 0 \end{bmatrix}$**.**

Yechim. Uchinchi satr — $0 = 0$, cheklov bermaydi; asosiy elementlar ikkita, noma’lumlar uchta. $z = t$ deb olaylik: $y + t = 3 \to y = 3 - t$; $x + (3 - t) + t = 4 \to x = 1$. Javob: $\begin{bmatrix} 1 & \\ 3 & t \\ t & \end{bmatrix}$. $t = 0$ da tekshirish: $\begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$ $4, 3, 0$ beradi — to‘g‘ri.

Mashq 12. Ziddiyatli satr. Tushuntiring, nega

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 2 \\ 1 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

tizimining yechimi yo‘q**.**

Yechim. Oxirgi satr $0 \cdot x + 0 \cdot y + 0 \cdot z = 1$ deb o‘qiladi, ya’ni $0 = 1$ — imkonsiz tenglik: chap tomondagi nol satr hech qanday x, y, z uchun o‘ng tomondagi nol bo‘lmagan qiymatni bera olmaydi. Tizim ziddiyatli.

Mashq 13. EP **va** PE **ni solishtiring.** $E = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}$ **va** $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ **uchun ikkala ko‘paytmani hisoblang.**

Yechim. EP: 1-satr $= \begin{bmatrix}0 & 1\end{bmatrix}$, 2-satr $= 4\cdot\begin{bmatrix}0 & 1\end{bmatrix} + \begin{bmatrix}1 & 0\end{bmatrix} =$ [1, 4], ya'ni $EP = \begin{bmatrix}0 & 1\\ 1 & 4\end{bmatrix}$. PE: 1-satr $= 1\cdot\begin{bmatrix}4 & 1\end{bmatrix} = \begin{bmatrix}4 & 1\end{bmatrix}$, 2-satr $= \begin{bmatrix}1 & 0\end{bmatrix}$, ya'ni $PE = \begin{bmatrix}4 & 1\\ 1 & 0\end{bmatrix}$. Ko'paytmalar har xil: «avval chiqarib tashlash, keyin almashtirish» va «avval almashtirish, keyin chiqarib tashlash» turli matritsalar beradi.

Mashq 14. Ustun fazosini tekshirish. $\begin{bmatrix}1 & 3\\ 2 & 6\end{bmatrix}\cdot\begin{bmatrix}x\\ y\end{bmatrix} = \begin{bmatrix}2\\ 5\end{bmatrix}$ **tizimini yechish mumkinmi?**

Yechim. Ikkala ustun ham ikkinchi komponenti birinchisining ikki barobariga ega: $\begin{bmatrix}1\\ 2\end{bmatrix}$ va $\begin{bmatrix}3\\ 6\end{bmatrix} = 3\cdot\begin{bmatrix}1\\ 2\end{bmatrix}$; $x\cdot\begin{bmatrix}1\\ 2\end{bmatrix} + y\cdot\begin{bmatrix}3\\ 6\end{bmatrix}$ har qanday kombinatsiyasi $\begin{bmatrix}s\\ 2s\end{bmatrix}$ ko'rinishida bo'ladi. $b = \begin{bmatrix}2\\ 5\end{bmatrix}$ uchun $5 \neq 2\cdot 2 = 4$ — b ustunlar fazosida yotmaydi. Javob: yo'q(birinchi tenglamaning ikkilangan versiyasi 2x + 6y $= 4 \neq 5$ — to'g'ri chiziqlar parallel).

Mashq 15. Muhim: almashtirish matritsasini tuzish. (R_1, R_2, R_3) **satrlarini** (R_3, R_1, R_2) **ga olib boradigan** P **ni tuzing.**

Yechim. Yangi 1-satr — eski R_3, yangi 2 — eski R_1, yangi 3 — eski R_2, shuning uchun P satrlari birlik matritsasi satrlari 3, 1, 2 tartibda: $P = \begin{bmatrix}0 & 0 & 1\\ 1 & 0 & 0\\ 0 & 1 & 0\end{bmatrix}$. $\begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 9\end{bmatrix}$ satrlarida tekshirish: $P\cdot A = \begin{bmatrix}7 & 8 & 9\\ 1 & 2 & 3\\ 4 & 5 & 6\end{bmatrix}$. Chapdan ko'paytirish satrlarni almashtiradi; $P^3 = I$.

2.4 Chiziqli algebra I, 5-hafta: mashqlar ISA — savollar va to'liq yechimlar

[Manba: 5-hafta Mustaqil o'quv topshiriqlari, mashqlar 1–15; yechimlar faylda yo'q — pastda to'liq yechimlar tekshiruv bilan
keltirilgan]

Mashq 1. Satr va ustun tasviri. $\begin{bmatrix}2 & -1\\ 1 & 1\end{bmatrix}\cdot\begin{bmatrix}x\\ y\end{bmatrix} = \begin{bmatrix}1\\ 5\end{bmatrix}$ **tizimini yeching. Ikkala geometrik tasviri ham tavsiflang. Har bir chiqarib tashlash bosqichini va javobni tekshiring.**

Yechim. Tenglamalarni qo'shamiz: 3x $= 6, x = 2, y = 3$. Tekshirish: $2\cdot 2 - 3 = 1, 2 + 3 = 5$. 2x $- y = 1$ va $x + y = 5$ to'g'ri chiziqlari (2, 3) da kesishadi; ustunlar: $2\cdot\begin{bmatrix}2\\ 1\end{bmatrix} + 3\cdot\begin{bmatrix}-1\\ 1\end{bmatrix} = \begin{bmatrix}1\\ 5\end{bmatrix}$. Javob: $x = 2, y = 3$.

Mashq 2. Uchta natija. $\begin{bmatrix}1 & 1\\ 2 & 2\end{bmatrix}\cdot\begin{bmatrix}x\\ y\end{bmatrix} = \begin{bmatrix}2\\ c\end{bmatrix}$ **tizimini** $c = 4$ **va** $c = 5$ **uchun tasniflang. Har bir chiqarib tashlash bosqichini va javobni tekshiring.**

Yechim. $R_2 \leftarrow R_2 - 2R_1$ [00|$c-4$] beradi. $c=4$ da: $0=0$, yechimlar $\begin{bmatrix} 2 & t \\ t & \end{bmatrix}$ (tekshirish: $(2-t)+t=2, 2(2-t)+$ 2t $=4$). $c=5$ da: $0=1$ — yechim yo‘q.

Mashq 3. Ko**‘paytchini toping.** $\left[\begin{array}{cc|c} 2 & 3 & 7 \\ 8 & 13 & 29 \end{array}\right]$ **da birinchi asosiy element ostidagini chiqarib tashlang; ikkala asosiy elementni toping. Har bir bosqichni va javobni tekshiring.**

Yechim. $\lambda_{21} = \frac{8}{2} = 4$; $R_2 \leftarrow R_2 - 4R_1$: [8, 13 | 29] $-4\cdot$[2, 3 | 7] = [0, 1 | 1]. Natija: $\left[\begin{array}{cc|c} 2 & 3 & 7 \\ 0 & 1 & 1 \end{array}\right]$, asosiy elementlar 2 va 1. Aylantirilishi bilan tekshirish: [0, 1 | 1] $+4\cdot$[2, 3 | 7] = [8, 13 | 29].

Mashq 4. Orqaga qaytish almashtirish. $\begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 2 \\ 0 & 0 & 4 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \\ 8 \end{bmatrix}$ **tizimini yeching. Har bir bosqichni va javobni tekshiring.**

Yechim. Pastdan yuqoriga: $z = \frac{8}{4} = 2$; 3y $+2\cdot 2 = 7, y = 1$; 2x $+1-2 = 3, x = 2$. Tekshirish: $\begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 2 \\ 0 & 0 & 4 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \\ 8 \end{bmatrix}$. Javob: $x = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$.

Mashq 5. To**‘liq chiqarib tashlash.**

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 3 & 1 & 10 \\ 1 & 2 & 3 & 14 \end{array}\right]$$

tizimini chiqarib tashlash usuli bilan yeching. Har bir bosqichni va javobni tekshiring.

Yechim. $R_2 \leftarrow R_2 - 2R_1$ va $R_3 \leftarrow R_3 - R_1$ $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -1 & -2 \\ 0 & 1 & 2 & 8 \end{array}\right]$ beradi; keyin $R_3 \leftarrow R_3 - R_2$ [0 0 3 | 10] beradi. Orqaga: $z = \frac{10}{3}, y = z - 2 = \frac{4}{3}, x = 6 - y - z = \frac{4}{3}$. Tekshirish: satrlar 6, 10, 14 beradi — to‘g‘ri. Javob: [4/3; 4/3; 10/3].

Mashq 6. Elementar matritsani aniqlang. 3×3 **matritsida** $R_3 \leftarrow R_3 - 4R_1$ **amalga oshiradigan** E **ni toping. Har bir bosqichni va javobni tekshiring.**

Yechim. $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$ — birlik matritsa $-4(3,1)o$‘rnida; EA ning uchinchi qatori (qator 3) $- 4\cdot$(qator 1). Tekshirish $\begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{bmatrix}$ uchun: qator $3 = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} - 4\cdot \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} -3 & -2 & -1 \end{bmatrix}$. Bekor qilish: $E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix}$.

Amaliyot 7. EA ko‘paytmasi. $E = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}, A = \begin{bmatrix} 3 & 1 \\ 6 & 5 \end{bmatrix}$ bo‘lsin. EA ni hisoblang va amalni tushuntiring. Har bir bosqichni ko‘rsating va javobni tekshiring.

Yechim. Qator 1: $\begin{bmatrix} 3 & 1 \end{bmatrix}$; qator 2: $\begin{bmatrix} 6 & 5 \end{bmatrix} - 2 \cdot \begin{bmatrix} 3 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 3 \end{bmatrix}$; $EA = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$ — 2-qatordan 1-qatorning ikkilanganini ayirish (kopaytuvchi 2). Tekshirish: $E^{-1} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} A$ ni qaytaradi.

Amaliyot 8. O‘zgartirilgan o‘ng tomon. $A = \begin{bmatrix} 3 & 1 \\ 6 & 5 \end{bmatrix}$ va $b = \begin{bmatrix} 4 \\ 17 \end{bmatrix}$ uchun $E = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$ dan foydalanib sistemani yeching. Har bir bosqichni ko‘rsating va javobni tekshiring.

Yechim. $Eb = [4;\ -2 \cdot 4 + 17] = \begin{bmatrix} 4 \\ 9 \end{bmatrix}$; $EA = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$; keyin 3y $= 9 \to y = 3$ va 3x $+3 = 4 \to x = \frac{1}{3}$. Asl sistemada tekshirish: $3 \cdot (\frac{1}{3}) + 3 = 4, 6 \cdot (\frac{1}{3}) + 5 \cdot 3 = 17$ — to‘g‘ri.

Amaliyot 9. Nol asosiy element. $\begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$ ni qatorlarni almashtirib yeching; P ni toping. Har bir bosqichni ko‘rsating va javobni tekshiring.

Yechim. $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$; $PA = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}, Pb = \begin{bmatrix} 7 \\ 4 \end{bmatrix}$; 2y $= 4 \to y = 2$, 3x $+2 = 7 \to x = \frac{5}{3}$. Tekshirish: $0 \cdot (\frac{5}{3}) + 2 \cdot 2 = 4, 3 \cdot (\frac{5}{3}) + 2 = 7$ — to‘g‘ri.

Amaliyot 10. Uchta qatorning almashtirilishi. $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 4 & 0 & 2 \end{bmatrix}$ matritsasi 1 va 3-qatorlarini almashtiruvchi P ni toping; PA ni hisoblang. Har bir bosqichni ko‘rsating va javobni tekshiring.

Yechim. $P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$; PA ning 1-qatori A ning 3-qatori va PA ning 3-qatori A ning 1-qatori, shuning uchun $PA = \begin{bmatrix} 4 & 0 & 2 \\ 0 & 1 & 3 \\ 1 & 2 & 0 \end{bmatrix}$. Tekshirish: $P^2 = I$, ya’ni $P \cdot PA = A$; $P^{-1} = P = P^T$.

Amaliyot 11. Ayniy, lekin mos keladigan. $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 0 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$ ni yeching. Har bir bosqichni ko‘rsating va javobni tekshiring.

Yechim. $R_2 \leftarrow R_2 - 2R_1$ $[000|6 - 2 \cdot 3] = [0\ 0\ 0 \mid 0]$ — qator yo‘qoldi, sistema mos, lekin ayniy. Qoladi $x+$ 2y $+z = 3$ va $y + z = 2$; $z = t$: $y = 2 - t, x = t - 1$. Javob: $\begin{bmatrix} t & 1 \\ 2 & t \\ t & \end{bmatrix}$. $t = 2$

da tekshirish: $\begin{bmatrix}1\\0\\2\end{bmatrix}$ $3,6,2$ beradi — to‘g‘ri.

Amaliyot 12. Ayniy va mos kelmaydigan. Oldingi sistemada o‘ng tomonni $b=\begin{bmatrix}3\\7\\2\end{bmatrix}$ ga o‘zgartiring. Sistema mos keladimi? Har bir bosqichni ko‘rsating va javobni tekshiring.

Yechim. $R_2 \leftarrow R_2 - 2R_1$ [000|7 − 2 · 3] = [0 0 0 | 1], ya’ni $0=1$ — ziddiyat. Sistema mos kelmaydi: zarur shart $b_2 = 2b_1 (7 \neq 6)$ buzilgan.

Amaliyot 13. Ustunlarning mos kelish talqini. $A=\begin{bmatrix}1&2\\2&4\end{bmatrix}$ uchun qaysi $b=$ [b_1; b_2] vektorlari $Ax=b$ tenglamani yechimga ega? Har bir bosqichni ko‘rsating va javobni tekshiring.

Yechim. $R_2 \leftarrow R_2 - 2R_1$ [12|b_1; 00|$b_2 - 2b_1$] ni qoldiradi; mos kelish uchun $b_2 = 2b_1$ talab qilinadi. Ustunlar $\begin{bmatrix}1\\2\end{bmatrix}$ va $\begin{bmatrix}2\\4\end{bmatrix} = 2 \cdot \begin{bmatrix}1\\2\end{bmatrix}$ $\begin{bmatrix}1\\2\end{bmatrix}$ ga karrali, shuning uchun Ax har doim $b_2 = 2b_1$ to‘g‘risida yotadi. Javob: $b = [b_1;\ 2b_1]$. Tekshirish: [4; 8] $(x=4, y=0)$ yechimga ega, $\begin{bmatrix}4\\9\end{bmatrix}$ esa yo‘q.

Amaliyot 14. Matritsa amallar tartibi. $E=\begin{bmatrix}1&0\\-2&1\end{bmatrix}$ va $P=\begin{bmatrix}0&1\\1&0\end{bmatrix}$ bo‘lsin. EP va PE ni solishtiring. Har bir bosqichni ko‘rsating va javobni tekshiring.

Yechim. EP: qatorlar $\begin{bmatrix}0&1\end{bmatrix}$ va [1, −2], ya’ni $EP=\begin{bmatrix}0&1\\1&-2\end{bmatrix}$. PE: qatorlar $\begin{bmatrix}-2&1\end{bmatrix}$ va $\begin{bmatrix}1&0\end{bmatrix}$, ya’ni $PE=\begin{bmatrix}-2&1\\1&0\end{bmatrix}$. Ko‘paytmalar farq qiladi — qator amallari kommutativ emas: EP da chiqarish almashtirilgan qatorlarga ta’sir qiladi, PE da esa asl qatorlarga.

Amaliyot 15. Murojaat: sistemani qurish. Qatorlarni almashtirishni talab qiladigan, uchta asosiy elementga ega va $x=\begin{bmatrix}1\\2\\3\end{bmatrix}$ yechimga ega bo‘lgan 3×3 sistemani qurib ko‘ring. Har bir bosqichni ko‘rsating va javobni tekshiring.

Yechim. $A=\begin{bmatrix}0&1&0\\1&0&0\\0&0&1\end{bmatrix}$ — $(1,1)o$‘rnida nol, pastda birlik; $b = A\cdot\begin{bmatrix}1\\2\\3\end{bmatrix} = \begin{bmatrix}2\\1\\3\end{bmatrix}$. 1 va 2-qatorlarni almashtirish ($P=\begin{bmatrix}0&1&0\\1&0&0\\0&0&1\end{bmatrix}$) birlik matritsani beradi: asosiy elementlar

1, 1, 1, yechim $x = \begin{bmatrix}1\\2\\3\end{bmatrix}$. Tekshirish: $A \cdot \begin{bmatrix}1\\2\\3\end{bmatrix} = \begin{bmatrix}2\\1\\3\end{bmatrix}$; almashtirmasdan birinchi ustun chiqarilmaydi.

3 6-hafta — Matritsa amallari, teskari matritsalar vaajratishlar (Chiziqli algebra I)

3.1 Chiziqli algebra I, 6-hafta: formulalar katalogi

[Manba: 6-hafta darsi prezentatsiyasi (Strang, Introduction to Linear Algebra, 4th ed., §§2.4–2.7; qo‘shimcha: Lay, §§1.4–1.6)]

3.5.1 5-haftani takrorlash: sistemaning ikki tasviri va chiqarish

Satr tasviri $(row$ picture): tenglamalar — kesishgan to‘g‘ri chiziqlar yoki tekisliklar. Ustun tasviri (column picture): $Ax = x_1 \cdot a_1 + ... + x_n \cdot a_n = b$. Misol: $\begin{bmatrix}2&1\\1&1\end{bmatrix} \cdot \begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}5\\3\end{bmatrix}$ $(x,\ y) = (2,\ 1)$ ni beradi; ikkala tasvir ham bir sistemani tasvirlaydi. Chiqarish: asosiy element (pivote)$a_{kk} \neq 0$ bo‘lganda, $m_{ik} = a_{ik}/a_{kk}$ ko‘paytuvchi olinadi va $R_i \leftarrow R_i - m_{ik} \cdot R_k$ amalga oshiriladi; shunday $\begin{bmatrix}1&2&1\\2&5&3\\1&3&3\end{bmatrix} \rightarrow \begin{bmatrix}1&2&1\\0&1&1\\0&1&2\end{bmatrix} \rightarrow \begin{bmatrix}1&2&1\\0&1&1\\0&0&1\end{bmatrix}$, keyin teskari almashtirib uchburchak sistemani yechamiz. Aynanlik: $\begin{bmatrix}1&2\\2&4\end{bmatrix} \cdot \begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}3\\6\end{bmatrix}$ cheksiz ko‘p yechimga ega, 6 ni 7 ga almashtirish $0 = 1$ ziddiyatini beradi. Kvadrat matritsalar uchun to‘liq asosiy elementlar to‘plami har qanday o‘ng tomondagi uchun yagona yechimni anglatadi.

3.5.$2Ko$‘paytirish qoidalari va AB ning to‘rt talqini

$$A(m \times n), B(n \times p) \rightarrow AB(m \times p), (AB)_{ij} = \sum_k a_{ik} \cdot b_{kj}$$

Misol: $\begin{bmatrix}1&2&0\\-1&3&4\end{bmatrix} \cdot \begin{bmatrix}2&1\\0&-1\\3&2\end{bmatrix} = \begin{bmatrix}2&-1\\10&4\end{bmatrix}$. AB ga to‘rtta qarash: (1) element bo‘yicha — A matritsasining i-qatorini B matritsasining j-ustuni ko‘paytirish; (2) ustunlar bo‘yicha — $AB = [Ab_1 ... Ab_p]$; (3) qatorlar bo‘yicha — AB har bir qatori B qatorlarining kombinatsiyasi; (4) tashki ko‘paytmalar bo‘yicha — $AB = \sum_k a_k \cdot r_k$($A$ matritsasining k-ustuni B matritsasining k-qatoriga ko‘paytirish).

$$(AB)C = A(BC), A(B + C) = AB + AC, (A + B)C = AC + BC, AI = IA = A$$

$$umumanolgandaAB \neq BA; (A + B)^2 = A^2 + AB + BA + B^2 \neq A^2 + 2AB + B^2$$

Nocommutativlikning aniq misoli: $A = \begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$, $B = \begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}$ $AB = \begin{bmatrix}2 & 1\\ 1 & 1\end{bmatrix}$, lekin $BA = \begin{bmatrix}1 & 1\\ 1 & 2\end{bmatrix}$ — gorizontal va vertikal siljishlar (shears) qo‘llanish tartibiga bog‘liq.

3.5.3 Maxsus matritsalar va chiqarib tashlash matritsalari

Birlik matritsa I vektorlarni o‘zgarishsiz qoldiradi; diagonal $D = diag(d_1, ..., d_n)$ koordinatalarni masshtablaydi. Yuqori uchburchak matritsalarda diagonal ostida nollar, pastki uchburchaklarda — diagonal ustida; uchburchak tizimlar to‘g‘ri/teskari almashtirish bilan yechiladi. Chiqarib tashlash amaliyoti — o‘zi matritsa: $R_2 \leftarrow R_2 - 3R_1 E = \begin{bmatrix}1 & 0\\ -3 & 1\end{bmatrix}$ ga to‘g‘ri keladi, va $E \cdot \begin{bmatrix}2 & 1\\ 6 & 5\end{bmatrix} = \begin{bmatrix}2 & 1\\ 0 & 2\end{bmatrix}$; o‘zgargan tizim EAx $= Eb$, va $E^{-1} = \begin{bmatrix}1 & 0\\ 3 & 1\end{bmatrix}$ asl qatorlarni tiklaydi.

3.5.4 Ters matritsa: ta’rif, mezonlar, qoidalar

$$AA^{-1} = A^{-1}A = I \rightarrow Ax = bbarchabuchunyechimgaegax = A^{-1}b$$

Ters matritsa chiziq o‘tkazuvchi aynalishni teskarisiga o‘zgartiradi. Agar $Ax = 0$ nol bo‘lmagan x uchun, u holda teskari matritsa mavjud emas. Kvadrat A uchun quyidagi bayonotlar teng: teskari matritsa mavjud $\Leftrightarrow$ barcha asosiy elementlar noldan farqli (mumkin bo‘lgan almashtirishdan keyin) $\Leftrightarrow$ ustunlar chiziqli mustaqil $\Leftrightarrow$ $Ax = 0$ faqat $x = 0$ da $\Leftrightarrow$ $Ax = b$ har qanday b uchun yagona yechimga ega.

$$(AB)^{-1} = B^{-1}A^{-1},$$
$$(A^T)^{-1} = (A^{-1})^T, (A^{-1})^{-1} = A, (A^k)^{-1} = (A^{-1})^k$$

Ko‘paytmani teskarisiga olishda ko‘paytuvchilar tartibi o‘zgaradi: $(AB)(B^{-1}A^{-1}) = I$.

3.5.52×2 teskari matritsa va Gauss-Jordan usuli

$$\begin{bmatrix}a & b\\ c & d\end{bmatrix} \cdot \begin{bmatrix}d & b\\ c & a\end{bmatrix} = (ad - bc) \cdot I \rightarrow \begin{bmatrix}a & b\\ c & d\end{bmatrix}^{-1} = (1/(ad - bc)) \cdot \begin{bmatrix}d & b\\ c & a\end{bmatrix} ad - bc \neq 0bo'lganda$$

Misol: $\begin{bmatrix}2 & 1\\ 5 & 3\end{bmatrix}^{-1} = \begin{bmatrix}3 & -1\\ -5 & 2\end{bmatrix}$, chunki $det = 1$. Gauss-Jordan usuli: $[A|I] \rightarrow [I|A^{-1}]$ ikki yarmini ham bir xil qator amaliyotlari bilan: $\begin{bmatrix}1 & 2 & 1 & 0\\ 3 & 7 & 0 & 1\end{bmatrix} \rightarrow \begin{bmatrix}1 & 2 & 1 & 0\\ 0 & 1 & -3 & 1\end{bmatrix} \rightarrow \begin{bmatrix}1 & 0 & 7 & -2\\ 0 & 1 & -3 & 1\end{bmatrix}$, ya’ni $\begin{bmatrix}1 & 2\\ 3 & 7\end{bmatrix}^{-1} = \begin{bmatrix}7 & -2\\ -3 & 1\end{bmatrix}$. Ters matritsaning ustunlari: agar $A^{-1} = [x_1 ... x_n]$, u holda $Ax_j = e_j$ — teskari matritsani topish n ta tizimni yechishni anglatadi; bitta tizim $Ax = b$ uchun uchburchak chiqarib tashlash odatda A^{-1} ni aniq hisoblashdan afzalroqdir.

3.5.$6LU$-ajratma va sistemalarni faktorlar orqali yechish

$$A = LU : chiqaruvchilar—birlikpastkiuchburchakLda,$$
$$chiqarishnatijasi—yuqoriuchburchakU$$

$$\begin{bmatrix} 2 & 1 \\ 6 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} (tekshirishko'paytirishbilan)$$

$$E_k \cdots E_2 E_1 A = U \;\rightarrow\; A = E_1^{-1} E_2^{-1} \cdots E_k^{-1} U = LU$$

”Satrning ko‘paytmasini ayirish” operatsiyasining teskari — ”bu ko‘paytmani qayta qo‘shish”; ko‘paytuvchilar tartibi muhim (ong operatsiya birinchi ta’sir ko‘rsatadi). 3×3 misol: $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 5 & 4 \\ -2 & 2 & 3 \end{bmatrix}$ ko‘paytuvchilar $m_{21} = 2, m_{31} = -1, m_{32} = 1$ bilan $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$ va $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 3 & 2 \\ 0 & 0 & 2 \end{bmatrix}$ beradi ($LU = Ani$ qator bo‘yicha tekshiring). Sistemani yechish: avval $Ly = b$ ni to‘g‘ri almashtirish bilan, keyin $Ux = y$ ni teskari; $L = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 5 \\ 19 \end{bmatrix}$ uchun $y = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$ va $x =$ [3/2; 2] hosil bo‘ladi. A ni bir marta ajratib, ko‘po‘ng tomonli sistemalarni yechish mumkin.

3.5.$7LDU$-ajratma, asosiy elementlar va determinant

$$U = D \cdot (D^{-1}U),$$
$$Dasosiyelementlarnio'zichigaoladiU; A = L \cdot D \cdot (D^{-1}U)birlikdiagonallarbilanLva$$

$$D^{-1}U$$

$$\begin{bmatrix} 2 & 1 \\ 6 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \cdot [1\frac{1}{2};$$
$$01]$$

Asosiy elementlar va determinant: $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ uchun $a \neq 0$ asosiy elementlar a va $d-$ cb/a ga teng, ularning ko‘paytmasi ad − bc $= detA$ ga teng. Satrlarni almashtirmasdan determinant asosiy elementlarning ko‘paytmasiga teng; normalangan LDU barcha asosiy elementlar noldan farqli bo‘lganda yagona bo‘ladi.

3.5.8 Permutatsiyalar, $PA = LU$ va qisman tanlov

Nima uchun satrlarni almashtirish kerak: $A = \begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}$ teskari, lekin birinchi asosiy element nolli; satrlarni almashtirish $PA = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}$ beradi — nolli asosiy element satrlarning noto‘g‘ri tartibidan, singanlikdan emas, balki signal bo‘lishi mumkin. Permutatsiya matritsalari I satrlarini almashtiradi; har bir qator va ustunda aynan bitta birlik bor. $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ satrlarni almashtiradi: PA — A satrlari, AP — A ustunlari; n satr uchun jami $n!$ permutatsiyalar. Misol: $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ $(x_1,\, x_2,\, x_3)^T$ ni $(x_2, x_3, x_1)^T$ ga o‘tkazadi va vektorlarning uzunliklari va skalyar ko‘paytmalarni saqlaydi.

$$P^{-1} = P^T (chunki P^T \cdot P = I); almashtirishlar PA = LU$$

Misol: $A = \begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}$, $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $L = I$, $U = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}$; almashtirishlarda L dagi ko‘paytuvchilarni to‘g‘ri kuzatib borish muhim. Asosiy elementning qisman tanlovi (partial pivoting): [10^{-8}1; 1 1] singan emas, lekin juda kichik birinchi asosiy element 10^8 ko‘paytuvchini beradi — satrlarni almashtirish budan qochadi; qoida — faqat ustundagi (asosiy element va uning ostidagilar) modul bo‘yicha eng katta elementni olish.

3.5.9 Transponiranіya, simmetrik matritsalar, LDL^T

$$(A^T)_{ij} = a_{ji}, (A+B)^T = A^T + B^T, (AB)^T = B^T \cdot A^T$$

Misol: $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$. Simmetrik matritsa $A^T = A$ shartni qanoatlantiradi. Agar simmetrik A matritsa almashtirmasiz LMD-ajralishga ega bo‘lsa, uning normalizlangan yuqori ko‘paytchisi L^T ga teng: $A = LDL^T$; masalan $\begin{bmatrix} 2 & 2 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.

3.5.10 Qo‘llanilishi: farqlar, yig‘indilar, uchdiagonal tizimlar

Farq matritsasi $A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$ $Ax = (x_1,\, x_2 - x_1,\, x_3 - x_2)^T$ beradi; uning teskari $A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ yig‘indilarni to‘plashni beradi — differensial va integrallashning diskret analogi. Uchdiagonal $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ eng yaqin qo‘shnilar aloqalarini (chekli farq tenglamalari) tasvirlaydi; uning omillari tor lentani saqlaydi va LU orqali yechish tuzilmadan foydalanib, zich teskari matritsadan qochadi.

3.5.11 Hafta xulosi

- • Matritsa ko‘paytirish bir nechta teng ma’noli talqinlarga ega, ammo

kommutativ emas.

- • Teskari matritsa faqoq noaniq kvadrat matritsalar uchun mavjud. • Istisqo $LMD = LMD$ ajralishida kodlangan.

- • Qatorlarni almashtirish P matritsasi bilan kodlanadi: $PA = LU$. • Transponirlash va almashtirish 7-hafta vektor fazolariga tayyorlik.

3.2 Chiziqli algebra I, 6-hafta: dars mashqlari — savollar va to‘liqyechimlar

[Manba: 6-hafta dars mashqlari (Yechilgan) — 20 masala yechimlari, 2×2 dan ilg‘or ajralishlargacha]

Mashq 1. 2×2 **teskari matritsa.** $A = \begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix}$ **uchun** A^{-1} **ni toping.** Ko**‘paytirish orqali tekshiring.**

Yechim. $detA = 2 \cdot 3 - 1 \cdot 2 = 4 \neq 0$. 2×2 formulasi bo‘yicha: $A^{-1} = (\frac{1}{4}) \cdot \begin{bmatrix} 3 & -1 \\ -2 & 2 \end{bmatrix}$ = [3/4 −1/4; −1/2 1/2]. Tekshirish: $A \cdot A^{-1} = \begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix}$ ·[3/4 −1/4; −1/2 1/2] = [$\frac{3}{2} - \frac{1}{2}, -\frac{1}{2} + \frac{1}{2}$; $\frac{3}{2} - \frac{3}{2}, -\frac{1}{2} + \frac{3}{2}$] $= I$.

Mashq 2. 2×2 **tizimni yechish.** $Ax = b$ **ni yeching, bunda** $A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}, b = \begin{bmatrix} 8 \\ 6 \end{bmatrix}$**.**

Yechim. Tizma: 3x + 2y $= 8, x+$ 4y $= 6$. Ikkinchi tenglamani 3 ga ko‘paytirib birinchisini ayirsak, 10y $= 10$, ya’ni $y = 1, x = 6 - 4 = 2$. Shunday qilib, $x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Tekshirish: $A \cdot x =$ [$3 \cdot 2 + 2$; $2 + 4$] $= \begin{bmatrix} 8 \\ 6 \end{bmatrix} = b$.

Mashq 3. Teskari 2×2**.** $A = \begin{bmatrix} 4 & 1 \\ 2 & 5 \end{bmatrix}$ **uchun** A^{-1} **ni toping.** Ko**‘paytirish orqali tekshiring.**

Yechim. $detA = 4 \cdot 5 - 1 \cdot 2 = 18 \neq 0$. $A^{-1} = (\frac{1}{18}) \cdot \begin{bmatrix} 5 & -1 \\ -2 & 4 \end{bmatrix}$ = [5/18 −1/18; −1/9 2/9]. Tekshirish: $A \cdot A^{-1} = I$ — masalan, (1,1) elementi $4 \cdot (\frac{5}{18}) - \frac{1}{9} = \frac{20}{18} - \frac{2}{18} = 1, (2, 2)$ elementi $-\frac{2}{18} + \frac{20}{18} = 1$, diagonal bo‘lmagan elementlar — nollar.

Mashq 4. 2×2 **tizmani yechish.** $Ax = b$ **ni yeching, bunda** $A = \begin{bmatrix} 2 & 2 \\ 1 & 6 \end{bmatrix}, b = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$**.**

Yechim. Tizma: 2x + 2y $= 4(yanix + y = 2), x+$ 6y $= 7$. Ayirish: 5y $= 5, y = 1, x = 1$. Shunday qilib, $x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Tekshirish: $A \cdot x = \begin{bmatrix} 2 & + & 2 \\ 1 & + & 6 \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix} = b$.

Mashq 5. 2×2 **ajratma.** $A=LU=LDU$ **ni toping** $A=\begin{bmatrix}2&1\\2&4\end{bmatrix}$ **uchun.**

Yechim. Asosiy element 2, ko‘paytchi $m_{21}=\frac{2}{2}=1$: $R_2\leftarrow R_2-R_1U=\begin{bmatrix}2&1\\0&3\end{bmatrix}$ beradi. Ko‘paytchni L ga yozamiz: $L=\begin{bmatrix}1&0\\1&1\end{bmatrix}, A=LU$(tekshirish: $\begin{bmatrix}2&1\end{bmatrix}+\begin{bmatrix}0&3\end{bmatrix}=\begin{bmatrix}2&4\end{bmatrix}$). Asosiy elementlar 2 va 3 ni $D=\begin{bmatrix}2&0\\0&3\end{bmatrix}$ ga chiqaramiz; U ni satrlarni asosiy elementlarga bo‘lish orqali normallasak, $D^{-1}U=$ [1 1/2; 0 1] hosil bo‘ladi. Shunday qilib, $A=LU=L\cdot D\cdot D^{-1}U$.

Mashq 6. 2×2 **ajratma.** $A=LU=LDU$ **ni toping** $A=\begin{bmatrix}3&1\\3&5\end{bmatrix}$ **uchun.**

Yechim. $m_{21}=\frac{3}{3}=1$: $R_2\leftarrow R_2-R_1U=\begin{bmatrix}3&1\\0&4\end{bmatrix}$ beradi. $L=\begin{bmatrix}1&0\\1&1\end{bmatrix}, A=LU$(tekshirish: $\begin{bmatrix}3&1\end{bmatrix}+\begin{bmatrix}0&4\end{bmatrix}=\begin{bmatrix}3&5\end{bmatrix}$). $D=\begin{bmatrix}3&0\\0&4\end{bmatrix}, D^{-1}U=$ [1 1/3; 0 1], va $A=L\cdot D\cdot D^{-1}U$.

Mashq 7. 2×2 **ajratma.** $A=LU=LDU$ **ni toping** $A=\begin{bmatrix}1&1\\4&3\end{bmatrix}$ **uchun.**

Yechim. $m_{21}=4$: $R_2\leftarrow R_2-4R_1U=\begin{bmatrix}1&1\\0&-1\end{bmatrix}$ beradi. $L=\begin{bmatrix}1&0\\4&1\end{bmatrix}, A=LU$(tekshirish: $4\cdot\begin{bmatrix}1&1\end{bmatrix}+\begin{bmatrix}0&-1\end{bmatrix}=\begin{bmatrix}4&3\end{bmatrix}$). Ikkinchi asosiy element manfiy — bu ruxsat etilgan: $D=\begin{bmatrix}1&0\\0&-1\end{bmatrix}, D^{-1}U=\begin{bmatrix}1&1\\0&1\end{bmatrix}$, va $A=L\cdot D\cdot D^{-1}U$.

Mashq 8. 2×2 **ajratma.** $A=LU=LDU$ **ni toping** $A=\begin{bmatrix}2&1\\5&4\end{bmatrix}$ **uchun.**

Yechim. $m_{21}=\frac{5}{2}$: $R_2\leftarrow R_2-(\frac{5}{2})R_1U=$ [2 1; 0 3/2] beradi ($4-\frac{5}{2}=\frac{3}{2}$). $L=$ [1 0; 5/2 1], $A=LU$(tekshirish: $(\frac{5}{2})\cdot\begin{bmatrix}2&1\end{bmatrix}+$ [0 3/2] $=\begin{bmatrix}5&4\end{bmatrix}$). $D=$ [2 0; 0 3/2], $D^{-1}U=$ [1 1/2; 0 1], va $A=L\cdot D\cdot D^{-1}U$.

Mashq 9. Chiqarish 3×3**.** $Ax=b$ **ni Gauss chiqarish usuli bilan yeching** $A=\begin{bmatrix}1&3&1\\2&5&2\\1&4&3\end{bmatrix}, b=\begin{bmatrix}6\\10\\6\end{bmatrix}$ **uchun.**

Yechim. Kengaytirilgan matritsa $\begin{bmatrix}1&3&1&6\\2&5&2&10\\1&4&3&6\end{bmatrix}$. 1-qadam ($m_{21}=2, m_{31}=1$): $R_2-2R_1=(0,-1,0|-2)$; $R_3-R_1=(0,1,2|0)$. 2-qadam ($m_{32}=-1$): $R_3+R_2=(0,0,2|-2)$. Uchburchak tizma: $x+$ 3y $+z=6, -y=-2$, 2z $=-2$. Orqali almashtirish: $z=-1, y=2, x=6-6+1=$

1. Shunday qilib, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Tekshirish: $A \cdot x = \begin{bmatrix} 1 & + & 6 & 1 \\ 2 & + & 10 & 2 \\ 1 & + & 8 & 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ 6 \end{bmatrix} = b$.

Mashq 10. 3×3 **chiqarib tashlash.** $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 4 \end{bmatrix}, b = \begin{bmatrix} 2 \\ 6 \\ 1 \end{bmatrix}$ **uchun** $Ax = b$ **tenglamani Gauss usulida chiqarib tashlash orqali yeching.**

Yechim. Kengaytirilgan matritsa $\begin{bmatrix} 1 & 1 & 1 & 2 \\ 2 & 3 & 2 & 6 \\ 1 & 2 & 4 & 1 \end{bmatrix}$. 1-qadam $(m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, 1, 0|2)$; $R_3 - R_1 = (0, 1, 3| - 1)$. 2-qadam $(m_{32} = 1)$: $R_3 - R_2 = (0, 0, 3| - 3)$. Uchburchak tizim: $x + y + z = 2, y = 2$, 3z $= -3$. Orqali almashtirish: $z = -1, y = 2, x = 2 - 2 + 1 = 1$. Shunday qilib, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Tekshirish: $A \cdot x = \begin{bmatrix} 1 & + & 2 & 1 \\ 2 & + & 6 & 2 \\ 1 & + & 4 & 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \\ 1 \end{bmatrix} = b$.

Mashq 11. 3×3 **chiqarib tashlash (noaniy holat).** $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 3 & 5 \end{bmatrix}, b = \begin{bmatrix} 4 \\ 8 \\ 2 \end{bmatrix}$ **uchun** $Ax = b$ **tenglamani Gauss usulida chiqarib tashlash orqali yeching.**

Yechim. Kengaytirilgan matritsa $\begin{bmatrix} 1 & 2 & 1 & 4 \\ 2 & 4 & 2 & 8 \\ 1 & 3 & 5 & 2 \end{bmatrix}$. 1-qadam $(m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, 0, 0|0)$ — ikkinchi tenglama birinchisining aniq ikki barobariga teng; $R_3 - R_1 = (0, 1, 4| - 2)$. 2 va 3-qatorlarni almashtirgandan keyin: $\begin{bmatrix} 1 & 2 & 1 & 4 \\ 0 & 1 & 4 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ — uchta noma'luf uchta ustunli element, bir o'zgaruvchi erkin. $z = t$ deb olamiz: $y = -2-$ 4t $(y + 4z = -2dan), x = 4-$ 2y $-z = 8+$ 7t. Umumiy yechim $x = \begin{bmatrix} 8 & + & 7t \\ -2 & & 4t \\ t & & \end{bmatrix} = \begin{bmatrix} 8 \\ -2 \\ 0 \end{bmatrix} + t \cdot \begin{bmatrix} 7 \\ -4 \\ 1 \end{bmatrix}$, $t \in R$ — cheksiz ko'p yechimlar; faylda ko'rsatilgan $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ — $t = -1$ bo'lganda maxsus yechim. Tekshirish: $A \cdot \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 & + & 4 & 1 \\ 2 & + & 8 & 2 \\ 1 & + & 6 & 5 \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \\ 2 \end{bmatrix} = b$. Geometrik jihatdan: birinchi ikki tenglama bitta tekislikni belgilaydi, uning uchinchi bilan kesishishi to'g'ri chiziq bo'ladi.

Mashq 12. 3×3 **chiqarib tashlash.** $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 4 & 4 \end{bmatrix}, b = \begin{bmatrix} 6 \\ 10 \\ 5 \end{bmatrix}$ **uchun** $Ax = b$

tenglamani Gauss usulida chiqarib tashlash orqali yeching.

Yechim. Kengaytirilgan matritsa $\begin{bmatrix}1&3&1&6\\2&5&2&10\\1&4&4&5\end{bmatrix}$. 1-qadam $(m_{21}=2, m_{31}=1)$: $R_2-2R_1=(0,-1,0|-2)$; $R_3-R_1=(0,1,3|-1)$. 2-qadam $(m_{32}=-1)$: $R_3+R_2=(0,0,3|-3)$. Uchburchak tizim: $x+$ 3y $+z=6, -y=-2$, 3z $=-3$. Orqali almashtirish: $z=-1, y=2, x=6-6+1=1$. Shunday qilib, $x=\begin{bmatrix}1\\2\\-1\end{bmatrix}$. Tekshirish: $A\cdot x=\begin{bmatrix}1&+&6&1\\2&+&10&2\\1&+&8&4\end{bmatrix}=\begin{bmatrix}6\\10\\5\end{bmatrix}=b$.

Mashq 13. LU **va** $LDU3\times 3$**.** $A=\begin{bmatrix}2&1&1\\2&3&2\\-2&1&1\end{bmatrix}$ **uchun** LU **va** LDU **ni toping.**

Yechim. Ko‘paytirgichlar $m_{21}=1, m_{31}=-1$: $R_2-R_1=(0,2,1)$; $R_3+R_1=(0,2,2)$. Ikkinchi ustun elementi $2, m_{32}=1$: $R_3-R_2=(0,0,1)$. Shunday qilib, $L=\begin{bmatrix}1&0&0\\1&1&0\\-1&1&1\end{bmatrix}$, $U=\begin{bmatrix}2&1&1\\0&2&1\\0&0&1\end{bmatrix}$; tekshirish: $LU=A$(uchinchi qator: $-\begin{bmatrix}2&1&1\end{bmatrix}+\begin{bmatrix}0&2&1\end{bmatrix}+\begin{bmatrix}0&0&1\end{bmatrix}=\begin{bmatrix}-2&1&1\end{bmatrix}$). $D=\begin{bmatrix}2&0&0\\0&2&0\\0&0&1\end{bmatrix}$, normalizatori yuqori ko‘paytma $D^{-1}U=$ [1 1/2 1/2; 0 1 1/2; 0 0 1]; $A=LU=L\cdot D\cdot D^{-1}U$.

Mashq 14. LU **va** $LDU3\times 3$**.** $A=\begin{bmatrix}2&1&1\\4&5&3\\-2&5&3\end{bmatrix}$ **uchun** LU **va** LDU **ni toping.**

Yechim. $m_{21}=2, m_{31}=-1$: $R_2-2R_1=(0,3,1)$; $R_3+R_1=(0,6,4)$. Ikkinchi ustun elementi $3, m_{32}=2$: $R_3-2R_2=(0,0,2)$. $L=\begin{bmatrix}1&0&0\\2&1&0\\-1&2&1\end{bmatrix}$, $U=\begin{bmatrix}2&1&1\\0&3&1\\0&0&2\end{bmatrix}$; tekshirish: LU ning uchinchi qatori $\begin{bmatrix}-2&5&3\end{bmatrix}$ ga teng. $D=\begin{bmatrix}2&0&0\\0&3&0\\0&0&2\end{bmatrix}$, $D^{-1}U=$ [1 1/2 1/2; 0 1 1/3; 0 0 1]; $A=LU=L\cdot D\cdot D^{-1}U$.

Mashq 15. LU **va** $LDU3\times 3$**.** $A=\begin{bmatrix}2&1&1\\2&5&2\\-2&11&5\end{bmatrix}$ **uchun** LU **va** LDU **ni toping.**

Yechim. $m_{21}=1, m_{31}=-1$: $R_2-R_1=(0,4,1)$; $R_3+R_1=(0,12,6)$. Ikkinchi ustun

elementi $4, m_{32} = 3$: $R_3 - 3R_2 = (0,0,3)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 3 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 4 & 1 \\ 0 & 0 & 3 \end{bmatrix}$; tekshirish: $-\begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + 3 \cdot \begin{bmatrix} 0 & 4 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} -2 & 11 & 5 \end{bmatrix}$. $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, $D^{-1}U =$ [1 1/2 1/2; 0 1 1/4; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

Mashq 16. LU **va** $LDU3 \times 3$**.** $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 4 & 3 \\ -2 & 1 & 1 \end{bmatrix}$ **uchun** LU **va** LDU **ni toping.**

Yechim. $m_{21} = 2, m_{31} = -1$: $R_2 - 2R_1 = (0,2,1)$; $R_3 + R_1 = (0,2,2)$. Ikkinchi ustun elementi $2, m_{32} = 1$: $R_3 - R_2 = (0,0,1)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$; tekshirish: ikkinchi qator $2 \cdot \begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 4 & 3 \end{bmatrix}$, uchinchisi $-\begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 1 \end{bmatrix}$. $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $D^{-1}U =$ [1 1/2 1/2; 0 1 1/2; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

Mashq 17. Qatorlarni almashtirish va PLU**.** $A = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 1 & 0 \\ 2 & 3 & 2 \end{bmatrix}$ **uchun** $PA = LU$ **tenglamani qanoatlantiruvchi qatorlarni almashtirish** P **va ko‘paytuvchilarni toping.**

Yechim. Birinchi diagonal element nol — 1 va 2-qatorlarni almashtiramiz: $P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $PA = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 2 & 3 & 2 \end{bmatrix}$. Chiqarish: (2, 1) element allaqachon nol $(m_{21} = 0)$; $m_{31} = 2$: $R_3 - 2R_1 = (2,3,2) - 2 \cdot (1,1,0) = (0,1,2)$; keyin $m_{32} = \frac{1}{2}$: $R_3 - (\frac{1}{2})R_2 = (0,1,2) - (0,1,\frac{1}{2}) = (0,0,\frac{3}{2})$. Shunday qilib, $L =$ [1 0 0; 0 1 0; 2 1/2 1], $U =$ [1 1 0; 0 2 1; 0 0 3/2]. Tekshirish: $LU = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 2 & 3 & 2 \end{bmatrix} = PA$(uchinchi qator: $2 \cdot (1,1,0) + (\frac{1}{2}) \cdot (0,2,1) + (0,0,\frac{3}{2}) = (2,3,2)$).

Demak, $PA = LU$: P tenglamalarni qayta tartiblab, mos asosiy element beradi; $L2$ va 1/2 ko‘paytirlarini asl belgilari bilan $o‘z$ ichiga oladi, garchi amallar ularni ayirib tashlasa.

Mashq 18. Transponiranіya va teskari. $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix}$ **uchun** A^T **hisoblang va** $(A^T)^{-1} = (A^{-1})^T$ **tekshiring.**

Yechim. Transponiraniya: $A^T = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$. $detA = 1\cdot(1\cdot1-1\cdot0)-2\cdot(0\cdot1-1\cdot2)+0 = 1+4 = 5$. Qo‘shimcha matritsa orqali $(A^T)^{-1} = (\frac{1}{5})\cdot\begin{bmatrix} 1 & 2 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 1 \end{bmatrix}$. A^T dan mustaqil ravishda $A^{-1} = (\frac{1}{5})\cdot\begin{bmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \\ -2 & 4 & 1 \end{bmatrix}$ hisoblaymiz; keyin $(A^{-1})^T = (\frac{1}{5})\cdot\begin{bmatrix} 1 & 2 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 1 \end{bmatrix} = (A^T)^{-1}$. Tenglik bajarilgan; tekshirish: $A^T\cdot(A^T)^{-1} = I$.

Mashq 19. Simmetrik LDL^T. Simmetrik $A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix}$ ni LDL^T ko‘rinishida ajrating.

Yechim. Asosiy element 2; $m_{21} = \frac{2}{2} = 1 (m_{31} = 0)$: $R_2 - R_1 = (0, 3, 1)$. Ikkinchi asosiy element 3; $m_{32} = \frac{1}{3}$: $R_3 - (\frac{1}{3})R_2 = (0, 1, 3) - (0, 1, \frac{1}{3}) = (0, 0, \frac{8}{3})$. Almashtirishsiz simmetrik A uchun normalizlangan yuqori ko‘paytirchi L^T ga teng, ya’ni $A = LDL^T$, bunda $L =$ [1 0 0; 1 1 0; 0 1/3 1], $D =$ [2 0 0; 0 3 0; 0 0 8/3]. Tekshirish: $L\cdot D =$ [2 0 0; 2 3 0; 0 1 8/3], va $(LD)\cdot L^T = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix} = A$.

Mashq 20. Masala-muhokama: uch diagonal model. $Ax = b$ ni LU orqali yeching $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}, b = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$. Eng yaqin qo‘shnilar tuzilishini talqin qiling.

Yechim. Ajratish: $m_{21} = -\frac{1}{2}$, shuning uchun $R_2 \leftarrow R_2 + (\frac{1}{2})R_1$: $(-1, 2, -1) + (1, -\frac{1}{2}, 0) = (0, \frac{3}{2}, -1)$; $m_{32} = -\frac{2}{3}$, shuning uchun $R_3 \leftarrow R_3 + (\frac{2}{3})R_2$: $(0, -1, 2) + (0, 1, -\frac{2}{3}) = (0, 0, \frac{4}{3})$. Shunday qilib, $L =$ [1 0 0; −1/2 1 0; 0 −2/3 1], $U =$ [2 −1 0; 0 3/2 −1; 0 0 4/3]. $To‘g‘$ri almashtirish $Ly = b$: $y_1 = 2$; $-(\frac{1}{2})\cdot 2 + y_2 = 0$, bundan $y_2 = 1$; $-(\frac{2}{3})\cdot 1 + y_3 = 2$, bundan $y_3 = \frac{8}{3}$. Orqaga almashtirish $Ux = y$: $(\frac{4}{3})x_3 = \frac{8}{3}, x_3 = 2$; $(\frac{3}{2})x_2 - 2 = 1, x_2 = 2$; $2x_1 - 2 = 2, x_1 = 2$. Shunday qilib, $x = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$. Tekshirish: $A\cdot x = \begin{bmatrix} 4 & & 2 \\ -2 & + & 4 \quad 2 \\ -2 & + & 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} = b$. Matritsa uch diagonal — nol bo‘lmagan elementlar faqat asosiy diagonal va unga qo‘shni pozitsiyalarda: har bir noma’lum faqat eng yaqin qo‘shnilar bilan bog‘langan.

3.3 Chiziqli algebra I, 6-hafta: amaliy mashq — savollar va to‘liq yechimlar

[Manba: 6-hafta Amaliy mashq mashqlari bilan Yechimlar — 20 masala; Tanlangan batafsil yechimlar — masalalar 9–12, 17,

20 ning batafsil versiyalari]

Mashq 1. 2×2 **teskari.** $A=\begin{bmatrix}3 & 2\\ 1 & 6\end{bmatrix}$ **uchun** A^{-1} **toping.** Ko**‘paytirish orqali tekshiring.**

Yechim. $detA = 3\cdot 6 - 2\cdot 1 = 16 \neq 0$. $A^{-1} = (\frac{1}{16})\cdot \begin{bmatrix}6 & -2\\ -1 & 3\end{bmatrix}$ = [3/8 −1/8; −1/16 3/16]. Tekshirish: $A\cdot A^{-1} = I$ — (1,1) elementi $\frac{9}{8} - \frac{2}{16} = 1, (2,2)$ elementi $-\frac{1}{8} + \frac{18}{16} = 1$, diagonal bo‘lmagan elementlar — nollar.

Mashq 2. 2×2 **tizimni yechish.** $Ax = b$ **ni yeching, bunda** $A=\begin{bmatrix}4 & 1\\ 2 & 3\end{bmatrix}, b=\begin{bmatrix}14\\ 12\end{bmatrix}$**.**

Yechim. Tizim: 4x $+y = 14$, 2x + 3y $= 12$. Birinchi tenglamani 3 ga ko‘paytirib, ikkinchisini ayirsak, 10x $= 30$, ya’ni $x = 3, y = 14 - 12 = 2$. Shunday qilib, $x = \begin{bmatrix}3\\ 2\end{bmatrix}$. Tekshirish: $A\cdot x = \begin{bmatrix}12 & + & 2\\ 6 & + & 6\end{bmatrix} = \begin{bmatrix}14\\ 12\end{bmatrix} = b$.

Mashq 3. 2×2 **teskari.** $A=\begin{bmatrix}2 & 2\\ 1 & 4\end{bmatrix}$ **uchun** A^{-1} **toping.** Ko**‘paytirish orqali tekshiring.**

Yechim. $detA = 2\cdot 4 - 2\cdot 1 = 6 \neq 0$. $A^{-1} = (\frac{1}{6})\cdot \begin{bmatrix}4 & -2\\ -1 & 2\end{bmatrix}$ = [2/3 −1/3; −1/6 1/3]. Tekshirish: $A\cdot A^{-1} = I$ — (1,1) elementi $\frac{4}{3} - \frac{1}{3} = 1, (2,2)$ elementi $-\frac{1}{3} + \frac{4}{3} = 1$, diagonal bo‘lmagan elementlar — nollar.

Mashq 4. 2×2 **tizimni yechish.** $Ax = b$ **ni yeching, bunda** $A=\begin{bmatrix}3 & 1\\ 2 & 5\end{bmatrix}, b=\begin{bmatrix}8\\ 14\end{bmatrix}$**.**

Yechim. Tizim: 3x $+y = 8$, 2x + 5y $= 14$. Birinchi tenglamani 5 ga ko‘paytirib, ikkinchisini ayirsak, 13x $= 26$, ya’ni $x = 2, y = 8 - 6 = 2$. Shunday qilib, $x = \begin{bmatrix}2\\ 2\end{bmatrix}$. Tekshirish: $A\cdot x = \begin{bmatrix}6 & + & 2\\ 4 & + & 10\end{bmatrix} = \begin{bmatrix}8\\ 14\end{bmatrix} = b$.

Mashq 5. 2×2 **ajratish.** $A = LU = LDU$ **ni toping** $A=\begin{bmatrix}3 & 1\\ 5 & 5\end{bmatrix}$ **uchun.**

Yechim. $m_{21} = \frac{5}{3}$: $R_2 \leftarrow R_2 - (\frac{5}{3})R_1 U$ = [3 1; 0 10/3] beradi $(5 - \frac{5}{3} = \frac{10}{3})$. L = [1 0; 5/3 1], $A = LU$(tekshirish: $(\frac{5}{3})\cdot \begin{bmatrix}3 & 1\end{bmatrix}$ + [0 10/3] $= \begin{bmatrix}5 & 5\end{bmatrix}$). D = [3 0; 0 10/3], $D^{-1}U$ = [1 1/3; 0 1], va $A = L\cdot D\cdot D^{-1}U$.

Mashq 6. 2×2 **ajratish.** $A = LU = LDU$ **ni toping** $A=\begin{bmatrix}1 & 1\\ 2 & 3\end{bmatrix}$ **uchun.**

Yechim. $m_{21} = 2$: $R_2 \leftarrow R_2 - 2R_1$ dan $U = \begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$ hosil bo‘ladi. $L = \begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix}, A = LU$(tekshirish: $2\cdot \begin{bmatrix}1 & 1\end{bmatrix} + \begin{bmatrix}0 & 1\end{bmatrix} = \begin{bmatrix}2 & 3\end{bmatrix}$). Ikkala asosiy element ham 1 ga teng,

shuning uchun $D = I$ va $D^{-1}U = U$: $A = \begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix} \cdot \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} \cdot \begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$.

Mashq 7. 2×2 **ajratish.** $A = LU = LDU$ **ni toping** $A = \begin{bmatrix}2 & 1\\ 3 & 4\end{bmatrix}$ **uchun.**

Yechim. $m_{21} = \frac{3}{2}$: $R_2 \leftarrow R_2 - (\frac{3}{2})R_1$ dan $U =$ [2 1; 0 5/2] $(4 - \frac{3}{2} = \frac{5}{2})$ hosil bo‘ladi. $L =$ [1 0; 3/2 1], $A = LU$(tekshirish: $(\frac{3}{2}) \cdot \begin{bmatrix}2 & 1\end{bmatrix} +$ [0 5/2] $= \begin{bmatrix}3 & 4\end{bmatrix}$). $D =$ [2 0; 0 5/2], $D^{-1}U =$ [1 1/2; 0 1], va $A = L \cdot D \cdot D^{-1}U$.

Mashq 8. 2×2 **ajratish.** $A = LU = LDU$ **ni toping** $A = \begin{bmatrix}3 & 1\\ 4 & 5\end{bmatrix}$ **uchun.**

Yechim. $m_{21} = \frac{4}{3}$: $R_2 \leftarrow R_2 - (\frac{4}{3})R_1$ dan $U =$ [3 1; 0 11/3] $(5 - \frac{4}{3} = \frac{11}{3})$ hosil bo‘ladi. $L =$ [1 0; 4/3 1], $A = LU$(tekshirish: $(\frac{4}{3}) \cdot \begin{bmatrix}3 & 1\end{bmatrix} +$ [0 11/3] $= \begin{bmatrix}4 & 5\end{bmatrix}$). $D =$ [3 0; 0 11/3], $D^{-1}U =$ [1 1/3; 0 1], va $A = L \cdot D \cdot D^{-1}U$.

Mashq 9. 3×3 **chiqarib tashlash.** $Ax = b$ **tenglamani Gauss chiqarib tashlash usuli bilan yeching** $A = \begin{bmatrix}1 & 1 & 1\\ 2 & 3 & 2\\ 1 & 2 & 4\end{bmatrix}, b = \begin{bmatrix}2\\ 6\\ 1\end{bmatrix}$ **uchun.**

Yechim. Tizma: $x + y + z = 2$, 2x + 3y + 2z $= 6, x+$ 2y + 4z $= 1$; kengaytirilgan matritsa $\begin{bmatrix}1 & 1 & 1 & 2\\ 2 & 3 & 2 & 6\\ 1 & 2 & 4 & 1\end{bmatrix}$. x ni chiqarib tashlaymiz $(m_{21} = 2, m_{31} = 1)$: yangi ikkinchi qator $(2, 3, 2|6) - 2 \cdot (1, 1, 1|2) = (0, 1, 0|2)$; yangi uchinchi qator $(1, 2, 4|1) - (1, 1, 1|2) = (0, 1, 3| - 1)$. y ni chiqarib tashlaymiz $(m_{32} = 1)$: $(0, 1, 3| - 1) - (0, 1, 0|2) = (0, 0, 3| - 3)$. Uchburchak tizma: $x + y + z = 2, y = 2$, 3z $= -3$. Orqali almashtirish: $z = -1, y = 2, x = 2 - 2 + 1 = 1$. Shunday qilib, $x = \begin{bmatrix}1\\ 2\\ -1\end{bmatrix}$. Tekshirish: $A \cdot x = \begin{bmatrix}1 & + & 2 & 1\\ 2 & + & 6 & 2\\ 1 & + & 4 & 4\end{bmatrix} = \begin{bmatrix}2\\ 6\\ 1\end{bmatrix} = b$.

Mashq 10. 3×3 **chiqarib tashlash (cheksiz ko‘**p **yechimlar).** $Ax = b$ **tenglamani Gauss chiqarib tashlash usuli bilan yeching** $A = \begin{bmatrix}1 & 2 & 1\\ 2 & 4 & 2\\ 1 & 3 & 5\end{bmatrix}, b = \begin{bmatrix}4\\ 8\\ 2\end{bmatrix}$ **uchun.**

Yechim. Tizma: $x+$ 2y $+z = 4$, 2x + 4y + 2z $= 8, x+$ 3y + 5z $= 2$; kengaytirilgan matritsa $\begin{bmatrix}1 & 2 & 1 & 4\\ 2 & 4 & 2 & 8\\ 1 & 3 & 5 & 2\end{bmatrix}$. x ni chiqarib tashlaymiz $(m_{21} = 2, m_{31} = 1)$: ikkinchi qator (0, 0, 0 | 0) beradi — $0 = 0$ tenglama cheklovlar qoymaydi (ikkinchi tenglama birinchisining aniq ikki barobaridir); uchinchi qator: (0, 1, 4 | −2). 2 va 3-qatorlarni almashtirib, bosqichli ko‘rinishga $\begin{bmatrix}1 & 2 & 1 & 4\\ 0 & 1 & 4 & -2\\ 0 & 0 & 0 & 0\end{bmatrix}$ kelib chiqadi — uchta noma’lum uchun ikkita asosiy element. $z = t, t \in R$ deb olaylik: ikkinchi qatordan $y = -2-$ 4t; birinchidan $x = 4-$ 2y

$-z = 8+$ 7t. Barcha yechimlar: $x = \begin{bmatrix} 8 & + & 7t \\ -2 & & 4t \\ t & & \end{bmatrix} = \begin{bmatrix} 8 \\ -2 \\ 0 \end{bmatrix} + t \cdot \begin{bmatrix} 7 \\ -4 \\ 1 \end{bmatrix}$, $t \in R$. Tekshirish: $(8+7t)+2(-2-4t)+t = 4$ va $(8+7t)+3(-2-4t)+$ 5t $= 2$; ikkinchi tenglama avtomatik ravishda birinchining ikki barobari sifatida bajariladi. Geometrik jihatdan: birinchi ikki tenglama bir xil tekislikni tasvirlaydi va uning uchinchi bilan kesishishi — to‘g‘ri chiziq (yechimlar oilasi). Qisqa javob fayli $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ — $t = -1$ da xususiy yechim: $A \cdot \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \\ 2 \end{bmatrix} = b$.

Mashq 11. 3×3 **chiqarib tashlash.** $Ax = b$ **tenglamani Gauss chiqarib tashlash usuli bilan yeching** $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 4 & 4 \end{bmatrix}, b = \begin{bmatrix} 6 \\ 10 \\ 5 \end{bmatrix}$ **uchun.**

Yechim. Tizma: $x+$ 3y $+z = 6$, 2x + 5y + 2z $= 10, x+$ 4y + 4z $= 5$; kengaytirilgan matritsa $\begin{bmatrix} 1 & 3 & 1 & 6 \\ 2 & 5 & 2 & 10 \\ 1 & 4 & 4 & 5 \end{bmatrix}$. x ni chiqarib tashlaymiz $(m_{21} = 2, m_{31} = 1)$: ikkinchi qator (0, −1, 0 | −2); uchinchi qator (0, 1, 3 | −1). Ikkinchi asosiy element −1; uning ostidagi elementni yo‘qotish uchun 2-qatorni 3-qatorga qo‘shamiz $(m_{32} = -1)$: $(0,1,3|-1) + (0,-1,0|-2) = (0,0,3|-3)$. Uchburchak tizma: $x+$ 3y $+z = 6, -y = -2,$ 3z $= -3$. Orqali almashtirish: $z = -1, y = 2, x = 6-6+1 = 1$. Shunday qilib, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Tekshirish: $A \cdot x = \begin{bmatrix} 1 & + & 6 & 1 \\ 2 & + & 10 & 2 \\ 1 & + & 8 & 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ 5 \end{bmatrix} = b$.

Mashq 12. 3×3 **chiqarib tashlash.** $Ax = b$ **tenglamani Gauss chiqarib tashlash usuli bilan yeching** $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 3 \end{bmatrix}, b = \begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix}$ **uchun.**

Yechim. Tizma: $x+y+z = 2$, 2x + 3y + 2z $= 6, x+$ 2y + 3z $= 2$; kengaytirilgan matritsa $\begin{bmatrix} 1 & 1 & 1 & 2 \\ 2 & 3 & 2 & 6 \\ 1 & 2 & 3 & 2 \end{bmatrix}$. x ni chiqarib tashlaymiz $(m_{21} = 2, m_{31} = 1)$: ikkinchi qator (0, 1, 0 | 2); uchinchi qator (0, 1, 2 | 0). y ni chiqarib tashlaymiz $(m_{32} = 1)$: $(0,1,2|0) - (0,1,0|2) = (0,0,2|-2)$. Uchburchak tizma: $x+y+z = 2, y = 2,$ 2z $= -2$. Orqali almashtirish: $z = -1, y = 2, x = 2-2+1 = 1$. Shunday qilib, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Tekshirish: $A \cdot x = \begin{bmatrix} 1 & + & 2 & 1 \\ 2 & + & 6 & 2 \\ 1 & + & 4 & 3 \end{bmatrix} =$

$$\begin{bmatrix}2\\6\\2\end{bmatrix} = b.$$

Mashq 13. LU **va** $LDU 3\times 3$**.** $A = \begin{bmatrix}2 & 1 & 1\\ 4 & 5 & 3\\ -2 & 5 & 3\end{bmatrix}$ **uchun** LU **va** LDU **ni toping.**

Yechim. $m_{21}=2, m_{31}=-1$: $R_2-2R_1=(0,3,1)$; $R_3+R_1=(0,6,4)$. Ikkinchi asosiy element $3, m_{32}=2$: $R_3-2R_2=(0,0,2)$. $L=\begin{bmatrix}1 & 0 & 0\\ 2 & 1 & 0\\ -1 & 2 & 1\end{bmatrix}, U=\begin{bmatrix}2 & 1 & 1\\ 0 & 3 & 1\\ 0 & 0 & 2\end{bmatrix}$; tekshirish: uchinchi qator $LU-\begin{bmatrix}2 & 1 & 1\end{bmatrix}+2\cdot\begin{bmatrix}0 & 3 & 1\end{bmatrix}+\begin{bmatrix}0 & 0 & 2\end{bmatrix}=\begin{bmatrix}-2 & 5 & 3\end{bmatrix}$. $D=\begin{bmatrix}2 & 0 & 0\\ 0 & 3 & 0\\ 0 & 0 & 2\end{bmatrix}, D^{-1}U =$ [1 1/2 1/2; 0 1 1/3; 0 0 1]; $A=LU=L\cdot D\cdot D^{-1}U$.

Mashq 14. LU **va** $LDU 3\times 3$**.** $A = \begin{bmatrix}2 & 1 & 1\\ 2 & 5 & 2\\ -2 & 11 & 5\end{bmatrix}$ **uchun** LU **va** LDU **ni toping.**

Yechim. $m_{21}=1, m_{31}=-1$: $R_2-R_1=(0,4,1)$; $R_3+R_1=(0,12,6)$. Ikkinchi asosiy element $4, m_{32}=3$: $R_3-3R_2=(0,0,3)$. $L=\begin{bmatrix}1 & 0 & 0\\ 1 & 1 & 0\\ -1 & 3 & 1\end{bmatrix}, U=\begin{bmatrix}2 & 1 & 1\\ 0 & 4 & 1\\ 0 & 0 & 3\end{bmatrix}$; tekshirish: $-\begin{bmatrix}2 & 1 & 1\end{bmatrix}+3\cdot\begin{bmatrix}0 & 4 & 1\end{bmatrix}+\begin{bmatrix}0 & 0 & 3\end{bmatrix}=\begin{bmatrix}-2 & 11 & 5\end{bmatrix}$. $D=\begin{bmatrix}2 & 0 & 0\\ 0 & 4 & 0\\ 0 & 0 & 3\end{bmatrix}, D^{-1}U =$ [1 1/2 1/2; 0 1 1/4; 0 0 1]; $A=LU=L\cdot D\cdot D^{-1}U$.

Mashq 15. LU **va** $LDU 3\times 3$**.** $A = \begin{bmatrix}2 & 1 & 1\\ 4 & 4 & 3\\ -2 & 1 & 1\end{bmatrix}$ **uchun** LU **va** LDU **ni toping.**

Yechim. $m_{21}=2, m_{31}=-1$: $R_2-2R_1=(0,2,1)$; $R_3+R_1=(0,2,2)$. Ikkinchi asosiy element $2, m_{32}=1$: $R_3-R_2=(0,0,1)$. $L=\begin{bmatrix}1 & 0 & 0\\ 2 & 1 & 0\\ -1 & 1 & 1\end{bmatrix}, U=\begin{bmatrix}2 & 1 & 1\\ 0 & 2 & 1\\ 0 & 0 & 1\end{bmatrix}$; tekshirish: $2\cdot\begin{bmatrix}2 & 1 & 1\end{bmatrix}+\begin{bmatrix}0 & 2 & 1\end{bmatrix}=\begin{bmatrix}4 & 4 & 3\end{bmatrix}$ va $-\begin{bmatrix}2 & 1 & 1\end{bmatrix}+\begin{bmatrix}0 & 2 & 1\end{bmatrix}+\begin{bmatrix}0 & 0 & 1\end{bmatrix}=\begin{bmatrix}-2 & 1 & 1\end{bmatrix}$. $D=\begin{bmatrix}2 & 0 & 0\\ 0 & 2 & 0\\ 0 & 0 & 1\end{bmatrix}, D^{-1}U =$ [1 1/2 1/2; 0 1 1/2; 0 0 1]; $A=LU=L\cdot D\cdot D^{-1}U$.

Mashq 16. LU **va** $LDU 3\times 3$**.** $A = \begin{bmatrix}2 & 1 & 1\\ 2 & 4 & 2\\ -2 & 5 & 3\end{bmatrix}$ **uchun** LU **va** LDU **ni toping.**

Yechim. $m_{21}=1, m_{31}=-1$: $R_2-R_1=(0,3,1)$; $R_3+R_1=(0,6,4)$. Ikkinchi asosiy element $3, m_{32}=2$: $R_3-2R_2=(0,0,2)$. $L=\begin{bmatrix}1&0&0\\1&1&0\\-1&2&1\end{bmatrix}$, $U=\begin{bmatrix}2&1&1\\0&3&1\\0&0&2\end{bmatrix}$; tekshirish: uchinchi qator $LU-\begin{bmatrix}2&1&1\end{bmatrix}+2\cdot\begin{bmatrix}0&3&1\end{bmatrix}+\begin{bmatrix}0&0&2\end{bmatrix}=\begin{bmatrix}-2&5&3\end{bmatrix}$. $D=\begin{bmatrix}2&0&0\\0&3&0\\0&0&2\end{bmatrix}$, $D^{-1}U=$ [1 1/2 1/2; 0 1 1/3; 0 0 1]; $A=LU=L\cdot D\cdot D^{-1}U$.

Mashq 17. Qatorlarning almashtirishi va PLU. $A=\begin{bmatrix}0&2&1\\1&1&0\\2&3&2\end{bmatrix}$ **uchun $PA=LU$ tenglikni qanoatlantiruvchi qatorlarning almashtirishi P va omillarni toping.**

Yechim. (1) Birinchi diagonal element 0 — uni asosiy element sifatida ishlatib bo‘lmaydi; 1 va 2-qatorlarni almashtiramiz. Mos almashtiruvchi matritsa ($Idabirxil$ almashtirish): $P=\begin{bmatrix}0&1&0\\1&0&0\\0&0&1\end{bmatrix}$; A ni chapdan P ga ko‘paytirish qatorlarni almashtiradi: $PA=\begin{bmatrix}1&1&0\\0&2&1\\2&3&2\end{bmatrix}$. (2) Birinchi asosiy element 1 dan pastni chiqaramiz: (2, 1) elementi allaqachon nol ($m_{21}=0$); $m_{31}=\frac{2}{1}=2$, va $R_3\leftarrow R_3-2R_1$: $(2,3,2)-2\cdot(1,1,0)=(0,1,2)$. (3) Ikkinchi asosiy element 2; $m_{32}=\frac{1}{2}$, va $R_3\leftarrow R_3-(\frac{1}{2})R_2$: $(0,1,2)-(0,1,\frac{1}{2})=(0,0,\frac{3}{2})$; $U=$ [1 1 0; 0 2 1; 0 0 3/2]. $(4)L$ ni omillardan quramiz: $L=$ [1 0 0; m_{21}10; $m_{31}m_{32}$1] = [1 0 0; 0 1 0; 2 1/2 1]. (5) Tekshirish: $LU=\begin{bmatrix}1&1&0\\0&2&1\\2&3&2\end{bmatrix}=PA$ — uchinchi qator $2\cdot(1,1,0)+(\frac{1}{2})\cdot(0,2,1)+(0,0,\frac{3}{2})=(2,3,2)$. Shunday qilib, $PA=LU$: P tenglamalarni qayta tartibga solib, mos asosiy elementni ta’minlaydi, chiqarish esa faktorizatsiyani yakunlaydi; $L2$ va 1/2 omillarini asl belgilari bilan $o‘z$ ichiga oladi, garchi amaliyotlar ularni ayiradi.

Mashq 18. Transponir va teskari. $A=\begin{bmatrix}1&2&0\\0&1&1\\2&0&1\end{bmatrix}$ **uchun A^T ni hisoblang va $(A^T)^{-1}=(A^{-1})^T$ tekshiring.**

Yechim. $A^T=\begin{bmatrix}1&0&2\\2&1&0\\0&1&1\end{bmatrix}$. $detA=1\cdot(1\cdot1-1\cdot0)-2\cdot(0\cdot1-1\cdot2)+0\cdot(0\cdot0-1\cdot2)=5$. $Qo‘$shni matritsa bo‘yicha $(A^T)^{-1}=(\frac{1}{5})\cdot\begin{bmatrix}1&2&-2\\-2&1&4\\2&-1&1\end{bmatrix}$. Alohida: $A^{-1}=(\frac{1}{5})\cdot\begin{bmatrix}1&-2&2\\2&1&-1\\-2&4&1\end{bmatrix}$,

bundan $(A^{-1})^T = (\frac{1}{5}) \cdot \begin{bmatrix} 1 & 2 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 1 \end{bmatrix} = (A^T)^{-1}$. Tenglik bajarilgan; tekshirish: $A^T \cdot (A^T)^{-1} = I$.

Mashq 19. Simmetrik LDL^T. Simmetrik $A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix}$ ni LDL^T ko‘rinishida ajrating.

Yechim. Asosiy element 2; $m_{21} = 1 (m_{31} = 0)$: $R_2 - R_1 = (0, 3, 1)$. Ikkinchi asosiy element 3; $m_{32} = \frac{1}{3}$: $R_3 - (\frac{1}{3})R_2 = (0, 0, \frac{8}{3})$. Almashtirishsiz simmetrik A uchun $A = LDL^T$, bu yerda L = [1 0 0; 1 1 0; 0 1/3 1], D = [2 0 0; 0 3 0; 0 0 8/3]. Tekshirish: $L \cdot D$ = [2 0 0; 2 3 0; 0 1 8/3], va $(LD) \cdot L^T = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix} = A$.

Mashq 20. Masala-muhokama: uch diagonal model. $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}, b = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix}$ uchun $Ax = b$ ni LU orqali yeching. Eng yaqin qo‘shnilarning tuzilishini tushuntiring.

Yechim. (1) Ajratish: $m_{21} = (-1)/2 = -\frac{1}{2}$, shuning uchun $R_2 \leftarrow R_2 + (\frac{1}{2})R_1$: $(-1, 2, -1) + (1, -\frac{1}{2}, 0) = (0, \frac{3}{2}, -1)$; $m_{32} = (-1)/(\frac{3}{2}) = -\frac{2}{3}$, shuning uchun $R_3 \leftarrow R_3 + (\frac{2}{3})R_2$: $(0, -1, 2) + (0, 1, -\frac{2}{3}) = (0, 0, \frac{4}{3})$. Shunday qilib, L = [1 0 0; −1/2 1 0; 0 −2/3 1], U = [2 −1 0; 0 3/2 −1; 0 0 4/3]; ko‘paytchilar L da boshlang‘ch ishoralar bilan, minuslarni ham hisobga olgan holda paydo bo‘ladi. $(2) To‘g‘$ri almashtirish $Ly = b$: $y_1 = 3$; $-(\frac{1}{2}) \cdot 3 + y_2 = 0, y_2 = \frac{3}{2}$; $-(\frac{2}{3}) \cdot (\frac{3}{2}) + y_3 = 2, y_3 = 3$. (3) Tegishli almashtirish $Ux = y$: $(\frac{4}{3})x_3 = 3, x_3 = \frac{9}{4}$; $(\frac{3}{2})x_2 - \frac{9}{4} = \frac{3}{2}, x_2 = (\frac{15}{4}) \cdot (\frac{2}{3}) = \frac{5}{2}$; $2x_1 - \frac{5}{2} = 3, x_1 = \frac{11}{4}$. Shunday qilib, x = [11/4; 5/2; 9/4]. Tekshirish: $2 \cdot (\frac{11}{4}) - \frac{5}{2} = 3$; $-(\frac{11}{4}) + 2 \cdot (\frac{5}{2}) - \frac{9}{4} = 0$; $-(\frac{5}{2}) + 2 \cdot (\frac{9}{4}) = 2$. (4) Eng yaqin qo‘shnilar tuzilishi: matritsa uchdiagonaldir ($nolbo$lmagan elementlar faqat asosiy diagonalda va uning ustida va ostida joylashgan); $x_2 x_1$ va x_3 bilan o‘zaro ta’sir qiladi, x_1–$x_3 o$‘rtasida to‘g‘ridan-to‘g‘ri bog‘liqlik yo‘$q (a_{13} = a_{31} = 0)$. Bunday tuzilma sterjen bo‘ylab issiqlik o‘tkazuvchanlik modellarida, prujinalar zanjirida va bir o‘lchamli chekli farqli tenglamalarda paydo bo‘ladi (A— qadam oralig‘i multiplikatori gacha bo‘lgan standart diskret manfiy ikkinchi hosila); chiqarib tashlash tor tuzilimni saqlaydi ($LvaU$ ikkidiagonal), shuning uchun sistema arzon hal qilinadi. $(5) LDU$: D = [2 0 0; 0 3/2 0; 0 0 4/3] va $D^{-1}U$ = [1 −1/2 0; 0 1 −2/3; 0 0 1], shuning uchun $A = L \cdot D \cdot D^{-1}U$.

3.4 Chiziqli algebra I, 6-hafta: mashqlar ISA — savollar va to‘liq yechimlar

[Manba: 6-hafta Mustaqil O‘qitish Topshirig‘i — 40 ta taklif qilingan masala (asosiy 2×2 dan tortib, qiyin masalagacha);

ko‘rsatmalar: chiqarish bosqichlarini ko‘rsatish, ko‘paytmalarni tekshirish va teskari matritsa yoki ajratish mavjudligini asoslash.

Yechimlar Python da tekshirilgan]

Mashq 1. 2×2 **teskari matritsa.** $A=\begin{bmatrix}2 & 2\\ 1 & 6\end{bmatrix}$ **uchun** A^{-1} **ni toping.** Ko**‘paytirish orqali tekshiring.**

Yechim. $detA = 2\cdot 6 - 2\cdot 1 = 10 \neq 0$. $A^{-1} = (\frac{1}{10})\cdot \begin{bmatrix}6 & -2\\ -1 & 2\end{bmatrix}$ = [3/5 −1/5; −1/10 1/5]. Tekshirish: $A\cdot A^{-1} = I$ — (1,1) elementi $\frac{6}{5}-\frac{1}{5}=1, (2,2)$ elementi $-\frac{1}{5}+\frac{6}{5}=1$, diagonal bo‘lmagan elementlar nol.

Mashq 2. 2×2 **sistemani yechish.** $Ax = b$ **sistemani yeching, bunda** $A = \begin{bmatrix}3 & 1\\ 2 & 3\end{bmatrix}, b = \begin{bmatrix}8\\ 10\end{bmatrix}$**.**

Yechim. Sistema: 3x $+y = 8$, 2x + 3y $= 10$. Birinchi tenglamani 3 ga ko‘paytirib, ikkinchisini ayirsak, 7x $= 14$, ya’ni $x = 2, y = (8-6)/1 = 2$. Shunday qilib, $x = \begin{bmatrix}2\\ 2\end{bmatrix}$. Tekshirish: $A\cdot x = \begin{bmatrix}6 & + & 2\\ 4 & + & 6\end{bmatrix} = \begin{bmatrix}8\\ 10\end{bmatrix} = b$.

Mashq 3. 2×2 **teskari matritsa.** $A=\begin{bmatrix}4 & 2\\ 1 & 4\end{bmatrix}$ **uchun** A^{-1} **ni toping.** Ko**‘paytirish orqali tekshiring.**

Yechim. $detA = 4\cdot 4 - 2\cdot 1 = 14 \neq 0$. $A^{-1} = (\frac{1}{14})\cdot \begin{bmatrix}4 & -2\\ -1 & 4\end{bmatrix}$ = [2/7 −1/7; −1/14 2/7]. Tekshirish: $A\cdot A^{-1} = I$ — (1,1) elementi $\frac{8}{7}-\frac{1}{7}=1, (2,2)$ elementi $-\frac{1}{7}+\frac{8}{7}=1$, diagonal bo‘lmagan elementlar nol.

Mashq 4. 2×2 **sistemani yechish.** $Ax = b$ **sistemani yeching, bunda** $A = \begin{bmatrix}2 & 1\\ 2 & 5\end{bmatrix}, b = \begin{bmatrix}4\\ 12\end{bmatrix}$**.**

Yechim. Sistema: 2x $+y = 4$, 2x + 5y $= 12$. Ikkinchi tenglamadan birinchisini ayirsak: 4y $= 8, y = 2$; keyin 2x $= 4-2, x = 1$. Shunday qilib, $x = \begin{bmatrix}1\\ 2\end{bmatrix}$. Tekshirish: $A\cdot x = \begin{bmatrix}2 & + & 2\\ 2 & + & 10\end{bmatrix} = \begin{bmatrix}4\\ 12\end{bmatrix} = b$.

Mashq 5. 2×2 **ajratish.** $A = LU = LDU$ **ni toping** $A = \begin{bmatrix}2 & 1\\ 5 & 4\end{bmatrix}$ **uchun.**

Yechim. $m_{21} = \frac{5}{2}$: $R_2 \leftarrow R_2 - (\frac{5}{2})R_1U$ = [2 1; 0 3/2] beradi. L = [1 0; 5/2 1], $A = LU$(tekshirish: $(\frac{5}{2})\cdot \begin{bmatrix}2 & 1\end{bmatrix}$ + [0 3/2] = $\begin{bmatrix}5 & 4\end{bmatrix}$). D = [2 0; 0 3/2], $D^{-1}U$ = [1 1/2; 0 1], va $A = L\cdot D\cdot D^{-1}U$.

Mashq 6. 2×2 **ajratish.** $A = LU = LDU$ **ni toping** $A = \begin{bmatrix}3 & 1\\ 2 & 5\end{bmatrix}$ **uchun.**

Yechim. $m_{21} = \frac{2}{3}$: $R_2 \leftarrow R_2 - (\frac{2}{3})R_1U$ = [3 1; 0 13/3] beradi $(5-\frac{2}{3}=\frac{13}{3})$. L = [1 0; 2/3 1], $A = LU$(tekshirish: $(\frac{2}{3})\cdot \begin{bmatrix}3 & 1\end{bmatrix}$ + [0 13/3] = $\begin{bmatrix}2 & 5\end{bmatrix}$). D = [3 0; 0 13/3], $D^{-1}U$ = [1 1/3; 0 1], va $A = L\cdot D\cdot D^{-1}U$.

Mashq 7. 2×2 **ajratish.** $A=LU=LDU$ **ni toping** $A=\begin{bmatrix}1 & 1\\ 3 & 5\end{bmatrix}$ **uchun.**

Yechim. $m_{21}=3$: $R_2\leftarrow R_2-3R_1U=\begin{bmatrix}1 & 1\\ 0 & 2\end{bmatrix}$ beradi. $L=\begin{bmatrix}1 & 0\\ 3 & 1\end{bmatrix}, A=LU$(tekshirish: $3\cdot\begin{bmatrix}1 & 1\end{bmatrix}+\begin{bmatrix}0 & 2\end{bmatrix}=\begin{bmatrix}3 & 5\end{bmatrix}$). $D=\begin{bmatrix}1 & 0\\ 0 & 2\end{bmatrix}$, $D^{-1}U=\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$, va $A=L\cdot D\cdot D^{-1}U$.

Mashq 8. 2×2 **ajratish.** $A=LU=LDU$ **ni toping** $A=\begin{bmatrix}2 & 1\\ 4 & 4\end{bmatrix}$ **uchun.**

Yechim. $m_{21}=2$: $R_2\leftarrow R_2-2R_1U=\begin{bmatrix}2 & 1\\ 0 & 2\end{bmatrix}$ beradi. $L=\begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix}, A=LU$(tekshirish: $2\cdot\begin{bmatrix}2 & 1\end{bmatrix}+\begin{bmatrix}0 & 2\end{bmatrix}=\begin{bmatrix}4 & 4\end{bmatrix}$). $D=\begin{bmatrix}2 & 0\\ 0 & 2\end{bmatrix}, D^{-1}U=$ [1 1/2; 0 1], va $A=L\cdot D\cdot D^{-1}U$.

Mashq 9. 3×3 **chiqarish.** $Ax=b$ **ni Gauss chiqarish usuli bilan yeching** $A=\begin{bmatrix}1 & 3 & 1\\ 2 & 5 & 2\\ 1 & 4 & 4\end{bmatrix}, b=\begin{bmatrix}6\\ 10\\ 5\end{bmatrix}$ **uchun.**

Yechim. Kengaytirilgan matritsa $\begin{bmatrix}1 & 3 & 1 & 6\\ 2 & 5 & 2 & 10\\ 1 & 4 & 4 & 5\end{bmatrix}$. 1-qadam $(m_{21}=2, m_{31}=1)$: $R_2-2R_1=(0,-1,0|-2)$; $R_3-R_1=(0,1,3|-1)$. 2-qadam $(m_{32}=-1)$: $R_3+R_2=(0,0,3|-3)$. Uchburchak tizim: $x+$ 3y $+z=6, -y=-2$, 3z $=-3$; bundan $z=-1, y=2, x=1$. Shunday qilib, $x=\begin{bmatrix}1\\ 2\\ -1\end{bmatrix}$. Tekshirish: $A\cdot x=\begin{bmatrix}1 & + & 6 & 1\\ 2 & + & 10 & 2\\ 1 & + & 8 & 4\end{bmatrix}=\begin{bmatrix}6\\ 10\\ 5\end{bmatrix}=b$.

Mashq 10. 3×3 **chiqarish.** $Ax=b$ **ni Gauss chiqarish usuli bilan yeching** $A=\begin{bmatrix}1 & 1 & 1\\ 2 & 3 & 2\\ 1 & 2 & 3\end{bmatrix}, b=\begin{bmatrix}2\\ 6\\ 2\end{bmatrix}$ **uchun.**

Yechim. Kengaytirilgan matritsa $\begin{bmatrix}1 & 1 & 1 & 2\\ 2 & 3 & 2 & 6\\ 1 & 2 & 3 & 2\end{bmatrix}$. 1-qadam $(m_{21}=2, m_{31}=1)$: $R_2-2R_1=(0,1,0|2)$; $R_3-R_1=(0,1,2|0)$. 2-qadam $(m_{32}=1)$: $R_3-R_2=(0,0,2|-2)$. Uchburchak tizim: $x+y+z=2, y=2$, 2z $=-2$; bundan $z=-1, y=2, x=1$. Shunday qilib, $x=\begin{bmatrix}1\\ 2\\ -1\end{bmatrix}$.

Tekshirish: $A\cdot x=\begin{bmatrix}1 & + & 2 & 1\\ 2 & + & 6 & 2\\ 1 & + & 4 & 3\end{bmatrix}=\begin{bmatrix}2\\ 6\\ 2\end{bmatrix}=b$.

Mashq 11. 3×3 **chiqarib tashlash (cheksiz ko‘**p **yechimlar).** $Ax=b$ **tenglamani**

Gaus usulida chiqarib tashlash orqali yeching $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 3 & 6 \end{bmatrix}, b = \begin{bmatrix} 4 \\ 8 \\ 1 \end{bmatrix}$ **uchun.**

Yechim. Kengaytirilgan matritsa $\begin{bmatrix} 1 & 2 & 1 & 4 \\ 2 & 4 & 2 & 8 \\ 1 & 3 & 6 & 1 \end{bmatrix}$. 1-qadam $(m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, 0, 0|0)$ — ikkinchi tenglama birinchisining ikki barobar katta va cheklovlar qo‘shmaydi; $R_3 - R_1 = (0, 1, 5| - 3)$. 2 va 3-qatorlarni almashtirgandan keyin: $\begin{bmatrix} 1 & 2 & 1 & 4 \\ 0 & 1 & 5 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ — uchta noma’lum uchun ikkita asosiy element. Ozod o‘zgaruvchi $z = t$: $y = -3-$ 5t, $x = 4-$ 2y $-z = 10+$ 9t. Cheksiz ko‘p yechimlar: $x = \begin{bmatrix} 10 & + & 9t \\ -3 & & 5t \\ t & & \end{bmatrix}, t \in R$; maxsus yechim $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ $t = -1$ ga to‘g‘ri keladi. Tekshirish: $A \cdot \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 & + & 4 & 1 \\ 2 & + & 8 & 2 \\ 1 & + & 6 & 6 \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \\ 1 \end{bmatrix} = b$.

Mashq 12. 3×3 **chiqarib tashlash.** $Ax = b$ **tenglamani Gaus usulida chiqarib tashlash orqali yeching** $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 4 & 3 \end{bmatrix}, b = \begin{bmatrix} 6 \\ 10 \\ 6 \end{bmatrix}$ **uchun.**

Yechim. Kengaytirilgan matritsa $\begin{bmatrix} 1 & 3 & 1 & 6 \\ 2 & 5 & 2 & 10 \\ 1 & 4 & 3 & 6 \end{bmatrix}$. 1-qadam $(m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, -1, 0| - 2)$; $R_3 - R_1 = (0, 1, 2|0)$. 2-qadam $(m_{32} = -1)$: $R_3 + R_2 = (0, 0, 2| - 2)$. Orqali almashtirish: $z = -1, y = 2, x = 6 - 6 + 1 = 1$. Shunday qilib, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Tekshirish: $A \cdot x = \begin{bmatrix} 1 & + & 6 & 1 \\ 2 & + & 10 & 2 \\ 1 & + & 8 & 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ 6 \end{bmatrix} = b$.

Mashq 13. LU **va** $LDU3 \times 3$**.** $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 4 & 3 \\ -2 & 1 & 1 \end{bmatrix}$ **uchun** LU **va** LDU **hisoblang.**

Yechim. $m_{21} = 2, m_{31} = -1$: $R_2 - 2R_1 = (0, 2, 1)$; $R_3 + R_1 = (0, 2, 2)$. Ikkinchi asosiy

element $2, m_{32} = 1$: $R_3 - R_2 = (0,0,1)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$; tekshirish: $LU = A$(uchinchi qator: $-\begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 1 \end{bmatrix}$). $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $D^{-1}U =$ [1 1/2 1/2; 0 1 1/2; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

Mashq 14. LU **va** $LDU3 \times 3$**.** $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 4 & 2 \\ -2 & 5 & 3 \end{bmatrix}$ **uchun** LU **va** LDU **hisoblang.**

Yechim. $m_{21} = 1, m_{31} = -1$: $R_2 - R_1 = (0,3,1)$; $R_3 + R_1 = (0,6,4)$. Ikkinchi asosiy element $3, m_{32} = 2$: $R_3 - 2R_2 = (0,0,2)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix}$; tekshirish: LU uchinchi qatori $-\begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + 2 \cdot \begin{bmatrix} 0 & 3 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 5 & 3 \end{bmatrix}$. $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, $D^{-1}U =$ [1 1/2 1/2; 0 1 1/3; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

Mashq 15. LU **va** $LDU3 \times 3$**.** $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 6 & 3 \\ -2 & 11 & 5 \end{bmatrix}$ **uchun** LU **va** LDU **hisoblang.**

Yechim. $m_{21} = 2, m_{31} = -1$: $R_2 - 2R_1 = (0,4,1)$; $R_3 + R_1 = (0,12,6)$. Ikkinchi asosiy element $4, m_{32} = 3$: $R_3 - 3R_2 = (0,0,3)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 3 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 4 & 1 \\ 0 & 0 & 3 \end{bmatrix}$; tekshirish: $-\begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + 3 \cdot \begin{bmatrix} 0 & 4 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} -2 & 11 & 5 \end{bmatrix}$. $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, $D^{-1}U =$ [1 1/2 1/2; 0 1 1/4; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

Mashq 16. LU **va** $LDU3 \times 3$**.** $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ -2 & 1 & 1 \end{bmatrix}$ **uchun** LU **va** LDU **hisoblang.**

Yechim. $m_{21} = 1, m_{31} = -1$: $R_2 - R_1 = (0,2,1)$; $R_3 + R_1 = (0,2,2)$. Ikkinchi asosiy element $2, m_{32} = 1$: $R_3 - R_2 = (0,0,1)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$; tekshirish: $LU = A$. $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $D^{-1}U =$ [1 1/2 1/2; 0 1 1/2; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

Mashq 17. Qatorlarni almashtirish va PLU. $A = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 1 & 0 \\ 2 & 3 & 2 \end{bmatrix}$ **uchun qatorlarni almashtirish P va $PA = LU$ tenglikni qanoatlantiruvchi omillarni toping.**

Yechim. Nol birinchi asosiy element — 1 va 2-qatorlarni almashtiring: $P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $PA = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 2 & 3 & 2 \end{bmatrix}$. Chiqarib tashlash: $m_{31} = 2(R_3 - 2R_1 = (0, 1, 2))$; $m_{32} = \frac{1}{2}(R_3 - (\frac{1}{2})R_2 = (0, 0, \frac{3}{2}))$. $L =$ [1 0 0; 0 1 0; 2 1/2 1], $U =$ [1 1 0; 0 2 1; 0 0 3/2]; tekshirish: $LU = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 2 & 3 & 2 \end{bmatrix} = PA$. Shunday qilib, $PA = LU$.

Mashq 18. Transponirovka va teskari. $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix}$ **uchun A^T ni hisoblang va $(A^T)^{-1} = (A^{-1})^T$ tenglikni tekshiring.**

Yechim. $A^T = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$; $detA = 1 \cdot (1-0) - 2 \cdot (0-2) + 0 = 5$. $(A^T)^{-1} = (\frac{1}{5}) \cdot \begin{bmatrix} 1 & 2 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 1 \end{bmatrix}$. Alohida: $A^{-1} = (\frac{1}{5}) \cdot \begin{bmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \\ -2 & 4 & 1 \end{bmatrix}$, bundan $(A^{-1})^T = (\frac{1}{5}) \cdot \begin{bmatrix} 1 & 2 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 1 \end{bmatrix} = (A^T)^{-1}$. Tenglik bajarilgan; tekshirish: $A^T \cdot (A^T)^{-1} = I$.

Mashq 19. Simmetrik LDL^T. **Simmetrik** $A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix}$ **ni LDL^T ko‘rinishida ajrating.**

Yechim. Ko‘paytuvchilar $m_{21} = 1, m_{31} = 0, m_{32} = \frac{1}{3}$. $L =$ [1 0 0; 1 1 0; 0 1/3 1], $D =$ [2 0 0; 0 3 0; 0 0 8/3]; simmetrik matritsa uchun almashtirishsiz $A = LDL^T$. Tekshirish: $L \cdot D \cdot L^T = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix} = A$.

Mashq 20. Masala-muhokama: uch diagonal modeli. $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}, b = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$ **uchun $Ax = b$ ni LU bilan yeching. Eng yaqin qo‘shnilar tuzilishini**

tushuntiring.

Yechim. LU: $m_{21} = -\frac{1}{2}$ va $m_{32} = -\frac{2}{3}$, shuning uchun L = [1 0 0; −1/2 1 0; 0 −2/3 1], U = [2 −1 0; 0 3/2 −1; 0 0 4/3]. $To‘g‘$ri almashtirish $Ly = b$: $y_1 = 2, y_2 = 1, y_3 = \frac{8}{3}$. Orq almashtirish $Ux = y$: $x_3 = 2, x_2 = 2, x_1 = 2$. Shunday qilib, $x = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$. Tekshirish: $A \cdot x = \begin{bmatrix} 4 & & 2 \\ -2 & + & 4 & 2 \\ -2 & + & 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} = b$. Matritsa uch diagonal: bog‘lanishlar faqat eng yaqin qo‘shnilar bilan (x_2—$x_1 va x_3$ bilan; x_1–$x_3 o‘$rtasida to‘$g‘$ri bog‘lanish yo‘$q, a_{13} = a_{31} = 0$); L va U ikki diagonal — tor tuzilish saqlanib qolgan.

Mashq 21. Teskari 2×2. $A = \begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix}$ uchun A^{-1} ni toping. $Ko‘$paytirish orqali tekshiring.

Yechim. $detA = 2 \cdot 3 - 1 \cdot 2 = 4 \neq 0$. $A^{-1} = (\frac{1}{4}) \cdot \begin{bmatrix} 3 & -1 \\ -2 & 2 \end{bmatrix}$ = [3/4 −1/4; −1/2 1/2]. Tekshirish: $A \cdot A^{-1} = \begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix}$ ·[3/4 −1/4; −1/2 1/2] = $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$.

Mashq 22. 2×2 tizimini yechish. $Ax = b$ ni yeching, bunda $A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}, b = \begin{bmatrix} 8 \\ 6 \end{bmatrix}$.

Yechim. Tizim: 3x + 2y $= 8, x+$ 4y $= 6$. Ikkinchi tenglamadan $x = 6-$ 4y; almashtirish: $3(6 - 4y)+$ 2y $= 8$, ya’ni 18 − 10y $= 8, y = 1, x = 2$. Shunday qilib, $x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Tekshirish: $A \cdot x = \begin{bmatrix} 6 & + & 2 \\ 2 & + & 4 \end{bmatrix} = \begin{bmatrix} 8 \\ 6 \end{bmatrix} = b$.

Mashq 23. Teskari 2×2. $A = \begin{bmatrix} 4 & 1 \\ 2 & 5 \end{bmatrix}$ uchun A^{-1} ni toping. $Ko‘$paytirish orqali tekshiring.

Yechim. $detA = 4 \cdot 5 - 1 \cdot 2 = 18 \neq 0$. $A^{-1} = (\frac{1}{18}) \cdot \begin{bmatrix} 5 & -1 \\ -2 & 4 \end{bmatrix}$ = [5/18 −1/18; −1/9 2/9]. Tekshirish: $A \cdot A^{-1} = I$ — (1,1) elementi $\frac{20}{18} - \frac{2}{18} = 1, (2, 2)$ elementi $-\frac{2}{18} + \frac{20}{18} = 1$, diagonal bo‘lmagan elementlar — nol.

Mashq 24. 2×2 tizimini yechish. $Ax = b$ ni yeching, bunda $A = \begin{bmatrix} 2 & 2 \\ 1 & 6 \end{bmatrix}, b = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$.

Yechim. Tizim: 2x + 2y $= 4(yanix + y = 2), x+$ 6y $= 7$. Ayirish: 5y $= 5, y = 1, x = 1$. Shunday qilib, $x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Tekshirish: $A \cdot x = \begin{bmatrix} 2 & + & 2 \\ 1 & + & 6 \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix} = b$.

Mashq 25. 2×2 ajratish. $A = LU = LDU$ ni $A = \begin{bmatrix} 2 & 1 \\ 2 & 4 \end{bmatrix}$ uchun toping.

Yechim. $m_{21}=1$: $R_2 \leftarrow R_2 - R_1U = \begin{bmatrix}2 & 1\\ 0 & 3\end{bmatrix}$ beradi. $L = \begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}, A = LU$(tekshirish: $\begin{bmatrix}2 & 1\end{bmatrix} + \begin{bmatrix}0 & 3\end{bmatrix} = \begin{bmatrix}2 & 4\end{bmatrix}$). $D = \begin{bmatrix}2 & 0\\ 0 & 3\end{bmatrix}, D^{-1}U =$ [1 1/2; 0 1], va $A = L \cdot D \cdot D^{-1}U$.

Mashq 26. 2×2 **ajratish.** $A = LU = LDU$ **ni** $A = \begin{bmatrix}3 & 1\\ 3 & 5\end{bmatrix}$ **uchun toping.**

Yechim. $m_{21}=1$: $R_2 \leftarrow R_2 - R_1U = \begin{bmatrix}3 & 1\\ 0 & 4\end{bmatrix}$ beradi. $L = \begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}, A = LU$(tekshirish: $\begin{bmatrix}3 & 1\end{bmatrix} + \begin{bmatrix}0 & 4\end{bmatrix} = \begin{bmatrix}3 & 5\end{bmatrix}$). $D = \begin{bmatrix}3 & 0\\ 0 & 4\end{bmatrix}, D^{-1}U =$ [1 1/3; 0 1], va $A = L \cdot D \cdot D^{-1}U$.

Mashq 27. 2×2 **ajratish.** $A = LU = LDU$ **ni** $A = \begin{bmatrix}1 & 1\\ 4 & 3\end{bmatrix}$ **uchun toping.**

Yechim. $m_{21}=4$: $R_2 \leftarrow R_2 - 4R_1U = \begin{bmatrix}1 & 1\\ 0 & -1\end{bmatrix}$ beradi. $L = \begin{bmatrix}1 & 0\\ 4 & 1\end{bmatrix}, A = LU$(tekshirish: $4\cdot\begin{bmatrix}1 & 1\end{bmatrix} + \begin{bmatrix}0 & -1\end{bmatrix} = \begin{bmatrix}4 & 3\end{bmatrix}$). $D = \begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}$ (manfiy asos elementi ruxsat etilgan), $D^{-1}U = \begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$, va $A = L \cdot D \cdot D^{-1}U$.

Mashq 28. 2×2 **ajratish.** $A = LU = LDU$ **ni** $A = \begin{bmatrix}2 & 1\\ 5 & 4\end{bmatrix}$ **uchun toping.**

Yechim. $m_{21} = \frac{5}{2}$: $R_2 \leftarrow R_2 - (\frac{5}{2})R_1U =$ [2 1; 0 3/2] beradi. $L =$ [1 0; 5/2 1], $A = LU$(tekshirish: $(\frac{5}{2})\cdot\begin{bmatrix}2 & 1\end{bmatrix} +$ [0 3/2] $= \begin{bmatrix}5 & 4\end{bmatrix}$). $D =$ [2 0; 0 3/2], $D^{-1}U =$ [1 1/2; 0 1], va $A = L \cdot D \cdot D^{-1}U$.

Mashq 29. 3×3 **chiqarib tashlash.** $A = \begin{bmatrix}1 & 3 & 1\\ 2 & 5 & 2\\ 1 & 4 & 3\end{bmatrix}, b = \begin{bmatrix}6\\ 10\\ 6\end{bmatrix}$ **uchun** $Ax = b$ **ni Gauss chiqarib tashlash usuli bilan yeching.**

Yechim. Kengaytirilgan matritsa $\begin{bmatrix}1 & 3 & 1 & 6\\ 2 & 5 & 2 & 10\\ 1 & 4 & 3 & 6\end{bmatrix}$. 1-qadam $(m_{21}=2, m_{31}=1)$: $R_2 - 2R_1 = (0,-1,0|-2)$; $R_3 - R_1 = (0,1,2|0)$. 2-qadam $(m_{32}=-1)$: $R_3 + R_2 = (0,0,2|-2)$. Orq almashtirish: $z=-1, y=2, x=1$. Shunday qilib, $x = \begin{bmatrix}1\\ 2\\ -1\end{bmatrix}$. Tekshirish: $A\cdot x = \begin{bmatrix}6\\ 10\\ 6\end{bmatrix} = b$.

Mashq 30. 3×3 **chiqarib tashlash.** $A = \begin{bmatrix}1 & 1 & 1\\ 2 & 3 & 2\\ 1 & 2 & 4\end{bmatrix}, b = \begin{bmatrix}2\\ 6\\ 1\end{bmatrix}$ **uchun** $Ax = b$ **ni Gauss chiqarib tashlash usuli bilan yeching.**

Yechim. Kengaytirilgan matritsa $\begin{bmatrix}1&1&1&2\\2&3&2&6\\1&2&4&1\end{bmatrix}$. 1-qadam $(m_{21}=2, m_{31}=1)$: $R_2-2R_1=(0,1,0|2)$; $R_3-R_1=(0,1,3|-1)$. 2-qadam $(m_{32}=1)$: $R_3-R_2=(0,0,3|-3)$. Orq almashtirish: $z=-1, y=2, x=1$. Shunday qilib, $x=\begin{bmatrix}1\\2\\-1\end{bmatrix}$. Tekshirish: $A\cdot x=\begin{bmatrix}2\\6\\1\end{bmatrix}=b$.

Mashq 31. 3×3 **chiqarib tashlash (cheksiz ko‘**p **yechimlar).** $A=\begin{bmatrix}1&2&1\\2&4&2\\1&3&5\end{bmatrix}, b=\begin{bmatrix}4\\8\\2\end{bmatrix}$ **uchun** $Ax=b$ **ni Gauss chiqarib tashlash usuli bilan yeching.**

Yechim. Kengaytirilgan matritsa $\begin{bmatrix}1&2&1&4\\2&4&2&8\\1&3&5&2\end{bmatrix}$. 1-qadam $(m_{21}=2, m_{31}=1)$: $R_2-2R_1=(0,0,0|0)$ — ikkinchi tenglama birinchisining ikki barobar katta; $R_3-R_1=(0,1,4|-2)$. 2 va 3-qatorlarni almashtirgandan keyin: $\begin{bmatrix}1&2&1&4\\0&1&4&-2\\0&0&0&0\end{bmatrix}$. Ozod o‘zgaruvchi $z=t$: $y=-2-4t$, $x=4-2y-z=8+7t$. Cheksiz ko‘p yechimlar: $x=\begin{bmatrix}8+7t\\-2-4t\\t\end{bmatrix}, t\in R$; maxsus yechim $x=\begin{bmatrix}1\\2\\-1\end{bmatrix}$ $(t=-1)$. Tekshirish: $A\cdot\begin{bmatrix}1\\2\\-1\end{bmatrix}=\begin{bmatrix}4\\8\\2\end{bmatrix}=b$.

Mashq 32. 3×3 **chiqarib tashlash.** $A=\begin{bmatrix}1&3&1\\2&5&2\\1&4&4\end{bmatrix}, b=\begin{bmatrix}6\\10\\5\end{bmatrix}$ **uchun** $Ax=b$ **ni Gauss chiqarib tashlash usuli bilan yeching.**

Yechim. Kengaytirilgan matritsa $\begin{bmatrix}1&3&1&6\\2&5&2&10\\1&4&4&5\end{bmatrix}$. 1-qadam $(m_{21}=2, m_{31}=1)$: $R_2-2R_1=(0,-1,0|-2)$; $R_3-R_1=(0,1,3|-1)$. 2-qadam $(m_{32}=-1)$: $R_3+R_2=(0,0,3|-3)$. Orqaga almashtirish: $z=-1, y=2, x=1$. Shunday qilib, $x=\begin{bmatrix}1\\2\\-1\end{bmatrix}$. Tekshirish: $A\cdot x=\begin{bmatrix}6\\10\\5\end{bmatrix}=b$.

Mashq 33. LU **va** $LDU 3\times 3$**.** $A=\begin{bmatrix}2&1&1\\2&3&2\\-2&1&1\end{bmatrix}$ **uchun** LU **va** LDU **ni hisoblang.**

Yechim. $m_{21}=1, m_{31}=-1, m_{32}=1$: $R_2-R_1=(0,2,1)$; $R_3+R_1=(0,2,2)$; $R_3-R_2=(0,0,1)$. $L=\begin{bmatrix}1&0&0\\1&1&0\\-1&1&1\end{bmatrix}$, $U=\begin{bmatrix}2&1&1\\0&2&1\\0&0&1\end{bmatrix}$; tekshirish: $LU=A$. $D=\begin{bmatrix}2&0&0\\0&2&0\\0&0&1\end{bmatrix}$, $D^{-1}U=$ [1 1/2 1/2; 0 1 1/2; 0 0 1]; $A=LU=L\cdot D\cdot D^{-1}U$.

Mashq 34. LU **va** $LDU 3\times 3$**.** $A=\begin{bmatrix}2&1&1\\4&5&3\\-2&5&3\end{bmatrix}$ **uchun** LU **va** LDU **ni hisoblang.**

Yechim. $m_{21}=2, m_{31}=-1, m_{32}=2$: $R_2-2R_1=(0,3,1)$; $R_3+R_1=(0,6,4)$; $R_3-2R_2=(0,0,2)$. $L=\begin{bmatrix}1&0&0\\2&1&0\\-1&2&1\end{bmatrix}$, $U=\begin{bmatrix}2&1&1\\0&3&1\\0&0&2\end{bmatrix}$; tekshirish: $LU=A$(uchinchi qator: $-\begin{bmatrix}2&1&1\end{bmatrix}+2\cdot\begin{bmatrix}0&3&1\end{bmatrix}+\begin{bmatrix}0&0&2\end{bmatrix}=\begin{bmatrix}-2&5&3\end{bmatrix}$). $D=\begin{bmatrix}2&0&0\\0&3&0\\0&0&2\end{bmatrix}$, $D^{-1}U=$ [1 1/2 1/2; 0 1 1/3; 0 0 1]; $A=LU=L\cdot D\cdot D^{-1}U$.

Mashq 35. LU **va** $LDU 3\times 3$**.** $A=\begin{bmatrix}2&1&1\\2&5&2\\-2&11&5\end{bmatrix}$ **uchun** LU **va** LDU **ni hisoblang.**

Yechim. $m_{21}=1, m_{31}=-1, m_{32}=3$: $R_2-R_1=(0,4,1)$; $R_3+R_1=(0,12,6)$; $R_3-3R_2=(0,0,3)$. $L=\begin{bmatrix}1&0&0\\1&1&0\\-1&3&1\end{bmatrix}$, $U=\begin{bmatrix}2&1&1\\0&4&1\\0&0&3\end{bmatrix}$; tekshirish: $LU=A$(uchinchi qator: $-\begin{bmatrix}2&1&1\end{bmatrix}+3\cdot\begin{bmatrix}0&4&1\end{bmatrix}+\begin{bmatrix}0&0&3\end{bmatrix}=\begin{bmatrix}-2&11&5\end{bmatrix}$). $D=\begin{bmatrix}2&0&0\\0&4&0\\0&0&3\end{bmatrix}$, $D^{-1}U=$ [1 1/2 1/2; 0 1 1/4; 0 0 1]; $A=LU=L\cdot D\cdot D^{-1}U$.

Mashq 36. LU **va** $LDU 3\times 3$**.** $A=\begin{bmatrix}2&1&1\\4&4&3\\-2&1&1\end{bmatrix}$ **uchun** LU **va** LDU **ni hisoblang.**

Yechim. $m_{21}=2, m_{31}=-1, m_{32}=1$: $R_2-2R_1=(0,2,1)$; $R_3+R_1=(0,2,2)$; $R_3-R_2=(0,0,1)$. $L=\begin{bmatrix}1&0&0\\2&1&0\\-1&1&1\end{bmatrix}$, $U=\begin{bmatrix}2&1&1\\0&2&1\\0&0&1\end{bmatrix}$; tekshirish: $LU=A$. $D=\begin{bmatrix}2&0&0\\0&2&0\\0&0&1\end{bmatrix}$, $D^{-1}U=$ [1 1/2 1/2; 0 1 1/2; 0 0 1]; $A=LU=L\cdot D\cdot D^{-1}U$.

Mashq 37. Qatorlarni almashtirish va PLU. $A = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 1 & 0 \\ 2 & 3 & 2 \end{bmatrix}$ **uchun qatorlarni almashtirish P va $PA = LU$ tenglikni qanoatlantiruvchi omillarni toping.**

Yechim. 1 va 2-qatorlarni almashtirish (birinchi asosiy element nol): $P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $PA = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 2 & 3 & 2 \end{bmatrix}$. Chiqarib tashlash: $m_{31} = 2(R_3 - 2R_1 = (0, 1, 2))$; $m_{32} = \frac{1}{2}(R_3 - (\frac{1}{2})R_2 = (0, 0, \frac{3}{2}))$. $L =$ [1 0 0; 0 1 0; 2 1/2 1], $U =$ [1 1 0; 0 2 1; 0 0 3/2]; tekshirish: $LU = PA$. Shunday qilib, $PA = LU$.

Mashq 38. Transponir va teskari. $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix}$ **uchun A^T ni hisoblang va $(A^T)^{-1} = (A^{-1})^T$ tenglikni tekshiring.**

Yechim. $A^T = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$; $detA = 5$. $(A^T)^{-1} = (\frac{1}{5}) \cdot \begin{bmatrix} 1 & 2 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 1 \end{bmatrix}$; bosh tomondan, $A^{-1} = (\frac{1}{5}) \cdot \begin{bmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \\ -2 & 4 & 1 \end{bmatrix}$, va $(A^{-1})^T = (\frac{1}{5}) \cdot \begin{bmatrix} 1 & 2 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 1 \end{bmatrix} = (A^T)^{-1}$. Tenglik bajarilgan; tekshirish: $A^T \cdot (A^T)^{-1} = I$.

Mashq 39. Simmetrik LDL^T. Simmetrik $A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix}$ ni LDL^T ko‘rinishida ajrating.

Yechim. Omillar $m_{21} = 1, m_{31} = 0, m_{32} = \frac{1}{3}$. $L =$ [1 0 0; 1 1 0; 0 1/3 1], $D =$ [2 0 0; 0 3 0; 0 0 8/3], va $A = LDL^T$. Tekshirish: $L \cdot D \cdot L^T = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix} = A$.

Mashq 40. Masala-muhokama: uch diagonallik modeli. $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}, b = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$ **uchun $Ax = b$ ni LU bilan yeching. Eng yaqin qo‘shnilar tuzilishini talqin qiling.**

Yechim. LU: $L =$ [1 0 0; −1/2 1 0; 0 −2/3 1], $U =$ [2 −1 0; 0 3/2 −1; 0 0 4/3] (omillar $m_{21} = -\frac{1}{2}, m_{32} = -\frac{2}{3}L$ da asl belgilar bilan saqlanadi). $To‘g‘ri$ almashtirish $Ly = by =$ [2;

1; 8/3] beradi; orqaga almashtirish $Ux = yx_3 = 2, x_2 = 2, x_1 = 2$ beradi. Shunday qilib, $x = \begin{bmatrix}2\\2\\2\end{bmatrix}$. Tekshirish: $A \cdot x = \begin{bmatrix}2\\0\\2\end{bmatrix} = b$. Matritsa uch diagonallik — har bir noma'lum faqat eng yaqin qo'shnilar bilan bog'langan; omillar L va U ikki diagonallik, va tor tuzilma yechimni arzon qiladi.

3.5 Chiziqli algebra I, 6-hafta: misollar bilan takrorlash materiali

[Manba: 6-haftoq qo'shimcha material — Misollar bilan qayta ko'rib chiqish: $I)LU, LDU$ va normalangan U qisqa darsi; $II)$

3×3 tizimlar uchun Gaussexclusion qisqa darsi; $III)$ Gausss-Jordan usuli orqali teskari matritsani topishning batafsil misoli]

II darsning asosiy nuqtalari (5-hafto takrorlash). Tizim uchta tasvirda ko'rib chiqilishi mumkin: qatorli (tenglamalar — tekisliklar), matritsali $(Ax = b)$ va ustunli (A ustunlarining b ni berishi uchun kombinatsiyasi). Qatorlar uchun uchta aylantiriladigan amaliyonga ruxsat beriladi: ikki tenglamani almashtirish $(R_i \leftrightarrow R_j)$, tenglamani noldan farqli songa ko'paytirish $(R_i \leftarrow c \cdot R_i, c \neq 0)$ va boshqa tenglamaning ko'paytmasini qo'shish $(R_i \leftarrow R_i + c \cdot R_j)$; har biri butun tenglamaga, o'ng qismni ham qatnashgan holda qo'llaniladi. 3×3 tizim uchun protsedura: kengaytirilgan matritsani yozing; birinchi nol bo'lmagan asosiy elementni tanlang (kerak bo'lsa qatorlarni almashtiring); uning ostidagi elementlarni chiqaring; yangilangan matritsada ikkinchi asosiy elementni tanlang; uning ostidagi elementni chiqaring; ziddiyatni yoki erkin o'zgaruvchilarni tekshiring; uchta asosiy element bo'lganda teskiri almashtirishni amalga oshiring; yechimni asl tizimda tekshiring. Teskiri almashtirish formulalari: $z = c_3/u_{33}, y = (c_2 - u_{23} \cdot z)/u_{22}, x = (c_1 - u_{12} \cdot y - u_{13} \cdot z)/u_{11}$(asosiy elementlar noldan farqli). Odatiy xatolar: o'ng qismni yangilashni unutish; o'zgarishidan keyin eski qatorni foydalanish; ko'paytirchilar munosabati yo'nalishini chalkashtirish; taklif qilingan nol asosiy elementni degeneratsiya isboti deb hisoblash.

Misol 1. LU**, normalash va** LDU**(dars** I**).** $A = \begin{bmatrix}2 & 4 & 2\\4 & 11 & 7\\2 & 7 & 8\end{bmatrix}$ **matritsasi** LU **va** LDU **ajratmalarini toping.**

Yechim. Birinchi asosiy element 2 ostidagi elementlarni chiqaramiz: $R_2 \leftarrow R_2 - 2R_1$ va $R_3 \leftarrow R_3 - R_1 \begin{bmatrix}2 & 4 & 2\\0 & 3 & 3\\0 & 3 & 6\end{bmatrix}$ beradi; keyin ikkinchi asosiy element 3 ostida: $R_3 \leftarrow R_3 - R_2 \begin{bmatrix}2 & 4 & 2\\0 & 3 & 3\\0 & 0 & 3\end{bmatrix}$ beradi. Ko'paytchilar $m_{21} = 2, m_{31} = 1, m_{32} = 1$, shuning uchun $L = \begin{bmatrix}1 & 0 & 0\\2 & 1 & 0\\1 & 1 & 1\end{bmatrix}$ va $A = LUU = \begin{bmatrix}2 & 4 & 2\\0 & 3 & 3\\0 & 0 & 3\end{bmatrix}$ bilan; L asl belgilar bilan ko'paytchilarni saqlaydi (ayirish amaliyati $2R_1$ ni ayiradi, lekin $L_{21} = +2$). Normalash: har bir U qa-

torini uning diagonal elementiga bo‘lamiz — asosiy elementlar $2, 3, 3 D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ ga chiqariladi, normalangan yuqori ko‘paytma $D^{-1}U = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$, va $A = L \cdot D \cdot (D^{-1}U)$; D^{-1} ga chapdan ko‘paytirish qatorlarni masshtablaydi. E’tibor berish: $A = LU$ da asosiy elementlar U diagonalida joylashgan (D ajratilmaydi, birlik diagonal — L da); $A = LDU$ da asosiy elementlar D da, birlik diagonallar esa L va uchinchi ko‘paytmada; ko‘p darsliklar $D^{-1}U$ ni U deb qayta belgilaydi.

Misol 2. Faktorlar bo‘yicha tizimni yechish (dars I). Misol 1 ajratmasidan foydalanib, $Ax = \begin{bmatrix} 8 \\ 22 \\ 17 \end{bmatrix}$ ni yeching.

Yechim. $To‘g‘$ri almashtirish $Ly = b$: $y_1 = 8$; $2y_1 + y_2 = 22$, bundan $y_2 = 6$; $y_1 + y_2 + y_3 = 17$, bundan $y_3 = 3$, ya’ni $y = \begin{bmatrix} 8 \\ 6 \\ 3 \end{bmatrix}$. Teskiri almashtirish $Ux = y$: $3x_3 = 3, x_3 = 1$; $3x_2 + 3x_3 = 6, x_2 = 1$; $2x_1 + 4x_2 + 2x_3 = 8, x_1 = 1$. Shunday qilib, $x = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. LDU bilan hisoblash uch bosqamga bo‘linadi: $Ly = b, Dz = y$(asosiy elementlarga bo‘lish), keyin $(D^{-1}U)x = z$. Qatorlarni almashtirish kerak bo‘lganda, standart shakl — $PA = LU$ yoki $PA = LDU$, bu erda P almashtirishlarni yozadi;

asosiy elementlarni chiqarish va bo‘lish ularning noldan farqliligini talab qiladi.

Misol 3. Gaussexlectionning to‘liq misoli (dars II). 2x $+y - z = 1$, 4x + 3y $+z = 13, -$2x $+y+$ 2z $= 6$ tizimini yeching.

Yechim. Bu erda $A = \begin{bmatrix} 2 & 1 & -1 \\ 4 & 3 & 1 \\ -2 & 1 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 13 \\ 6 \end{bmatrix}$. Bosqich 1: birinchi asosiy element 2, ko‘paytchi $m_{21} = \frac{4}{2} = 2$; $R_2 \leftarrow R_2 - 2R_1$: $(4x + 3y + z) - 2(2x + y - z) = 13 - 2 \cdot 1$, ya’ni $y+$ 3z $= 11$. Bosqich 2: $m_{31} = -\frac{2}{2} = -1$; $R_3 \leftarrow R_3 + R_1$: $(-2x + y + 2z) + (2x + y - z) = 6 + 1$, ya’ni 2y $+z = 7$. Bosqich 3: ikkinchi asosiy element 1 (yangilangan 2-tenglamadagi y koeffitsienti), $m_{32} = 2$; $R_3 \leftarrow R_3 - 2R_2$: $(2y + z) - 2(y + 3z) = 7 - 22$, ya’ni $-$5z $= -15$. Asosiy elementlar 2, 1, -5 — ikkinchi va uchinchisi yangilangan tenglamalardan, asl diagonal emasdan olinadi. Teskiri almashtirish: $z = 3, y = 2, x = 1$, ya’ni $(x, y, z) = (1, 2, 3)$. Tekshirish: $2 \cdot 1 + 2 - 3 = 1$; $4 \cdot 1 + 6 + 3 = 13$; $-2 \cdot 1 + 2 + 6 = 6$. Kengaytirilgan matritsa bilan: $\begin{bmatrix} 2 & 1 & -1 & 1 \\ 4 & 3 & 1 & 13 \\ -2 & 1 & 2 & 6 \end{bmatrix} \to \begin{bmatrix} 2 & 1 & -1 & 1 \\ 0 & 1 & 3 & 11 \\ 0 & 2 & 1 & 7 \end{bmatrix} \to \begin{bmatrix} 2 & 1 & -1 & 1 \\ 0 & 1 & 3 & 11 \\ 0 & 0 & -5 & -15 \end{bmatrix}$.

Misol 4. Exclusionni matritsa ko‘paytirishi sifatida (dars II). Misol 3 amaliyotlarini exclusion matritsalari sifatida yozing.

Yechim. $R_2 \leftarrow R_2 - 2R_1$ uchun: $E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ ($E_{21} \cdot A$ko paytmaning ikkinchi qatori $-2 \cdot (A$ ning 1-qatori$) + (A$ ning 2-qatriga) teng); 1-qatorni 3-qatorga qo‘shish uchun: $E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$; $R_3 \leftarrow R_3 - 2R_2$ uchun: $E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}$. To‘liq chiqarib tashlash: $E_{32} \cdot E_{31} \cdot E_{21} \cdot A = U$ va $E_{32} \cdot E_{31} \cdot E_{21} \cdot b = c$, bunda $U = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & -5 \end{bmatrix}$, $c = \begin{bmatrix} 1 \\ 11 \\ -15 \end{bmatrix}$; eng o‘ngdagi operatsiya birinchi ta’sir ko‘rsatadi. Qoidalar: $(BC)_{ij} = \sum_k b_{ik} \cdot c_{kj}$; assotsiativlik $(BC)D = B(CD)$; kommutativsizlik $BC \neq CB$ — shuning uchun qator operatsiyalari tartibi muhim.

5-misol. Nol taklif etilgan asosiy element (II dars). $y + z = 5, x + y + z = 6$**, 2x +** 3**y +** 4**z** $= 20$ **tenglamalar sistemasini yeching.**

Yechim. $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 2 & 3 & 4 \end{bmatrix}$, $b = \begin{bmatrix} 5 \\ 6 \\ 20 \end{bmatrix}$; birinchi taklif etilgan asosiy element nol — 1 va 2-qatorlarni almashtiramiz (almashtirish matritsasi $P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, qayta tartiblangan sistema PAx $= Pb$): $\begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 1 & 5 \\ 2 & 3 & 4 & 20 \end{bmatrix}$. Keyin $R_3 \leftarrow R_3 - 2R_1$: (0, 1, 2 | 8); keyin $R_3 \leftarrow R_3 - R_2$: (0, 0, 1 | 3). Orqaga almashtirish: $z = 3, y = 2, x = 1$. Taklif etilgan asosiy pozitsiyadagi nol avtomatik ravishda yagona yechimning yo‘qligini anglatmaydi; sonli hisoblar uchun qisman tanlov joriy ustunning modul bo‘yicha eng katta mavjud elementni oladi, chiqarib tashlash ko‘paytmalarini nazorat qiladi.

6-misol. Sistemaning nechta yechimi bor (II dars). $x + y + z = 6$**, 2x +** 3**y +** 4**z** $= 20$**, 3x +** 4**y +** 5**z** $= k$ **oilasini ko‘rib chiqing. qanday** k **uchun yechimlar yo‘**q**, qanday** k **uchun cheksiz ko‘**p**?**

Yechim. $R_2 \leftarrow R_2 - 2R_1$ va $R_3 \leftarrow R_3 - 3R_1$ qo‘llaymiz: $y +$ 2z $= 8$ va $y +$ 2z $= k - 18$ olinadi. 2-tenglamani 3-tenglamadan ayiramiz: $0 = k - 26$. Agar $k \neq 26$ — yechimlar yo‘q($k = 27bo$lganda $0 = 1$ — tenglamalar mos emas). Agar $k = 26$ — oxirgi tenglama $0 = 0$ bo‘lib, cheklov qo‘shmaydi: erkin o‘zgaruvchi $z = t$ uchun $y = 8 -$ 2t va $x = 6 - y - z = t - 2$, ya’ni $(x, y, z) = (t - 2, 8 - 2t, t), t \in R$ — cheksiz ko‘p yechimlar. 3×3 uchun natijalar jadvali: uchta asosiy element — yagona yechim; [$000|d$] qatori, $d \neq 0$ — yechimlar yo‘q; uchta asosiy elementdan kam, ziddiyatli qator yo‘q — cheksiz ko‘p yechimlar (bitta nol qati o‘z-o‘zidan moslikni kafolatlamaydi — boshqa qatorlarni tekshirish kerak).

7-misol. LU**-faktorizatsiya va qayta foydalanish (II dars). 3-misol sistemasining** $A = LU$ **ni yozing va** $Lc = b, Ux = c$ **ni yeching.**

Yechim. 3-misol ko‘paytmalari: $m_{21} = 2, m_{31} = -1, m_{32} = 2$; ularni diagonal ostiga joylaymiz: $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix}$, va $A = LUU = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & -5 \end{bmatrix}$ bilan; L dagi belgilar o‘zlari

ko‘paytmalardir, chiqarib tashlash operatsiyalari ularning manfiy qiymatlaridan foydalanadi. Ikkita bosqichda yechamiz: $Lc = bc_1 = 1, c_2 = 11, c_3 = -15$ beradi ($togri$ almashtirish); keyin $Ux = c$ orqaga almashtirish orqali (1, 2, 3) olinadi. Afzallik bir nechta sistemalarda bir A matritsasi, lekin turli o‘ng tomonlar bo‘lganda namoyon bo‘ladi: A bir marta ajratiladi, L va U qayta foydalaniladi. Agar qator almashtirishlar kerak bo‘lsa, standart faktorizatsiya — $PA = LU$.

8-misol. A **va** B **mashqlari** (II **dars). Sistemalarni yeching:** $(A)x + y + z = 6$**, 2x** $+$ **3y** $+z = 11, x+$ **2y** $+$ **3z** $= 14$**;** $(B)y+$ **2z** $= 8, x + y + z = 6$**, 2x** $+$ **3y** $+z = 11$**.**

Yechim. $(A)x$ ni chiqaramiz: $R_2 - 2R_1y - z = -1$ beradi; $R_3 - R_1y+$ 2z $= 8$ beradi. Ayirish: 3z $= 9, z = 3, y = z - 1 = 2, x = 6 - 2 - 3 = 1$. Javob (1, 2, 3); tekshirish: $1 + 2 + 3 = 6, 2 + 6 + 3 = 11, 1 + 4 + 9 = 14$. (B) Birinchi tenglamada x yo‘$q(nol$ taklif etilgan asosiy element) — x ni chiqarishdan oldin birinchi ikki qatorni almashtiramiz: $x + y + z = 6, y+$ 2z $= 8$, 2x + 3y $+z = 11$. x ni chiqaramiz: $R_3 - 2R_1y - z = -1$ beradi; $y+$ 2z $= 8$ bilan birga ayirish 3z $= 9, z = 3, y = 2, x = 1$ beradi. Javob (1, 2, 3); tekshirish: $2 + 2 \cdot 3 = 8, 1 + 2 + 3 = 6, 2 + 6 + 3 = 11$.

9-misol. C **va** D **mashqlari** (II **dars).** (C) **qanday** k **uchun** $x + y + z = 6$**, 2x** $+$ **3y** $+$ **4z** $= 20$**, 3x** $+$ **4y** $+$ **5z** $= k$ **sistemasida yechimlar bo‘lmaydi, qanday** k **uchun cheksiz ko‘**p **yechimlar borligini aniqlang.** $(D)R_3 \leftarrow R_3 - 4R_1$ **operatsiyasining matritsasi topiling.**

Yechim. $(C)Bu6$-misol oilasi: uchinchi tenglamani chiqarib tashlagandan so‘ng $u0 = k - 26$ ga aylanadi, shuning uchun $k \neq 26$ bo‘lganda yechim yo‘$q, k = 26$ bo‘lganda cheksiz ko‘p yechim mavjud: $(x, y, z) = (t - 2, 8 - 2t, t), t \in R$; yagona yechim beradigan k qiymati mavjud emas (koeffitsientlar matritsida faqat ikkita asosiy element bor). (D) Amaliyot faqat 3-qatorni o‘zgartiradi, undan $4 \cdot (1-$qator) ayirilib: $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$; $E \cdot A$ ko‘paytmasining uchinchi qatori (A ning 3-qatori) $- 4 \cdot (A$ ning 1-qatori) ga teng, qolgan qatorlar o‘zgarmaydi.

10-misol. Gauss-Jordan usuli bilan teskari matritsa (III **dars).** $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$ **matritsasining teskari matritsini toping, birlik matritsa qo‘shib va chap to‘liq birlik matritsaga keltiring.**

Yechim. Nazariya: $n \times nA$ matritsasining teskarisi A^{-1} matritsasi bo‘lib, $AA^{-1} = A^{-1}A = I$; qatorlardagi har bir elementar amaliyot chapdan elementar matritsaga ko‘paytirishga teng, va agar $E_k \cdots E_2E_1A = I$ bo‘lsa, u holda $A^{-1} = E_k \cdots E_2E_1$ — shuning uchun I ga qo‘llanadigan xuddi shu amaliyotlar A^{-1} ni beradi: $[A|I] \to [I|A^{-1}]$; har bir amaliyot butun kengaytirilgan qatorga ta’sir qiladi. 1-qadam: $[A|I] = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{bmatrix}$. 2-qadam (birinchi asosiy element ostini chiqarish): $R_2 \leftarrow R_2 - R_1$: $\begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{bmatrix}$. 3-qadam (ikkinchi ostida): $R_3 \leftarrow R_3 - R_2$: $\begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{bmatrix}$.

4-qadam (uchinchisida ustida): $R_2 \leftarrow R_2 - R_3$: $\begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 2 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{bmatrix}$. 5-qadam (ikkinchisida ustida): $R_1 \leftarrow R_1 - R_2$: $\begin{bmatrix} 1 & 0 & 0 & 3 & -2 & 1 \\ 0 & 1 & 0 & -2 & 2 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{bmatrix}$. Chap to‘liq birlik matritsaga aylandi, shuning uchun $A^{-1} = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{bmatrix}$. Ko‘paytirish orqali tekshirish: $A \cdot A^{-1} = I$(masalan, (2, 1) elementi: $1 \cdot 3 + 2 \cdot (-2) + 1 \cdot 1 = 0$).

11-misol. Qachon usul ishlamaydi $(III$ **dars).** $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ **matritsasining teskarisi yo‘qligini ko‘rsating.**

Yechim. Kvadrat matritsa faqat $[A|I]$ dagi chap blok birlik matritsaga keltirilgagina teskariga ega. $R_2 \leftarrow R_2 - 2R_1$ ni qo‘llaymiz: $\begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & -2 & 1 \end{bmatrix}$. Chap blokda nol qator paydo bo‘ldi — I ni hosil qilish uchun yetarlicha asosiy elementlar yo‘q; matritsa singulyar (notekis), teskarisi mavjud emas. Eslatma: Gauss chiqarib tashlash yuqori uchburchak matritsada to‘xtaydi, Gauss-Jordan esa har bir asosiy element 1 ga, va har bir asosiy element ustidagi va ostidagi barcha elementlar nolga aylanguncha davom ettiradi.

4 7-hafta — Transponirlash, almashtirishlar va vektorli-fazolar (Chiziqli algebra I)

4.1 Chiziqli algebra I, 7-hafta: formulalar katalogi

[Manba: 7-hafta ma’ruzasi ”Transponirlar, almashtirishlar, vektorli fazolar va kichik fazolar” (Strang, Introduction to Linear

Algebra, 4th ed., §2.7 pp. 108–120, §3.1 pp. 121–132)]

4.5.1 6-haftani takrorlash: $A = LU, LDU, PA = LU$

Gauss chiqarishi $E_k \cdots E_1 A = U$ beradi. Qatorlarni almashtirmasdan, chiqarish ko‘paytmalari pastburchakli L(bosh diagonalda birlar) to‘planadi va $A = LU$: masalan, $A = \begin{bmatrix} 2 & 1 \\ 4 & 5 \end{bmatrix}, R2 \leftarrow R2 - 2R1$ bosqichi $U = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$, $L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ ni beradi. Bosh elementlarni ajratib, $A = LD$ŕ hosil qilamiz: $D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ — bosh elementlardan diagonal, Ū (U chiziq bilan) = [1 1/2; 0 1] — bosh diagonalda birlar bilan yuqori burchakli.

$$A = LU, A = LD; almashtirishlaruchun — PA = LU$$

Agar bosh element nol bo‘lsa ($A = \begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}$), qatorlar almashtiriladi: $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} PA =$

$\begin{bmatrix}3 & 4\\0 & 2\end{bmatrix}$ ni beradi; barcha qator almashtirishlari P da yoziladi.

4.5.2 Transponirlash: ta’rif va asosiy identifikatlar

Agar A — $m \times n$ matritsa bo‘lsa, $A^T o$‘lchami $n \times m$ bo‘ladi va elementlari $(A^T)_{ij} = a_{ji}$ bo‘ladi: ustunlar satrlarga aylanadi. Misol: $A = \begin{bmatrix}1 & 2 & 3\\4 & 5 & 6\end{bmatrix}$ $A^T = \begin{bmatrix}1 & 4\\2 & 5\\3 & 6\end{bmatrix}$ ni beradi.

$$(A^T)^T = A, (A+B)^T = A^T + B^T, (cA)^T = cA^T$$

Element bo‘yicha isbot (yigindiga uchun): $((A+B)^T)_{ij} = (A+B)_{ji} = a_{ji}+b_{ji} = (A^T+B^T)_{ij}$.

$$(AB)^T = B^T A^T$$

Ko‘paytuvchilar tartibi teskarisiga o‘zgaradi ($A^T B^T$ hatto aniqlanmagan bo‘lishi mumkin). Isbot: $((AB)^T)_{ij} = (AB)_{ji} = \Sigma_k a_{jk} b_{ki} = \Sigma_k (B^T)_{ik}(A^T)_{kj} = (B^T A^T)_{ij}$. Dars tekshiruvi: $A = \begin{bmatrix}1 & 2\\0 & 1\end{bmatrix}$, $B = \begin{bmatrix}2 & 0\\3 & 1\end{bmatrix}$: $(AB)^T = \begin{bmatrix}8 & 3\\2 & 1\end{bmatrix} = B^T A^T$.

$$(A^{-1})^T = (A^T)^{-1}$$

Invertirilgan A uchun transponir qilamiz $AA^{-1} = I$: $(A^{-1})^T A^T = I$ ni olamiz; xuddi shunday $A^T (A^{-1})^T = I$. Demak, transponir va teskari amallar o‘zaro almashtiriladi.

4.5.3 Simmetrik va antimetrik matritsalar; $A^T A$ ko‘paytmasi

Matritsa S simmetrik, agar $S^T = S (s_{ij} = s_{ji})$; misol: $S = \begin{bmatrix}2 & -1\\-1 & 3\end{bmatrix}$. Matritsa K antimetrik, agar $K^T = -K$; misol: $K = \begin{bmatrix}0 & 4\\-4 & 0\end{bmatrix}$; asosiy diagonal nolli: $k_{ii} = -k_{ii} \Rightarrow k_{ii} = 0$.

$$A = (A + A^T)/2 + (A - A^T)/2 = S + K, S^T = S, K^T = -K$$

Ixtiyoriy kvadrat matritsa shunday ajraladi; dars misoli: $A = \begin{bmatrix}1 & 4\\2 & 3\end{bmatrix} \to S = \begin{bmatrix}1 & 3\\3 & 3\end{bmatrix}$, $K = \begin{bmatrix}0 & 1\\-1 & 0\end{bmatrix}$; tekshirish: $S + K = A$.

$$(A^T A)^T = A^T A, x^T A^T A x = (Ax)^T (Ax) = ||Ax||^2 \geq 0$$

$A^T A$ ko‘paytmasi har doim simmetrikdir ($(AB)^T = B^T A^T$ maxsus holati), va $x^T A^T A x = ||Ax||^2 \geq 0$ har qanday x uchun. Misol: $A = \begin{bmatrix}1 & 2\\0 & 1\end{bmatrix}$ $A^T A = \begin{bmatrix}1 & 2\\2 & 5\end{bmatrix}$ ni beradi; $R^T R$ matritsalar — eng kichik kvadratlar asosi. Skalyar va tashqi ko‘paytma: ustunlar $u, v \in R^n$ uchun $u^T v$ soni skalyar ko‘paytma, uv^T esa $n \times n$ matritsadir. Misol: $u = \begin{bmatrix}1\\2\end{bmatrix}, v = \begin{bmatrix}3\\4\end{bmatrix}$: $u^T v = 11$, $uv^T = \begin{bmatrix}3 & 4\\6 & 8\end{bmatrix}$.

4.5.4 Perestanovkalar: $P^T = P^{-1}$, oltita 3×3 matritsa, qisman elementlar almashtirish

Perestanovka matritsi birlik matritsaning satrlari almashtirilishidan hosil bo‘ladi: har bir satr va ustunda aynan bitta birlik bo‘ladi. Misol: $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ [x_1; x_2; x_3] ni [x_2; x_3; x_1] ga o‘tkazadi.

$$PA — matritsa A bilan almashtirilgan qatorlar; AQ — almashtirilgan ustunlar bilan$$

Chapdan ko‘paytirish satrlarni almashtiradi: $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$, $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \rightarrow PA = \begin{bmatrix} 3 & 4 \\ 5 & 6 \\ 1 & 2 \end{bmatrix}$. O‘ngdan ko‘paytirish ustunlarni almashtiradi: $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, Q = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \rightarrow AQ = \begin{bmatrix} 2 & 3 & 1 \\ 5 & 6 & 4 \end{bmatrix}$. Umuman olganda $PA \neq AP$.

$$P^T P = I, ya'ni P^{-1} = P^T; kelib chiqishi : ||Px|| = ||x||, detP = \pm 1$$

P ustunlari — turli koordinata birlik vektorlari, shuning uchun ularning skalyar ko‘paytmalari I ni hosil qiladi: $P^T P = I$, bundan $P^{-1} = P^T$. Natijalar: $||Px|| = ||x||$ va $detP = \pm 1$. Bir qatorlarni almashtirish determinant belgisini o‘zgartiradi: $det \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = -1$, shuning uchun $det(PA) = -detA$; ikki almashtirish belgisini qaytaradi. n tartibli almashtirishlar soni aynan $n!$; $n = 3$ bo‘lganda ular oltita (Strang, §2.7) — uchtasi juft ($detP = +1$) va uchtasi toq ($detP = -1$); $P^{-1} = P^T$, shuning uchun $(P_{32}P_{21})^{-1} = P_{21}P_{32}$:

Almashtirish	**Matritsa (qator bo‘yicha)**	$\det P$	$P^{-1} = P^T$
I	$\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$	$+1$	I
P_{21} (1, 2-qatorlarni almashtirish)	$\begin{bmatrix}0&1&0\\1&0&0\\0&0&1\end{bmatrix}$	-1	P_{21}
P_{32} (2, 3-qatorlarni almashtirish)	$\begin{bmatrix}1&0&0\\0&0&1\\0&1&0\end{bmatrix}$	-1	P_{32}
P_{31} (1, 3-qatorlarni almashtirish)	$\begin{bmatrix}0&0&1\\0&1&0\\1&0&0\end{bmatrix}$	-1	P_{31}
$P_{32}P_{21}$ (tsiklik)	$\begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix}$	$+1$	$P_{21}P_{32}$
$P_{21}P_{32}$ (tsiklik)	$\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}$	$+1$	$P_{32}P_{21}$

Asosiy elementning qisman tanlovi (partial pivoting): agar asosiy element nol bo‘lsa, qatorlar almashtiriladi, joriy ustundagi moduli eng katta bo‘lgan nomzod tanlanadi. Misol: $A = [10^{-5}1;\ 1\ 1]$: almashtirishsiz ko‘paytuvchi $\frac{1}{10}^{-5} = 10^5$, almashtirishdan keyin — $10^{-5}/1 = 10^{-5}$. Almashtirishlar P da yoziladi: $PA = LU$.

4.5.5 Vektorli makonlar: ta’rif va misollar

Vektorli makon R ustida — vektorlar to‘plami qo‘shish va skalyar ko‘paytirish amaliyotlari bilan, vektorli makon aksiomalatiga javob beradi: nol vektor, qarama-qarshi elementlar, assotsiativlik, distributivlik va $h.k.$ Dars misollari: R^n, haqiqiy matritsalar, ko‘phadlar, uzluksiz funksiyalar — ”vektor” har qanday ob’ekt bo‘lishi mumkin.

4.5.6 Alt makonlar va yopilishni tekshirish

$$W \subseteq V — kichik fazolar \Leftrightarrow 0 \in W;$$
$$u + v \in W barchau, v \in W uchun; cu \in W barchau \in W, c \in R uchun$$

Teng ravishda: bo‘sh bo‘lmagan W au + bv kombinatsiyalariga nisbatan yopiq. Tekshirish: (1) nol vektor; (2) yig‘indi; (3) skalyar ko‘paytirish — har qanday bandning buzilishi inkor etadi.

- • Boshdan o‘tuvchi to‘g‘ri chiziq: $W = \{(t,\ 2t)\} = span\{(1,\ 2)\}$: $u = (s,\ 2s), v$

$= (t,\ 2t)$ uchun au + bv $= (as + bt, 2(as + bt))$ yana W da — alt makon.

- • Siljigan to‘g‘ri chiziq (kontremisol): $S = \{(x, y)$: $y = 2x + 1\}$ — nol vektor

tenglamaga javob bermaydi; alt makon koordinata boshidan o‘tishi shart.

- • Boshlang‘ich nuqtadan o‘tuvchi tekislik: $W = \{(x, y, z)\colon\ x + y + z = 0\} =$

$span\{(1,\ 0,\ -1),\ (0,\ 1,\ -1)\}$ $(z = -x - ydan : (x, y, z) = x(1, 0, -1) + y(0, 1, -1))$; $x + y + z = 1$ tekisligi — kichik fazo emas $(nolyoq)$.

- • R^3 geometriyasi: kichik fazolar — faqat $\{0\}$, 0 dan o‘tuvchi to‘g‘ri chiziqlar, 0

dan o‘tuvchi tekisliklar va butun R^3.

- • Kesishish va birlashma: $U \cap W$ — kichik fazo; birlashma — albatta emas: x

va yo‘qlari R^2 ning kichik fazolari, lekin $(1, 0) + (0, 1) = (1, 1)$ hech qanday oqda joylashmagan.

- • Matritsalar va ko‘phadlar: simmetrik 2×2 — barcha 2×2 matritsalar kichik

fazosi $(yig$indilar va ko‘paytmalar simmetrik); $p(0) = 0$ bo‘lgan ko‘phadlar — kichik fazo, $p(0) = 1$ bo‘lganlari — yo‘q.

4.5.7 Chiziqli birikmalar va chiziqli qoplamalar $(span)$

$$spanv_1, ..., v_k = c_1v_1 + ... + c_kv_k : c_i \in R$$

Chiziqli qoplama — berilgan vektorlarning barcha birikmalarining to‘plami; u har doim kichik fazodir: agar $u = \Sigma a_i v_i$ va $v = \Sigma b_i v_i$ bo‘lsa, su + tv = $\Sigma(sa_i + tb_i)v_i$ — yana bir birikma. Dars misoli: $(3, 5)span\{(1,\ 1),\ (1,\ -1)\}$ da joylashganmi? $c_1 + c_2 = 3, c_1 - c_2 = 5$ tenglamalarini yechamiz $\to c_1 = 4, c_2 = -1$ — ha.

4.5.8 Ustunlar fazosi $C(A)$

$$A = [a_1a_2...a_n] \to Ax = x_1a_1 + ... + x_na_n;$$
$$C(A) = Col(A) = spana_1, ..., a_n$$

Ax ko‘paytmasi — A ustunlarining chiziqli birikmasi. Ustunlar fazosi $C(A) = \text{Col}(A)$ — ustunlarning chiziqli qoplamasi, $m \times n$ matritsa A uchun R^m ning kichik fazosi. Misol: $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 0 & 0 \end{bmatrix}$ — ustunlar proporsional, $\text{Col}(A) = span\{(1,\ 2,\ 0)\}$ — R^3 dagi to‘g‘ri chiziq.

$$tizimAx = byechimgaega \Leftrightarrow b \in C(A)$$

$Ax = b$ tizimida yechim bo‘lishi shart va shartli ravishda b ustunlar fazosida joylashgan bo‘lishi bilan bog‘liq: yechim xb ni beradigan ustunlar birikmasining koeffitsientlarini tanlaydi. Yechim mavjudligining asosiy mezonini bilish kerak.

4.5.9 Nol fazosi $N(A)$

$$N(A) = Null(A) = x : Ax = 0$$

Nol fazosi — bir xil tizimning yechimlari to‘plami; bu R^n ning kichik fazosi: agar $Au = Av = 0$ bo‘lsa, $A(su + tv) = 0$. Misol: $A = \begin{bmatrix}1 & 2\end{bmatrix}$: $x_1 + 2x_2 = 0 x_1 = -2x_2$ beradi, shuning uchun Null$(A) = span\{(-2, 1)\}$ — to‘g‘ri chiziq.

$$x : Ax = 0 — hardoim kichik fazolar;$$
$$x : Ax = b, b \neq 0 — kichik fazolar emas (0 nio' zi chiga olmaydi)$$

Bir xil tizimning yechimlari kichik fazoni hosil qiladi; bir xil bo‘lmagan $(b \neq 0)$ tizimning yechimlari — yo‘q: nol vektor ularga tegishli emas. Agar $Ax = b$ tizimi yechimga ega bo‘lsa, uning yechimlari nol fazosining siljishi: $x = x_p + x_n$, bunda x_p — maxsus yechim, $x_n \in$ Null(A)(affin to‘plami). 8-hafta o‘rtasiga qadar: A^T va $(AB)^T$ ni hisoblash, $P^{-1} = P^T$ va $PA = LU$ dan foydalanish, yopiq holatni tekshirish, qoplamalarni va tegishlikni $C(A)$ topish.

4.2 Chiziqli algebra I, 7-hafta: dars mashqlari — savollar va to‘liq yechimlar

[Manba: 7-hafta dars mashqlari (Yechilgan), Chiziqli algebra $I(Kuz2026)$ — 20 dars mashqlasi; arifmetika pythonda
tekshirilgan]

Mashq 1. 2×2 **matritsani transponirovka qilish.** $A = \begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}$ **uchun** A^T **ni hisoblang.**

Yechim. Transponirovka qatorlar va ustunlarni almashtiradi: $(A^T)_{ij} = a_{ji}$. Shuning uchun $A^T = \begin{bmatrix}1 & 3\\ 2 & 4\end{bmatrix}$; ayniqsa, $(A^T)_{12} = a_{21} = 3$, diagonal o‘zgarmaydi.

Mashq 2. Yig‘indining transponirovkasi. $A = \begin{bmatrix}1 & 0\\ 2 & 3\end{bmatrix}, B = \begin{bmatrix}2 & 1\\ 0 & -1\end{bmatrix}$ **uchun** $(A+B)^T$ **ni hisoblang.**

Yechim. Avval qo‘shamiz: $A + B = \begin{bmatrix}3 & 1\\ 2 & 2\end{bmatrix}$. Keyin transponirovka qilamiz: $(A + B)^T = \begin{bmatrix}3 & 2\\ 1 & 2\end{bmatrix}$. Tekshirish: $A^T + B^T = \begin{bmatrix}1 & 2\\ 0 & 3\end{bmatrix} + \begin{bmatrix}2 & 0\\ 1 & -1\end{bmatrix} = \begin{bmatrix}3 & 2\\ 1 & 2\end{bmatrix}$ — mos keldi.

Mashq 3. A^TA **ko‘paytmasi.** $A = \begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix}$ **uchun** A^TA **ni toping.**

Yechim. $A^TA = \begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix} \cdot \begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 2\\ 2 & 5\end{bmatrix}$ — natija simmetrik, har doimgidek $(ISA26-$mashq).

Mashq 4. Ko‘**paytmaning transponirovkasi.** $A = \begin{bmatrix}1 & 1\\ 0 & 2\end{bmatrix}, B = \begin{bmatrix}2 & 0\\ 1 & 3\end{bmatrix}$ **uchun** $(AB)^T$ **ni toping.**

Yechim. $AB = \begin{bmatrix} 3 & 3 \\ 2 & 6 \end{bmatrix}$, shuning uchun $(AB)^T = \begin{bmatrix} 3 & 2 \\ 3 & 6 \end{bmatrix}$. Tenglik xuddi shuni beradi: $B^T A^T = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 3 & 6 \end{bmatrix}$ — ko‘paytuvchilar teskari tartibda.

Mashq 5. Simmetriklikni tekshirish. $A = \begin{bmatrix} 2 & -3 \\ -3 & 1 \end{bmatrix}$ **matritsasi simmetrikmi?**

Yechim. $A^T = \begin{bmatrix} 2 & -3 \\ -3 & 1 \end{bmatrix} = A$. *Ha*, simmetrik: mos keladigan diagonal tashqari elementlar bir xil ($a_{12} = a_{21} = -3$), diagonal transponirovlanganda o‘zgarmaydi.

Mashq 6. Simmetrik va antimetrik qismlar. $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ **ni simmetrik va antimetrik qismlarga ajrating.**

Yechim. $A^T = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$; $S = (A + A^T)/2 = \begin{bmatrix} 1 & 3 \\ 3 & 3 \end{bmatrix}$(diagonal tashqari elementlar o‘rtacha), $K = (A - A^T)/2 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. Tekshirish: $S + K = A, S^T = S, K^T = -K$.

Mashq 7. Qatorlarni almashtirish: PA**.** $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ **uchun** PA **ni toping.**

Yechim. $PA = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$: ko‘paytmaning birinchi qatori A ning ikkinchi qatori va aksincha. Chapdan ko‘paytirish 1 va 2-qatorlarni almashtiradi.

Mashq 8. Ustunlarni almashtirish: AP**.** $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ **uchun** AP **ni toping.**

Yechim. $AP = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$: ko‘paytmaning birinchi ustuni A ning ikkinchi ustuni, va aksincha. O‘ngdan ko‘paytirish 1 va 2 ustunlarini almashtiradi; shu bilan birga $AP \neq PA$.

Masala 9. Permutatsiya matritsasining teskari matritsasi. $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ **uchun** P^{-1} **ni toping.**

Yechim. Permutatsiya matritsasi uchun $P^{-1} = P^T = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$. Tekshirish: $P^T P = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} = I$ — P ning ustunlariortonormalangan.

Masala 10. 1 va 3-qatorlarni almashtirish. 3×3 **matritsaning 1 va 3-qatorlarini almashtiruvchi permutatsiya matritsasi** P **ni toping.**

Yechim. 1 va 3-qatorlarni almashtirish, I_3 ga qo‘llanilganda: $P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$. Har qanday A uchun ko‘paytma PA uning birinchi va uchinchi qatorlarini almashtiradi; masalan, $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ uchun $\begin{bmatrix} 7 & 8 & 9 \\ 4 & 5 & 6 \\ 1 & 2 & 3 \end{bmatrix}$ hosil bo‘ladi.

Masala 11. $PA = LU$ **nol bosh elementi bilan.** $A = \begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}$ **uchun** P, L, U **ni toping.**

Yechim. Bosh element $a_{11} = 0$ — almashtirmasdan bo‘lmaydi. Qatorlarni almashtiramiz: $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $PA = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}$. Bosh element 3 ostida 0 — chiqarish tugadi, ko‘paytchilar yo‘q: $L = I_2, U = PA = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}$; natija: $PA = LU$.

Masala 12. LU **qatorlarni almashtirmasdan.** $A = \begin{bmatrix} 2 & 1 \\ 6 & 5 \end{bmatrix}$ **uchun** L, U **ni qatorlarni almashtirmasdan toping.**

Yechim. Bosh element $2 \neq 0$, almashtirish kerak emas; ko‘paytchi $m_{21} = \frac{6}{2} = 3$. Qadam $R2 \leftarrow R2 - 3R1$: $\begin{bmatrix} 6 & 5 \end{bmatrix} - 3 \cdot \begin{bmatrix} 2 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 2 \end{bmatrix}$, shuning uchun $U = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$, $L = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$. Tekshirish: $LU = \begin{bmatrix} 2 & 1 \\ 6 & 5 \end{bmatrix} = A$.

Masala 13. Altbo‘shlik: boshdan o**‘tuvchi to‘**g**‘ri chiziq.** $W =$ **{**(x, y)**:** $y =$ **3x}** R^2 **ning altbo‘shligimi?**

Yechim. Ha. $W = \{t(1, 3)\text{: } t \in R\}$: nol vektort $= 0$ ga mos keladi. $u = (s,$ 3s$), v = (t,$ 3t) va har qanday a, b uchun: au + bv $= (as + bt, 3(as + bt)) \in W$ — yopiqlik mavjud; W — altbo‘shlik (boshdan o‘tuvchi to‘g‘ri chiziq).

Masala 14. Misol: siljigan to‘g**‘ri chiziq.** $S =$ **{**(x, y)**:** $y =$ **3x** $+2$**} altbo‘shlikmi?**

Yechim. $Yo‘q$. Nol vektor tenglamani qanoatlantirmaydi: $3 \cdot 0 + 2 = 2 \neq 0$. Nol vektor testi inkor uchun yetarli: altbo‘shlik koordinata boshini $o‘z$ ichiga olishi shart.

Masala 15. Altbo‘shlik: tekislik. $W =$ **{**(x, y, z)**:** $x + y - z = 0$**} altbo‘shlikmi?**

Yechim. Ha. $f(x, y, z) = x + y - z$ va $u, v \in W$ uchun: $f(au + bv) = a \cdot f(u) + b \cdot f(v) = 0$, demak au + bv $\in W$; va $f(0, 0, 0) = 0$. W — altbo‘shlik (boshdan o‘tuvchi tekislik).

Masala 16. Vektorlarning chiziqli birikmasi. (4, 1) ni (1, 1) va (1, −1) vektorlarining birikmasi sifatida ifodalang.

Yechim. $c_1(1, 1) + c_2(1, -1) = (4, 1)$ ni qidiramiz: $c_1 + c_2 = 4, c_1 - c_2 = 1$. $Qo‘$shamiz: $2c_1 = 5, c_1 = \frac{5}{2}, c_2 = \frac{3}{2}$. Tekshirish: $(\frac{5}{2} + \frac{3}{2}, \frac{5}{2} - \frac{3}{2}) = (4, 1)$.

Masala 17. Proporsional vektorlarning qobig‘i**. (1, 2, 0) va (2, 4, 0) vektorlarining chiziqli qobig‘ini toping.**

Yechim. Ikkinchi vektor birinchisining ikki barobar katta: $(2,4,0)=2(1,2,0)$, shuning uchun har qanday birlik (1, 2, 0) ga karrali. Natija: $span${(1, 2, 0), (2, 4, 0)} = $\{t(1,2,0)\colon\ t\in R\}$ — R^3 dagi to‘g‘ri chiziq.

Masala 18. 1×3 **matritsaning nol bo‘shlig‘i.** $A=\begin{bmatrix}1 & 2 & -1\end{bmatrix}$ **uchun Null**(A) **ni toping.**

Yechim. $Ax=0$ tenglama: $x_1+2x_2-x_3=0$. Erkin o‘zgaruvchilar $x_2=s, x_3=t$, unda $x_1=-2$s $+t$ va $x=s(-2,1,0)+t(1,0,1)$. Natija: Null$(A)=span${(−2, 1, 0), (1, 0, 1)} — R^3 dagi tekislik. Tekshirish: $1\cdot(-2)+2\cdot 1-0=0$ va $1\cdot 1+2\cdot 0-1=0$.

Masala 19. Ustunlar bo‘shlig‘i. $A=\begin{bmatrix}1 & 2\\ 2 & 4\\ 0 & 0\end{bmatrix}$ **uchun Col**(A) **ni toping; unga (3, 6, 0) vektori tegishlimi?**

Yechim. Ustunlar $a_1=(1,2,0)$ va $a_2=(2,4,0)=2a_1$ proporsional, shuning uchun Col$(A)=span${(1, 2, 0)} — R^3 dagi to‘g‘ri chiziq. $(3,6,0)=3(1,2,0)$ ekanligidan, vektor shu to‘g‘ri chiziqda joylashgan: ha (masalan, $A\cdot\begin{bmatrix}3\\0\end{bmatrix}=(3,6,0)$).

Masala 20. Transponir va 3×3 **determinant.** $A=\begin{bmatrix}1 & 2 & 0\\ 0 & 1 & 1\\ 0 & 0 & 2\end{bmatrix}$ **uchun** A^T **va** $det(A)$ **ni toping.**

Yechim. $A^T=\begin{bmatrix}1 & 0 & 0\\ 2 & 1 & 0\\ 0 & 1 & 2\end{bmatrix}$. A yuqori uchburchakli, shuning uchun $detA=1\cdot 1\cdot 2=2$(diagonal elementlar ko‘paytmasi); transponir determinantni o‘zgartirmaydi: $det(A^T)=2$.

4.3 Chiziqli algebra I, 7-hafta: mashqlar ISA — savollar va to‘liq yechimlar

[Manba: 7-hafta Mustaqil o‘qitish topshiriqlari, Chiziqli algebra $I(Kuz2026)$ — 50 ta masala yechimsiz; yechimlar python tili

yordamida tuzilgan va tekshirilgan]

Mustaqil o‘qitish topshiriqlari — baholanmaydigan mustaqil amaliyot (korsatmalar: vektor-ustunlar, oraliq bosqichlar, javoblar asosida kichik fazolar). Fayl qismlari: A(1–12) — transponiranіye, B(13–20) — almashtirishlar va pivoting, C(21–32) — kichik fazolar mezonlari, D(33–42) — chiziqli qobiqlar va matritsa kichik fazolari, E(43–46) — aralash qo‘llanilishi, F(47–50) — murakkab masalalar. Mashqlar 1–20 auditoriya masalalari 1–20 bilan bir xil (pastda — ixcham yechimlar, to‘liqlari yuqorida); qolgan 30 tasi to‘liq yechilgan.

Mashq 1. $A=\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}$ **uchun** A^T **ni hisoblang.**

Yechim. $A^T=\begin{bmatrix}1 & 3\\ 2 & 4\end{bmatrix}$ (satrlar ustunlarga aylandi); batafsil — yuqoridagi 1-masala.

Mashq 2. $A = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}, B = \begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix}$ **uchun** $(A+B)^T$ **ni hisoblang.**

Yechim. $A+B = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$, shuning uchun $(A+B)^T = \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix} = A^T + B^T$.

Mashq 3. $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ **uchun** $A^T A$ **ni toping.**

Yechim. $A^T A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$ — natija simmetrik.

Mashq 4. $A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}, B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$ **uchun** $(AB)^T$ **ni toping.**

Yechim. $AB = \begin{bmatrix} 3 & 3 \\ 2 & 6 \end{bmatrix}$, demak $(AB)^T = \begin{bmatrix} 3 & 2 \\ 3 & 6 \end{bmatrix} = B^T A^T$.

Mashq 5. $A = \begin{bmatrix} 2 & -3 \\ -3 & 1 \end{bmatrix}$ **matritsasi simmetrikmi?**

Yechim. Ha: $A^T = \begin{bmatrix} 2 & -3 \\ -3 & 1 \end{bmatrix} = A$ — diagonal tashqari elementlar bir xil ($a_{12} = a_{21} = -3$).

Mashq 6. $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ **ni simmetrik va antisimmetrik qismlarga ajrating.**

Yechim. $S = (A + A^T)/2 = \begin{bmatrix} 1 & 3 \\ 3 & 3 \end{bmatrix}$, $K = (A - A^T)/2 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$; tekshirish: $S + K = A$.

Mashq 7. $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ **uchun** PA **ni toping.**

Yechim. $PA = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$ — chapdan ko‘paytirish 1 va 2-qatorlarni almashtirdi.

Mashq 8. A **va** P **uchun** AP **ni toping** ($A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$)**.**

Yechim. $AP = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$ — o‘ngdan ko‘paytirish 1 va 2-ustunlarni almashtirdi; shu bilan birga $AP \neq PA$.

Mashq 9. $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ **uchun** P^{-1} **ni toping.**

Yechim. $P^{-1} = P^T = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$; tekshirish: $P^T P = I$.

Mashq 10. 3×3 **matritsasi 1 va 3-qatorlarni almashtiruvchi** P **almashtirish matritsasini toping.**

Yechim. $P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ — I_3 da 1 va 3-qatorlarni almashtirish; PA har qanday matritsaning birinchi va uchinchi qatorlarini almashtiradi.

Mashq 11. $A = \begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}$ **uchun** P, L, U **ni toping.**

Yechim. Nol bosh element almashtirishni talab qiladi: $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $PA = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}$; allaqachon yakunlangan, shuning uchun $L = I_2, U = PA$; natija: $PA = LU$.

Mashq 12. $A = \begin{bmatrix} 2 & 1 \\ 6 & 5 \end{bmatrix}$ **uchun qatorlarni almashtirmasdan** L, U **ni toping.**

Yechim. $m_{21} = \frac{6}{2} = 3$; $R2 \leftarrow R2 - 3R1$ bosqichi $U = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ ni beradi; $L = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$; tekshirish: $LU = A$.

Mashq 13. $W =$ **{**(x, y)**:** $y =$ **3x}** R^2 **ning kichik fazomiy?**

Yechim. Ha: $W = \{t(1, 3)$: $t \in R\}$, nol vektor $t = 0$ da, va $a(s,$ 3s$) + b(t,$ 3t$) =$ (as + bt, $3(as + bt)) \in W$.

Mashq 14. $S =$ **{**(x, y)**:** $y =$ **3x** $+2$**} kichik fazomiy?**

Yechim. $Yo'q$: (0, 0) tenglamani qanoatlantirmaydi $(0 \neq 3 \cdot 0 + 2)$ — nol vektor yo'q.

Mashq 15. $W =$ **{**(x, y, z)**:** $x + y - z = 0$**} kichik fazomiy?**

Yechim. Ha: $f(x, y, z) = x + y - z$ chiziqli, $f(0, 0, 0) = 0$ va $f(au + bv) = a \cdot f(u) + b \cdot f(v) = 0 u, v \in W$ uchun.

Mashq 16. (4, 1) ni (1, 1) va (1, −1) vektorlari kombinatsiyasi sifatida ifodalang.

Yechim. $c_1 + c_2 = 4, c_1 - c_2 = 1 \rightarrow c_1 = \frac{5}{2}, c_2 = \frac{3}{2}$; tekshirish: $(\frac{5}{2})(1, 1) + (\frac{3}{2})(1, -1) = (4, 1)$.

Mashq 17. (1, 2, 0) va (2, 4, 0) vektorlarining chiziqli qobig'ini toping.

Yechim. $(2, 4, 0) = 2(1, 2, 0)$, shuning uchun $span$**{**(1, 2, 0), (2, 4, 0)**}** $= \{t(1, 2, 0)$: $t \in R\}$ — to'g'ri chiziq.

Mashq 18. $A = \begin{bmatrix} 1 & 2 & -1 \end{bmatrix}$ **uchun Null**(A) **ni toping.**

Yechim. $x_1 + 2x_2 - x_3 = 0 \rightarrow x_1 = -2$s $+t, x_2 = s, x_3 = t$; Null$(A) = span$**{**(−2, 1, 0), (1, 0, 1)**}**.

Mashq 19. $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 0 & 0 \end{bmatrix}$ **uchun Col**(A) **ni toping; (3, 6, 0) unga tegishlimi?**

Yechim. Ustunlar proporsional: Col$(A) = span$**{**(1, 2, 0)**}**; $(3, 6, 0) = 3(1, 2, 0)$ bo'lgani uchun, vektor Col(A) ga tegishli.

Mashq 20. $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$ **uchun** A^T **va** $det(A)$ **ni toping.**

Yechim. $A^T = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix}$; A yuqori uchburchakli, shuning uchun $detA = 1 \cdot 1 \cdot 2 = 2 = det(A^T)$.

Mashq 21. O**‘lchamlar.** $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ **uchun** A^T **ni hisoblang va** o**‘lchamlarni ko‘rsating.**

Yechim. A — 2×3; $A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$ — 3×2: o‘lchamlar almashtiriladi, $(m \times n)^T = n \times m$.

Mashq 22. Isbot $(yig$**indi).** $(A+B)^T = A^T + B^T$ **ni element bo‘yicha isbotlang.**

Yechim. $A = (a_{ij}), B = (b_{ij}) m \times no$‘lchamli va har bir i, j uchun: $((A+B)^T)_{ij} = (A+B)_{ji} = a_{ji} + b_{ji} = (A^T)_{ij} + (B^T)_{ij} = (A^T + B^T)_{ij}$. Elementlar bir xil — matritsalar teng.

Mashq 23. Isbot $(ko$**paytma).** $(AB)^T = B^T A^T$ **tenglamani mos matritsalar uchun isbotlang.**

Yechim. A — $m \times n, B$ — $n \times p$ bo‘lsin. Har bir i, j uchun: $((AB)^T)_{ij} = (AB)_{ji} = \Sigma_k a_{jk} b_{ki} = \Sigma_k (B^T)_{ik} (A^T)_{kj} = (B^T A^T)_{ij}$. Demak, $(AB)^T = B^T A^T$: ko‘paytuvchilar tartibi o‘zgaradi.

Mashq 24. 3×2 **uchun** $A^T A$ **ko‘paytmasi.** $A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 3 \end{bmatrix}$ **uchun** $A^T A$ **ni hisoblang (faylda matritsa** 3×2 **qator bo‘yicha yozilgan).**

Yechim. $A^T A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 2 & 10 \end{bmatrix}$ — natija simmetrik, nazariya bashorat qilgandek (mashq 26).

Mashq 25. Simmetrik va antimetrik qismlar. $A = \begin{bmatrix} 1 & 3 \\ -2 & 4 \end{bmatrix}$ **uchun simmetrik va antimetrik qismlarini toping.**

Yechim. $A^T = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$. $S = (A + A^T)/2 =$ [1 1/2; 1/2 4], $K = (A - A^T)/2 =$ [0 5/2; −5/2 0]. Tekshirish: $S + K = A, S^T = S, K^T = -K$.

Mashq 26. $A^T A$ **simmetrikligi. Ixtiyoriy haqiqiy matritsa** A **uchun** $A^T A$ **simmetrik ekanligini isbotlang.**

Yechim. $(A^T A)^T = A^T (A^T)^T = A^T A ((AB)^T = B^T A^T$ qoidasi): matritsa o‘ziga transponir bo‘lgani uchun simmetrikdir ($Am \times no$lchamli bo‘lganda $A^T A$ ko‘paytmasi $n \times n$ kvadrat matritsadir).

Mashq 27. Skalyar va tashqi ko‘paytma. $u=(2,-1)^T, v=(3,4)^T$ **uchun** u^Tv **va** $\mathbf{uv}^T$ **ni hisoblang.**

Yechim. $u^Tv = 2\cdot 3+(-1)\cdot 4 = 2$ — son (skalyar ko‘paytma); $\mathrm{uv}^T = \begin{bmatrix}2\\-1\end{bmatrix}\cdot\begin{bmatrix}3 & 4\end{bmatrix} = \begin{bmatrix}6 & 8\\-3 & -4\end{bmatrix}$ — 2×2 matritsa (tashqi ko‘paytma).

Mashq 28. 1 va 2-qatorlarni almashtirish. I_3 **matritsasining 1 va 2-qatorlarini almashtiruvchi permutatsiya matritsasini tuzing.**

Yechim. $P=\begin{bmatrix}0&1&0\\1&0&0\\0&0&1\end{bmatrix}$ — I_3 matritsasi 1 va 2-qatorlari almashtirilgan. Har qanday uch qatorli A uchun PA birinchi ikki qatorni almashtiradi: $P\cdot\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix} = \begin{bmatrix}4&5&6\\1&2&3\\7&8&9\end{bmatrix}$.

Mashq 29. PA **va** AP **solishtirish.** $P=\begin{bmatrix}0&1\\1&0\end{bmatrix}$ **va** $A=\begin{bmatrix}2&5\\7&1\end{bmatrix}$ **uchun** PA **va** AP **ni solishtiring.**

Yechim. $PA=\begin{bmatrix}0&1\\1&0\end{bmatrix}\cdot\begin{bmatrix}2&5\\7&1\end{bmatrix}=\begin{bmatrix}7&1\\2&5\end{bmatrix}$ — qatorlar almashtirilgan; $AP=\begin{bmatrix}2&5\\7&1\end{bmatrix}\cdot\begin{bmatrix}0&1\\1&0\end{bmatrix}=\begin{bmatrix}5&2\\1&7\end{bmatrix}$ — ustunlar almashtirilgan. $PA\neq AP$: chap ko‘paytirish qatorlarga, o‘ng ko‘paytirish esa ustunlarga ta’sir qiladi.

Mashq 30. $P^TP=I$ **tekshiruvi.** $P=\begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix}$ **uchun** $P^TP=I$ **ekanligini tekshiring.**

Yechim. $P^T=\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}$; $P^TP=\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}\cdot\begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix}=I$. P^T qatorlari — P ustunlari (e_2,e_3,e_1), ularning skalyar ko‘paytmalari δ_{ij} ni beradi; natijada: $P^{-1}=P^T$.

Mashq 31. Permutatsiya matritsasining determinantini topish. 1 va 3-qatorlarni almashtiruvchi permutatsiya matritsasining determinantini toping.

Yechim. $P=\begin{bmatrix}0&0&1\\0&1&0\\1&0&0\end{bmatrix}$; bitta qatorlarni almashtirish belgini o‘zgartiradi: $detP=-1$.

Birinchi qator bo‘yicha yoyish orqali tekshirish: $1\cdot(-1)^{1+3}\cdot det\begin{bmatrix}0&1\\1&0\end{bmatrix}=1\cdot 1\cdot(-1)=-1$.

Mashq 32. 2×2 **uchun** $PA=LU$**.** $A=\begin{bmatrix}0&1\\4&2\end{bmatrix}$ **uchun** P,L,U **ni toping.**

Yechim. Bosh element $a_{11}=0$ — almashtirish kerak: $P=\begin{bmatrix}0&1\\1&0\end{bmatrix}$, $PA=\begin{bmatrix}4&2\\0&1\end{bmatrix}$. Bosh element 4 ostida 0 bor, chiqarish tugallandi: $L=I_2, U=\begin{bmatrix}4&2\\0&1\end{bmatrix}$; natija: $PA=LU$.

Mashq 33. Qisman bosh elementni tanlash. Qisman bosh elementni tanlash (partial pivoting) usulini $A=\begin{bmatrix}0.001&1\\2&1\end{bmatrix}$ ga qo‘llang; almashtirish oldingi va keyingi ko‘paytuvchilarni solishtiring.

Yechim. Almashtirmasdan: bosh element 0.001, ko‘paytuvchi $m_{21}=\frac{2}{0}.001=2000$ — juda katta ($2000\cdot$(birinchi qator) ayirish xatolarni kuchaytirardi). Qisman tanlov birinchi ustundagi modul bo‘yicha eng kandidatni ($son 2$) oladi; almashtirgandan keyin ($P=\begin{bmatrix}0&1\\1&0\end{bmatrix}$, $PA=\begin{bmatrix}2&1\\0.001&1\end{bmatrix}$) ko‘paytuvchi $m_{21}=0.\frac{001}{2}=0.0005$. Almashtirish ko‘paytuvchini 2000 dan 0.0005 gacha kamaytirdi ($4\cdot10^6$ marta).

Mashq 34. Altbo‘shmi? $x+2y=0$. $W=\{(x,y)$: $x+2y=0\}$ to‘plamining altbo‘shlik ekanligini tekshiring.

Yechim. Ha. $(1)0+2\cdot0=0$ — nol vektor W da. $(2)x+2y=0$ va $x'+2y'=0$ dan $(x+x')+2(y+y')=0$ kelib chiqadi. $(3)(cx)+2(cy)=c(x+2y)=0$. Natija: $W=span\{(-2, 1)\}$ — koordinata boshidan o‘tgan to‘g‘ri chiziq.

Mashq 35. Altbo‘shmi? $x+2y=3$. $W=\{(x,y)$: $x+2y=3\}$ to‘plamining altbo‘shlik ekanligini tekshiring.

Yechim. $Yo‘q$: $0+2\cdot0=0\neq3$ — nol vektor W da yo‘q, altbo‘shlik 0 ni $o‘z$ ichiga olishi kerak (siljigan to‘g‘ri chiziq koordinata boshidan o‘tmaydi).

Mashq 36. Altbo‘shmi? $x\geq0$. $W=\{(x,y)$: $x\geq0\}$ to‘plamining altbo‘shlik ekanligini tekshiring.

Yechim. $Yo‘q$. Nol vektor W da joylashgan va W qo‘shish nisbatan yopiq (manfiy bo‘lmagan abssissalar yig‘indisi manfiy bo‘lmagan). Ammo skalyarlar nisbatidan yopiqlik buzilgan: $(1,0)\in W$, lekin $(-1)\cdot(1,0)=(-1,0)\notin W$. Yarim tekislik — altbo‘shlik emas.

Mashq 37. Untor? $x-y+z=0$. Tekshirish kerakki, $W=\{(x,y,z)$: $x-y+z=0\}$ untor bo‘lishini.

Yechim. Ha: $f(x,y,z)=x-y+z$ chiziqli, $f(0,0,0)=0$ va $f(au+bv)=a\cdot f(u)+b\cdot f(v)=0 u,v\in W$ uchun. Aniq: $x=y-z(x,y,z)=y(1,1,0)+z(-1,0,1)$ beradi, ya’ni $W=span\{(1, 1, 0), (-1, 0, 1)\}$ — koordinata boshidan o‘tgan tekislik.

Mashq 38. Chiziqli birikma. (6, 2) ni (1, 1) va (1, −1) vektorlarining birikmasi sifatida ifodalash.

Yechim. $c_1+c_2=6, c_1-c_2=2\to c_1=4, c_2=2$. Tekshirish: $4(1,1)+2(1,-1)=(6,2)$.

Mashq 39. Qobiq geometriyasi. $span${(1, 2, 0), (0, 1, 2)} ni geometrik tasvirlang.

Yechim. Vektorlar proporsional emas — qobiq ikki o‘lchovli: koordinata boshidan o‘tgan tekislik. Uning nuqtalari: $s(1,2,0)+t(0,1,2)=(s, 2s+t, 2t)$; parametrlarni chiqarib tashlab ($s=x, t=z/2$), $4x-2y+z=0$ hosil bo‘ladi. Tekshirish: $4\cdot1-2\cdot2+0=0$ va $4\cdot0-2\cdot1+2=0$.

Mashq 40. Nol-untor 2×3. $A=\begin{bmatrix}1&1&1\\0&1&1\end{bmatrix}$ (2×3 matritsa, qator bo‘yicha yozilgan) uchun Null(A) ni toping.

Yechim. 2-qatoridan: $x_2=-x_3$; 1-qatoridan: $x_1=-x_2-x_3=0$. Erkin o‘zgaruvchi $x_3=t$: $x=t(0,-1,1)$. Natija: Null$(A)=span${(0, −1, 1)} — R^3 dagi to‘g‘ri chiziq. Tekshirish: $\begin{bmatrix}1&1&1\\0&1&1\end{bmatrix}\cdot\begin{bmatrix}0\\-1\\1\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix}$.

Mashq 41. Ustunlar bo‘shlig‘i 3×2. $A=\begin{bmatrix}1&2\\2&4\\3&6\end{bmatrix}$ uchun Col(A) ni toping.

Yechim. Ustunlar $a_1=(1,2,3)$ va $a_2=(2,4,6)=2a_1$ proporsional, shuning uchun har qanday birikma a_1 ning karrali: Col$(A)=span${(1, 2, 3)} — R^3 dagi to‘g‘ri chiziq ((t, 2t, 3t) vektorlari).

Mashq 42. Ustunlar bo‘shlig‘iga tegishlilik. (1, 1, 1) vektori $A=\begin{bmatrix}1&0\\0&1\\1&1\end{bmatrix}$ (3×2 matritsa) matritsasi uchun Col(A) ustunlar bo‘shligiga tegishlimi?

Yechim. $c_1(1,0,1)+c_2(0,1,1)=(1,1,1)$ ni qidiraylik: birinchi ikki koordinata $c_1=c_2=1$ beradi, lekin uchinchisi $2\neq1$ — ziddiyat. Natija: $(1,1,1)\notin$ Col$(A)(Col(A)$—$z=x+y$ tekisligi, $1\neq1+1$).

Mashq 43. Untorlarning kesishishi. Ikkita untorning kesishmasi untor ekanligini isbotlang.

Yechim. $U,W\subseteq V$ untorlari uchun: $(1)0\in U\cap W$. (2) Agar $u,v\in U\cap W, u+v\in U$ va $u+v\in W(har$ birining yopiq bo‘lishi), demak $u+v\in U\cap W$. (3) Agar $u\in U\cap W$, cu $\in U$ va cu $\in W$, demak cu $\in U\cap W$. Test bajarildi.

Mashq 44. Untorlarning birlashmasi. Birlashmasi untor bo‘lmagan ikki untor keltiring.

Yechim. R^2 da o‘qlarni $U=\{(x,0)\}$ va $W=\{(0,y)\}$ deb olaylik — ikkalasi ham untor. Birlashmada (1, 0) va (0, 1) bor, lekin ularning yig‘indisi (1, 1) hech qaysi o‘qda joylashmagan. Birlashma qo‘shishga nisbatan yopiq emas — untor emas.

Mashq 45. Simmetrik 3×3 matritsalar. 3×3 barcha haqiqiy matritsalar fazasining simmetrik 3×3 matritsalar to‘plami untormi?

Yechim. Ha. (1) Nol matritsa simmetrik. (2) Simmetrik S,T uchun $(S+T)^T=S^T+T^T=S+T$. $(3)(cS)^T=c\cdot S^T=$ cS. Yopiq bo‘lish bajarildi — untor.

Mashq 46. $p(0)=1$ bo‘lgan ko‘phadlar. $p(0)=1$ shartini qanoatlantiruvchi ko‘phadlar to‘plami untormi?

Yechim. Yo‘q. Nol ko‘phad $p(0)=0\neq1$ beradi — nol vektori yo‘q. Bundan tashqari, $p(x)=1$ va $q(x)=1+x$ uchun: $(p+q)(0)=2\neq1$ — qo‘shishning yopiq bo‘lishi buzilgan. ($p(0)=0bo$lgan ko‘phadlar, aksincha, — untor.)

Mashq 47. Murakkab masala. Ixtiyoriy haqiqiy A matritsasi uchun Null$(A^TA)=$ Null(A) ni isbotlang.

Yechim. Ikkki kiritish. (1) Agar $Ax = 0, u$ holda $A^TAx = A^T(Ax) = 0$, ya'ni Null$(A) \subseteq$ Null(A^TA). $(2)A^TAx = 0$ deb faraz qilaylik; chapdan x^T ko'paytiramiz: $x^TA^TAx = ||Ax||^2 = 0$. Ax koordinatalari kvadratlari yig'indisi nolga teng, demak, $Ax = 0$. Natija: Null$(A^TA) =$ Null(A); xulosa — A va A^TA bir xil nol-untor va rangga ega $(eng$ kichik kvadratlar fakti).

Mashq 48. Murakkab masala. P — almashtiruvchi matritsa bo'lsa, Col$(PA) = P$ Col(A) ni isbotlang.

Yechim. Ta'rifiga ko'ra Col$(PA) =$ {PAx: $x \in R^n$ }, P Col$(A) =$ {Pb: $b \in$ Col(A)}. (1) Agar $y =$ PAx, u holda $y = P(Ax)Ax \in$ Col(A) bilan — demak $y \in P$ Col(A). (2) Agar $y = Pbb = Ax \in$ Col(A) bilan, u holda $y =$ PAx $\in$ Col(PA). Natija: Col$(PA) = P$ Col(A) — qatorlarni almashtirish ustunlar bo'shlig'ining har bir nuqtasini bir xil almashtirish bilan o'zgartiradi $(b \to Pb$ birma-bir, P teskari).

Mashq 49. Murakkab masala. Agar W — untor bo'lsa, $span(W) = W$ ni isbotlang.

Yechim. $span(W)$ — barcha chekli kombinatsiyalar $c_1w_1 + ... + c_kw_k$ to'plami, bunda $w_i \in W$. (1) Har bir kombinatsiya W da joylashgan: $c_iw_i \in W$(skalyarlar), W elementlari yig'indisi W da $(qo$shish). (2) Har bir $w \in W1 \cdot w$ kombinatsiyasidir. Natija: $span(W) = W$ — bu kichik fazo o'zining barcha kombinatsiyalarini $o'z$ ichiga oladi.

Mashq 50. Murakkab masala. $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ **uchun Col(A) va Null(A) ni tasvirlang.**

Yechim. Ustunlar $a_1 = (1, 0, 0), a_2 = (2, 1, 0), a_3 = (0, 1, 0) = a_2 - 2a_1$; uchinchi qator nolli, shuning uchun barcha ustunlar $z = 0$ tekisligida joylashgan, a_1, a_2 esa mustaqil. Natija: Col$(A) = span${(1, 0, 0), (2, 1, 0)} = {$(s + 2t, t, 0)$} — bu $z = 0$ tekisligi. Nol-fazo: 2-qatordan $x_2 = -x_3$, 1-qatordan $x_1 = -2x_2 = 2x_3$; $x_3 = t$: $x = t(2, -1, 1)$. Natija: Null$(A) = span${(2, −1, 1)} — to'g'ri chiziq. Tekshirish: $A \cdot \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.